18.708: Topics in Algebra

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Contents

How strange to actually have to see the path of your journey in order to make it.

—Neal Shusterman, [Shu16]

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Minicourse 1

de Rham Cohomology in Mixed Characteristic

These talks were given by Alexander Petrov.

1.1 February 2

Here we go.

1.1.1 Algebraic de Rham Cohomology

Let's begin by describing what we mean by de Rham cohomology. We will consider a smooth variety X over an algebraically closed field F.

Definition 1.1 (smooth)**.** We say that a variety X over a field F is *smooth* if and only if $\Omega_{X/F}$ is a vector bundle of rank $\dim X$ on each connected component. Here, on an affine open subset $U \subseteq X$, recall that $\Omega_{X/F}(U)$ is spanned by symbols of the form $f\,dg$, where the symbol d is (as usual) F-linear and satisfies the Leibniz rule.

Definition 1.2 (algebraic de Rham cohomology)**.** Fix a smooth variety X over a field F. Then one can iterate the F-linear map $d\colon \mathcal{O}_X \to \Omega_{X/F}$ to a map $d\colon \Omega^i_{X/F} \to \Omega^{i+1}_{X/F}$ for each i, where $\Omega^i_{X/F} := \wedge^i \Omega_{X/F}$. We now define the *de Rham complex* to the complex

$$\Omega^\bullet_{X/F}\colon 0 \to \mathcal{O}_X \xrightarrow{d} \Omega^1_{X/F} \xrightarrow{d} \cdots,$$

and we define the *de Rham cohomology* $\mathrm{H}^n_{\mathrm{dR}}(X/F)$ to be the nth hypercohomology of $\Omega^\bullet_{X/F}$. Here, hypercohomology means the total cohomology of some produced acyclic double complex which resolves the complex (e.g., a Čech resolution). Note that this hypercohomology is merely a vector space over F.

Example 1.3. The map $d\colon \Omega^1_{X/F} \to \Omega^2_{X/F}$ is given by $d(f\,dg) = df \wedge dg$.

Example 1.4. Suppose that X is affine. Then vector bundles are already acyclic, so the hypercohomology does nothing. Thus,

$$\mathrm{H}^n_{\mathrm{dR}}(X/F) = \mathrm{H}^n\left(X; 0 \to \mathcal{O}_X \xrightarrow{d} \Omega^1_{X/F} \xrightarrow{d} \cdots\right).$$

As usual, this is $\ker\left(d|_{\Omega^n}\right) / \operatorname{im}\left(d|_{\Omega^{n-1}}\right)$.

Remark 1.5. If X is affine and $i > \dim X$, then $\Omega^i_{X/F}$ vanishes, so the algebraic de Rham cohomology also vanishes.

Remark 1.6. A different definition is required for non-smooth X. Roughly speaking, one should embed into a smooth variety and take cohomology there.

Here is one way to convince ourselves that this is a reasonable cohomology theory.

Theorem 1.7 (Grothendieck)**.** Suppose that X is a smooth variety over $\mathbb{C}$. Then there is a canonical isomorphism

$$\mathrm{H}^n_{\mathrm{B}}(X(\mathbb{C}); \mathbb{C}) \to \mathrm{H}^n_{\mathrm{dR}}(X/\mathbb{C}).$$

Here, the left-hand side is Betti cohomology (also called singular cohomology).

Sketch. We argue in the case that X is affine. Then $X(\mathbb{C})$ already has a notion of $\Omega^{i,\mathrm{an}}_{X/\mathbb{C}}$ given by the holomorphic forms. Algebraic forms embed into holomorphic ones, which produces a morphism

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}_X(X) \arrow[r] \arrow[d, hook] & \Omega^1_{X/\mathbb{C}}(X) \arrow[r] \arrow[d, hook] & \Omega^2_{X/\mathbb{C}}(X) \arrow[r] \arrow[d, hook] & \cdots \\
0 \arrow[r] & \mathcal{O}^{\mathrm{an}}_X(X) \arrow[r] & \Omega^{1,\mathrm{an}}_{X/\mathbb{C}}(X) \arrow[r] & \Omega^{2,\mathrm{an}}_{X/\mathbb{C}}(X) \arrow[r] & \cdots
\end{tikzcd}$$

of complexes. It then turns out that this is an isomorphism on cohomology, so we reduce to comparing analytic de Rham cohomology with singular cohomology.

This is now a problem of analysis. One can pass from holomorphic differentials to smooth differentials via a similar process, which produces another morphism

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}^{\mathrm{an}}_X(X) \arrow[r] \arrow[d, hook] & \Omega^{1,\mathrm{an}}_{X/\mathbb{C}}(X) \arrow[r] \arrow[d, hook] & \Omega^{2,\mathrm{an}}_{X/\mathbb{C}}(X) \arrow[r] \arrow[d, hook] & \cdots \\
0 \arrow[r] & C^\infty(X(\mathbb{C}), \mathbb{C}) \arrow[r] & \Omega^1_{C^\infty}(X(\mathbb{C})) \arrow[r] & \Omega^2_{C^\infty}(X(\mathbb{C})) \arrow[r] & \cdots
\end{tikzcd}$$

of complexes, which is also an isomorphism on complexes. We are now reduced to the setting of de Rham's theorem for real manifolds. ■

Example 1.8. Consider $X := \mathbb{A}^1_{\mathbb{C}} \setminus \{0\} = \operatorname{Spec} k[t, 1/t]$.

- Our differential map $d\colon \mathbb{C}[t, 1/t] \to \mathbb{C}[t, 1/t]\, dt$ sends t^n to $nt^{n-1}\, dt$. Thus, $\mathrm{H}^0_{\mathrm{dR}}(X)$ is one-dimensional given by the constants, and $\mathrm{H}^1_{\mathrm{dR}}(X)$ is one-dimensional spanned by dt/t.
- The point above works also for holomorphic differentials. The interesting bit is in degree 1, where the point is that there is no global antiderivative for dx/x.
- On the other hand, $X(\mathbb{C}) = \mathbb{C} \setminus \{0\}$ is homotopy equivalent to the circle, so we expect its singular cohomology to be supported in degrees 0 and 1, where it should be one-dimensional.

Corollary 1.9 (Artin vanishing)**.** If X is an affine algebraic complex smooth variety, then $\mathrm{H}^n(X(\mathbb{C});\mathbb{C}) = 0$ for $n > \dim X$.

Proof. The algebraic de Rham cohomology complex vanishes above $\dim X$. ■

Corollary 1.10. Fix a smooth variety X over $\mathbb{C}$. Then $\mathrm{H}^n_{\mathrm{dR}}(X/\mathbb{C})$ is finite-dimensional.

Proof. Pass to singular cohomology. ■

Remark 1.11. This corollary still admits algebraic proofs in characteristic zero by working with holonomic $\mathcal{D}$-modules. Pavel Etingof claims that there is an algebraic proof using the fact that the direct image of a holonomic $\mathcal{D}$-module is a holonomic $\mathcal{D}$-module.

We would like to point out that our de Rham cohomology is algebraic but still interesting.

Remark 1.12. Suppose that X is smooth over $\mathbb{Q}$. Base-changing by a field is exact, so

$$\mathrm{H}^n_{\mathrm{dR}}(X/\mathbb{Q})_{\mathbb{C}} \cong \mathrm{H}^n_{\mathrm{dR}}(X_{\mathbb{C}}/\mathbb{C}).$$

However, Theorem 1.7 grants an isomorphism to $\mathrm{H}^n(X(\mathbb{C});\mathbb{C}) \cong \mathrm{H}^n_{\mathrm{B}}(X(\mathbb{C});\mathbb{Z})_{\mathbb{C}}$. Notably, we then find a lattice and a rational structure over in some complex vector space, but the comparison between the two is quite interesting mathematically (and amounts to the study of periods).

Example 1.13. In the case that $X = \mathbb{A}^1_{\mathbb{Q}} \setminus \{0\}$, the comparison between $\mathrm{H}^1_{\mathrm{dR}}(X/\mathbb{Q})_{\mathbb{C}}$ and $\mathrm{H}^1_{\mathrm{B}}(X(\mathbb{C});\mathbb{Z})$ is mediated by a constant $2\pi i$. Indeed, once unwinds the de Rham theorem, this amounts to the statement that a contour integral of dx/x going once around the origin is $2\pi i$.

1.1.2 Frobenius Structure

We now pass to positive characteristic. Let k be a perfect field of positive characteristic p, and we may still consider a smooth variety X.

Remark 1.14. If k is perfect, then $\Omega^1_{X/k} = \Omega^1_{X/\mathbb{F}_p}$ by doing some thinking about inseparable extensions. The moral is that

$$y^{1/p}\,dy = d\left((y^{1/p})^p\right),$$

so the coefficients can be brought down when everything is a pth power.

This cohomology is rather strangely behaved.

Example 1.15. Take $X := \mathbb{A}^1_k$. The de Rham cohomology still lives in degrees zero and one, so we would like to study the kernel and cokernel of the k-linear map $d\colon k[t] \to k[t]\,dt$ given by $t^n \mapsto nt^{n-1}$.

- We see that $\mathrm{H}^0_{\mathrm{dR}}(\mathbb{A}^1_k/k) = \ker d$ is spanned by t^{pi} for each i.
- We see that $\mathrm{H}^1_{\mathrm{dR}}(\mathbb{A}^1_k/k) = \operatorname{im} d$ is infinite-dimensional because the differentials $t^{mp-1}\,dt$ fail to be in the image. In fact, these classes form a basis.

Let's try to view these infinite-dimensional groups as a feature instead of a bug. Indeed, it turns out that the de Rham complex has some extra structure. The de Rham complex

$$0 \to \mathcal{O}_X \xrightarrow{d} \Omega^1_{X/k} \xrightarrow{d} \Omega^2_{X/k} \xrightarrow{d} \cdots$$

is merely made of sheaves of k-vector spaces over X. In characteristic zero, this is all the structure present, but in characteristic p, we have more structure.

Notation 1.16. Fix a variety X over a field k of characteristic p. For a sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules, we define

$$\mathcal{F}^p := \{f^p : f \in \mathcal{O}_X\}$$

to locally be given by the pth powers.

The moral is that $d(f^p) = 0$ always, so the de Rham complex is in fact $\mathcal{O}_X^p$-linear! Let's attempt to codify this.

Definition 1.17 (relative Frobenius)**.** Fix a scheme X over a field k of characteristic p. Then there is an *absolute Frobenius* $F_{\text{abs}} \colon X \to X$ which is the identity on topological spaces and the pth power on sheaves. This is a morphism of schemes but not of k-schemes (in general). The *relative Frobenius* $F \colon X \to X^{(1)}$ is the morphism fitting into the following diagram.

$$\begin{tikzcd}
X \arrow[dr, dashed] \arrow[drr, bend left, "F_{\text{abs}}"] \arrow[ddr, bend right] & & \\
& X^{(1)} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d] \\
& k \arrow[r, "F_{\text{abs}}"] & k
\end{tikzcd}$$

Remark 1.18. Note that $X^{(p)}$ is isomorphic to X as a scheme but not as a k-scheme! However, we now benefit because the relative Frobenius F is morphism of k-schemes.

Remark 1.19. The relative Frobenius $F \colon X \to X^{(1)}$ is finite flat of degree $p^{\dim X}$

Example 1.20. If $X = \operatorname{Spec} k[t_1, \ldots, t_n]$, then $X^{(1)} = \operatorname{Spec} k\left[t_1^p, \ldots, t_n^p\right]$. Thus, we see that the embedding

$$k\left[t_1^p, \ldots, t_n^p\right] \subseteq k[t_1, \ldots, t_n]$$

is indeed finite flat of degree p^n.

We now see that

$$0 \to F_*\mathcal{O}_X \xrightarrow{d} F_*\Omega^1_{X/k} \xrightarrow{d} F_*\Omega^2_{X/F} \to \cdots$$

is a complex of quasicoherent sheaves on $X^{(1)}$. In fact, because F is finite flat, these are all vector bundles: $F_*\mathcal{O}_X$ has rank $p^{\dim X}$ and $F_*\Omega^i_{X/k}$ has rank $p^{\dim X}\binom{\dim X}{i}$. Because $\mathcal{O}_{X^{(1)}} = (F_*\mathcal{O}_X)^p$, we see that this complex is in fact $\mathcal{O}_{X^{(1)}}$-linear.

Example 1.21. Take $X = \operatorname{Spec} k[t]$. Then $X^{(1)} := \operatorname{Spec} k\left[t^p\right]$, and $d \colon k[t] \to k[t]\,dt$ is $k\left[t^p\right]$-linear! Thus, $\mathrm{H}^i_{\mathrm{dR}}(X/k)$ was required to be given by $k\left[t^p\right]$-modules, which explains why we received vector spaces of infinite dimension.

Note that passing through F_* is not going to adjust the underlying k-vector spaces, so

$$\mathrm{H}^n_{\mathrm{dR}}(X/k) = \mathbb{H}^n_{\mathrm{Zar}}\left(X^{(1)}; 0 \to F_*\mathcal{O}_X \xrightarrow{d} F_*\Omega^1_{X/k} \xrightarrow{d} F_*\Omega^2_{X/k} \xrightarrow{d} \cdots\right).$$

To see why this has globalized the $\mathcal{O}_X^p$-linearity, we need the Cartier isomorphism.

Theorem 1.22 (Cartier isomorphism)**.** Fix a smooth variety X over a perfect field k. Then there is a canonical isomorphism

$$\mathcal{H}^i(F_*\Omega^\bullet_X) \cong \Omega^i_{X^{(1)}}.$$

Here, the left-hand side is a coherent $\mathcal{O}_{X^{(1)}}$-module.

Remark 1.23. This is a reason why characteristic p may be more convenient than characteristic 0: one could still try to understand $\mathcal{H}^i(\Omega^\bullet_{X/k})$ when $\operatorname{char} k = 0$, but this has no easy answer.

Example 1.24. Consider $X = \mathbb{A}^1_k$. Then $\mathcal{H}^1$ is given by the module

$$\frac{k[t]\,dt}{d(k[t])},$$

which our formalism now remembers is a $k[t^p]$-module. And indeed, we can show that this is isomorphic to $k[t^p] \cdot t^{p-1}\,dt$. Setting $s := t^p$, we know that $\Omega^1_{X^{(1)}/k}$ is given by the module $k[s]\,ds$, so our isomorphism of modules is given by sending ds to $t^{p-1}\,dt$. One can even check that this isomorphism is canonical in the sense that it will not change under automorphisms of $\mathbb{A}^1$.

We will prove Theorem 1.22 later after a detour.

1.1.3 Crystalline Cohomology

We continue with our perfect field k of positive characteristic p. Our story so far has taken a variety X over a field k, and then we have produced some (total) complex $\mathrm{R}\Gamma_{\mathrm{dR}}(X/k)$ in the derived category $D(\mathrm{Vec}_k)$. Crystalline cohomology will allow us to produce an answer in characteristic 0 instead of characteristic p. The idea is to "choose" a lift to characteristic p and then check that the answer is independent of the lift.

The correct formalism for this lifting is that of a "formal scheme."

Definition 1.25 (Witt ring)**.** Fix a perfect field k of characteristic p. Then there is a ring $W(k)$ satisfying that

- $W(k)$ is p-torsion-free,
- $W(k)/p \cong k$, and
- $W(k)$ is the limit of the $W(k)/p^n$ as $n \to \infty$.

This ring $W(k)$ turns out to be unique up to unique isomorphism. We may write $W_n(k) := W(k)/p^n$.

Example 1.26. One can see that $W(\mathbb{F}_p) = \mathbb{Z}_p$ and $W(\overline{\mathbb{F}}_p)$ is its unramified closure.

Remark 1.27. There is a completely explicit construction of $W(k)$, but it is rather involved: given a p-torsion-free ring R, we identify $W(R) := R^{\mathbb{N}}$ but with ring structure chosen so that

$$(a_0, a_1, a_2, \ldots) \mapsto a_0^{p^n} + pa_1^{p^{n-1}} + \cdots + p^n a_n$$

is a ring homomorphism $W(R) \to R$. It turns out that this ring structure is given by some polynomials (called "ghost coordinates"), so we are allowed to define $W(k)$. From a higher level, it turns out that $W(k)$ is the unique deformation of k, which exists because $\Omega^1_{k/\mathbb{F}_p} = 0$.

Definition 1.28 (formal scheme)**.** Fix a perfect field k of characteristic p. A *p-adic formal scheme* X is a collection of schemes X_n over $W_n(k)$ equipped with isomorphisms

$$X_{n+1} \times_{W_{n+1}(k)} W_n(k) \to X_n.$$

The structure sheaf $\widehat{\mathcal{O}}_X$ is the inverse limit of the $\mathcal{O}_{X_n}$s.

Example 1.29. Given a scheme Y over $W(k)$, we can produce a formal scheme $\widehat{Y}$ with $\widehat{Y}_n := Y \times_{W(k)} W_n(k)$ and the induced internal isomorphisms.

Remark 1.30. If X_1 is affine, then all the nilpotent thickenings are affine, so we may say that the full formal scheme is affine.

Remark 1.31. We can even define $\widehat{\Omega} X^1_{\widetilde{X}}$.

We can now describe crystalline cohomology.

Theorem 1.32. Fix a perfect field k of positive characteristic p. Then there is a functor sending smooth k-varieties X to a complex $\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k))$ in the derived category $D(\mathrm{Mod}_{W(k)})$ satisfying the following.

(a) There is a quasi-isomorphism $\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k)) \otimes^{\mathbb{L}}_{W(k)} k \cong \mathrm{R}\Gamma_{\mathrm{dR}}(X/k)$.

(b) If $\widetilde{X}$ is a smooth formal scheme over $W(k)$ (meaning that $\widetilde{X}_n$ is smooth over $W_n(k)$ for all n), then the scheme $X := \widetilde{X} \times_{W(k)} k$ has

$$\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k)) \cong \mathrm{R}\Gamma_{\mathrm{dR}}\left(X; \widehat{\mathcal{O}}_{\widetilde{X}} \xrightarrow{d} \widehat{\Omega}^1_{\widetilde{X}} \xrightarrow{d} \cdots\right).$$

This right-hand side can be thought of as $\mathrm{R}\Gamma_{\mathrm{dR}}(\widetilde{X}/W(k))$.

Remark 1.33. Here, (a) immediately tells us that the cohomology of $\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k))$ is not expected to be finitely generated.

Remark 1.34. There is something remarkable here, which is that choosing two different lifts of X to a smooth formal scheme produces the same cohomology!

Remark 1.35. It turns out that flatness is equivalent to smoothness in this context.

1.2 February 4

Today we continue our discussion of crystalline cohomology.

1.2.1 Example of Crystalline Cohomology

We continue working over a perfect field k of characteristic p. Let's run an example.

Remark 1.36. If our formal scheme $\widetilde{X}$ is smooth and affine, then $\mathrm{R}\Gamma_{\mathrm{dR}}(X/W(k))$ can be computed directly as the complex

$$0 \to \mathcal{O}(\widetilde{X}) \xrightarrow{d} \Omega^1(\widetilde{X}) \xrightarrow{d} \Omega^2(\widetilde{X}) \xrightarrow{d} \cdots.$$

Example 1.37. Take $X = \mathbb{A}^1_k$, which we can deform to a formal scheme $\widetilde{X} = \widehat{\mathbb{A}}^1_k$. Then

$$\mathcal{O}(\widetilde{X}) = \lim W_n[t],$$

which we may refer to as $W\langle t\rangle$. Here, $W\langle t\rangle$ contains those power series whose terms are more and more divisible by p. Similarly, we can compute that $\Omega^1(\widetilde{X}) = W\langle t\rangle\, dt$. For example, we now see that W has characteristic zero, so the kernel of the differential $d\colon W\langle t\rangle \to W\langle t\rangle\, dt$ is exactly the constants. On the other hand, Theorem 1.32 tells us that $\mathrm{H}^1_{\mathrm{dR}}(\widetilde{X}/W(k))$ is required to be quite interesting because the de Rham complex of $\mathbb{A}^1_k$ is interesting.

Remark 1.38. One can extend Theorem 1.32 slightly: even if some k-scheme X can only be deformed to a smooth scheme X_n over $W_n(k)$, then we still have

$$\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k)) \otimes^{\mathbb{L}}_{W(k)} W_n(k) \cong \mathrm{R}\Gamma_{\mathrm{dR}}(X_n/W_n(k)).$$

The above example falls under the paradigm where the formal scheme comes from a genuine scheme. Here are some motivational remarks about this case.

Example 1.39. Let's consider the case where the formal scheme comes from a genuine scheme. Given a smooth scheme Y over $W(k)$ with formal completion $\widehat{Y}$, then $\mathrm{R}\Gamma_{\mathrm{dR}}(\widehat{Y}/W(k))$ is the derived p-adic completion of $\mathrm{R}\Gamma_{\mathrm{dR}}(Y/W(k))$. Explicitly, one takes a complex representing $\mathrm{R}\Gamma_{\mathrm{dR}}(Y/W(k))$ and takes a p-completion of each module. (This turns out to be a well-defined operation on the derived category.)

Example 1.40. We continue the previous example. Suppose further that Y is smooth and proper. Then it turns out that $\mathrm{R}\Gamma_{\mathrm{dR}}(Y/W(k))$ can be represented by finitely generated projective $W(k)$-modules, so the p-completion does nothing, so

$$\mathrm{R}\Gamma_{\mathrm{dR}}(\widehat{Y}/W(k)) \cong \mathrm{R}\Gamma_{\mathrm{dR}}(Y/W(k)).$$

1.2.2 The Mysterious Functor

Crystalline cohomology produces the following strange functoriality.

Corollary 1.41. Fix a smooth formal scheme $\widetilde{X}$ over $W(k)$. Given an endomorphism f of $\widetilde{X}$ such that f is the identity $\pmod p$, then f^* is the identity on the cohomology groups $\mathrm{H}^n_{\mathrm{dR}}(\widetilde{X}/W(k))$.

Proof using Theorem 1.32. By Theorem 1.32, one may pass to crystalline cohomology of the reduction $X := X \times_k W(k)$, where the result has no content. ■

Theorem 1.32 admits the following version in characteristic zero, where we heuristically replace the deformation $W(k) \to k$ with $\mathbb{C}[[t]] \to \mathbb{C}$.

Proposition 1.42. Given a smooth formal scheme $\widetilde{X}$ over $\mathbb{C}[[t]]$ with reduction X over $\mathbb{C}$. Then

$$\mathrm{R}\Gamma_{\mathrm{dR}}(\widetilde{X}/\mathbb{C}[[t]]) \cong \mathrm{R}\Gamma_{\mathrm{dR}}(X/\mathbb{C}) \otimes_{\mathbb{C}} \mathbb{C}[[t]].$$

Sketch. The idea of the proof is to pass through the larger de Rham cohomology $\mathrm{R}\Gamma_{\mathrm{dR}}(\widetilde{X}/\mathbb{C})$; by definition, this is the inverse limit of the de Rham cohomology of the reductions $\widetilde{X}_n$. By the Poincaré lemma, it turns out that $\mathrm{R}\Gamma_{\mathrm{dR}}(\widetilde{X}/\mathbb{C})$ is quasi-isomorphic to $\mathrm{R}\Gamma_{\mathrm{dR}}(X/\mathbb{C})$. (Approximately speaking, this is saying that a tubular neighborhood of X is homotopic to X.) The result now follows by extending scalars. ■

Remark 1.43. This proof does not work in our mixed characteristic situation $W(k) \to k$ because there is no direct way to geometrically link $\widetilde{X}$ and X via a Poincaré lemma.

To prove Theorem 1.32, it will turn out that the key ideas can be used to instead prove Corollary 1.41.

Proof of Corollary 1.41 when $p > 2$. We attempt to give a version of Poincaré's lemma. The idea there is to use Cartan's formula, which arises by viewing differentials as living in flows of vector fields.

1. Our analog of this will be
$$\log f := \sum_{n \geq 1} (-1)^{n+1} \frac{(f - \mathrm{id})^{\circ n}}{n},$$
viewed as a W-linear endomorphism of $\mathcal{O}_{\widetilde{X}}$. To see that this makes sense, note that $f - \mathrm{id} = pA$ for some endomorphism A, so it is at least true that the terms of the series are divisible by larger and larger powers of p. Thus, the series descends to a compatible sequence of endomorphisms for each $\mathcal{O}_{\widetilde{X}_n}$ and hence an endomorphism of $\mathcal{O}_{\widetilde{X}}$.

2. It further turns out that $\log f$ is a derivation of $\mathcal{O}_{\widetilde{X}}$, meaning that
$$(\log f)(ab) = (\log f)(a)b + a(\log f)(b).$$
This is a formal consequence of the fact that f is a ring map, and $\log f$ takes products to sums. (Indeed, one can already find this in characteristic zero.)

3. To continue, we take $p > 2$ for simplicity. Indeed, then one can see $f = \exp(\log f)$, meaning that
$$f \stackrel{?}{=} \sum_{n \geq 0} \frac{(\log f)^{\circ n}}{n!}.$$
The right-hand sum makes sense because $\log f$ is divisible by p, which then implies that the sum converges. Thus, we may write $\log f = pD$ for some derivation D. The equality now follows from some formal calculation of the series.

4. We now pass to the de Rham complex. We would like to check that the endomorphism $f^* \colon \Omega^\bullet_{\widetilde{X}} \to \Omega^\bullet_{\widetilde{X}}$ is a quasi-isomorphism. Well, there is a "Lie derivative" L_D for which f^* acts on $\Omega^i_{\widetilde{X}}$ by
$$\sum_{n \geq 0} \frac{p^n}{n!} L_D^{\circ n}.$$

Let's explain what this L_D is, which we may define affine-locally. For any $W(k)$-algebra A, a derivation $W\colon A \to A$ induces an endomorphism L_D on differentials as follows. Indeed, $\mathrm{id} + \varepsilon D$ is an automorphism of the thickening $A[\varepsilon]/\left(\varepsilon^2\right)$, so $\mathrm{id} + \varepsilon D$ acts on $\Omega^i_{A[\varepsilon]/(\varepsilon^2)}$. Certainly it must reduce to identity modulo ε, so we conclude that $(\mathrm{id} + \varepsilon D)^* = \mathrm{id} + \varepsilon L_D$ for some endomorphism L_D on differentials!

5. The previous step now tells us that $f^* = \mathrm{id} + L_D \circ G$ for some G. We will be done as soon as we can check that the endomorphism L_D of the de Rham complex is homotopic to zero. In other words, we are on the hunt for maps $\iota_D\colon \Omega^i_{\widetilde{X}} \to \Omega^{i-1}_{\widetilde{X}}$ fitting into the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}_{\widetilde{X}} \arrow[r] \arrow[d, dashed, "L_D"] & \Omega^1_{\widetilde{X}} \arrow[r] \arrow[d, dashed, "L_D"] \arrow[dl, dashed, "\iota_D"'] & \Omega^2_{\widetilde{X}} \arrow[r] \arrow[d, dashed, "L_D"] \arrow[dl, dashed, "\iota_D"'] & \cdots \\
0 \arrow[r] & \mathcal{O}_{\widetilde{X}} \arrow[r] & \Omega^1_{\widetilde{X}} \arrow[r] & \Omega^2_{\widetilde{X}} \arrow[r] & \cdots
\end{tikzcd}$$

so that $L_D = d\iota_D + \iota_D D$. This is precisely Cartan's formula, which admits an algebraic proof. (Namely, one takes ι_D to be a contraction whose vector field is D.) To show this, one can show directly that $d\iota_D + \iota_D D$ is a derivation, and then one can check that there is an equality on functions and 1-forms. ■

Remark 1.44. The construction of L_D may appear ad-hoc. The intuition here is that our objects are not only functorial under endomorphisms but also under derivations.

It turns out that all the interesting algebra to prove Theorem 1.32 is already contained in the proof of Corollary 1.41, though this should not be obvious yet.

1.2.3 The Cartier Isomorphism

Recall that we wanted to prove Theorem 1.22.

Theorem 1.22 (Cartier isomorphism)**.** Fix a smooth variety X over a perfect field k. Then there is a canonical isomorphism

$$\mathcal{H}^i(F_*\Omega^\bullet_X) \cong \Omega^i_{X^{(1)}}.$$

Here, the left-hand side is a coherent $\mathcal{O}_{X^{(1)}}$-module.

Proof. Let's start with $i = 0$. Note that there is a map

$$\mathcal{O}_{X^{(1)}} \to \ker\left(d\colon F_*\mathcal{O}_X \to F_*\Omega^1_X\right)$$

given by sending a function f to f^p. We would like to check that this is an isomorphism. Well, it is enough to check this on an open cover, but because X is smooth, we may pass étale-locally along some $f\colon U \to \mathbb{A}^n_k$ (where $U \subseteq X$ is some open subset) because the de Rham complex is immune to such deformations: indeed,

$$\left(F_*\Omega^0_X\right)|_{U^{(1)}} \cong f^{(1)*}\left(F_*\Omega^0_{\mathbb{A}^n_k}\right).$$

Thus, we may pass to $X = \mathbb{A}^n$.

We now work in general. By working locally, we may assume that X is affine and equal to some $\operatorname{Spec} A$. Then we would like an isomorphism

$$\mathrm{H}^i_{\mathrm{dR}}(X/k) \overset{?}{\cong} \Omega^i_{A^{(1)}/k}$$

of A-modules. Well, consider $\mathrm{R}\Gamma_{\mathrm{cris}}(A/W_2(k)) := \mathrm{R}\Gamma_{\mathrm{cris}}(A/W(k)) \otimes_{W(k)} W_2(k)$, which we know to be isomorphic to $\mathrm{R}\Gamma_{\mathrm{dR}}(\widetilde{X}/W_2(k))$ for any deformation $\widetilde{X}$ of X to $W_2(k)$. Accordingly, we receive a distinguished triangle

$$\mathrm{R}\Gamma_{\mathrm{dR}}(A/k) \to \mathrm{R}\Gamma_{\mathrm{cris}}(A/W_2(k)) \to \mathrm{R}\Gamma_{\mathrm{dR}}(A/k)$$

induced by the lifting $k \to W_2(k) \to k$, where the map $k \to W_2(k)$ is multiplication by p. Thus, we receive a long exact sequence

$$\cdots \to \mathrm{H}^i_{\mathrm{dR}}(A/k) \to \mathrm{H}^i_{\mathrm{dR}}(A/k) \xrightarrow{\beta_i} \mathrm{H}^{i+1}_{\mathrm{dR}}(A/k) \to \cdots .$$

It turns out that the graded map $\bigoplus_i \beta_i$ on the de Rham cohomology ring $\bigoplus_{i \geq 0} \mathrm{H}^i_{\mathrm{dR}}(X/k)$ is a derivation; this is just some exercise in the homological algebra. Explicitly, one has to check that

$$\beta_{i+j}(x_i \wedge x_j) = \beta_i(x_i) \wedge x_j + (-1)^i x_i \wedge \beta_j(x_j)$$

for any x_i and x_j of degrees i and j, respectively. It further turns out that $\beta_{i+1} \circ \beta_i = 0$.

Now that we have found a derivation, we can make differentials appear. The universal property of differentials produce morphisms

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}_{A^{(1)}} \arrow[r, "d"] \arrow[d, "c_0"] & \Omega^1_{A^{(1)}} \arrow[r, "d"] \arrow[d, "c_1"] & \Omega^2_{A^{(1)}} \arrow[r, "d"] \arrow[d, "c_2"] & \cdots \\
0 \arrow[r] & \mathrm{H}^0_{\mathrm{dR}}(A/k) \arrow[r, "\beta"] & \mathrm{H}^1_{\mathrm{dR}}(A/k) \arrow[r, "\beta"] & \mathrm{H}^2_{\mathrm{dR}}(A/k) \arrow[r, "\beta"] & \cdots
\end{tikzcd}$$

where f_0 is induced by the degree-zero argument, and all the relevant morphisms intertwine β and d and produce maps of graded rings. For example, one finds that we need $c_1(df) = \beta_0(c_0(f))$ for functions f. One now checks that this in fact an isomorphism on the level of modules, which is checked étale-locally and then passed to affine spaces. ■

Remark 1.45. We can afford to be a little more explicit about our construction of the maps $c_\bullet$. Fix a lift $\widetilde{A}$ over $W_2(k)$ as well as a lift of the Frobenius $\widetilde{F}$. Given $\omega \in \Omega^i_{A^{(1)}/k}$, we pull it back along $\widetilde{F}$ to get a differential divisible by p in $\mathrm{H}^i_{\mathrm{dR}}(\widetilde{A}/W_2(k))$. Then one can check that the class $\widetilde{F}^*(\omega)/p$ is well-defined in $\mathrm{H}^i_{\mathrm{dR}}(A/k)$.

Remark 1.46. Crystalline cohomology is not technically necessary because we could choose a lift $\widetilde{X}$ by hand using Remark 1.45. However, one needs to check that the constructed map is independent of the lift. This is not impossible (such lifts are well-understood in some cohomology group by deformation theory), but it is a little difficult.

1.3 February 9

Today we say more about the construction of crystalline cohomology.

1.3.1 $\mathcal{D}$-modules

We are going to use $\mathcal{D}$-modules to glue our de Rham complexes together.

Definition 1.47 (D-module)**.** Fix a smooth scheme X over a ring R. Then we define the quasicoherent sheaf $\mathcal{D}_{X/R}$ of associative algebras on X defined explicitly as

$$\mathcal{O}_X\{\partial_v : v \in T_X\},$$

where the relations ∂_v satisfies the relations $\partial_{v_1+v_2} = \partial_{v_1} + \partial_{v_2}$, $\partial_{fv} = f\partial_v$, $[\partial_v, f] = L_v(f)$, and $[\partial_v, \partial_w] = \partial_{[v,w]}$. Here, T_X is the tangent bundle, which is the dual of $\Omega^1_{X/R}$ (equivalently, the sheaf of derivations $\mathcal{O}_X \to \mathcal{O}_X$). A $\mathcal{D}$-*module* is a quasicoherent sheaf for which the $\mathcal{O}_X$-action extends to the action of a module for $\mathcal{D}_{X/R}$.

Remark 1.48. Precisely, we have given a definition of an associative ring on affine opens, which then glue together on affines as a sheaf on a base.

Remark 1.49. There is a natural action of $\mathcal{D}_{X/R}$ on $\mathcal{O}_X$, where functions act by multiplication, and the derivations act by taking derivatives. If R contains $\mathbb{Q}$, then this natural map $\mathcal{D}_{X/R} \to \mathrm{End}_R(\mathcal{O}_X)$ is injective.

Example 1.50. In positive characteristic p, the action map $\mathcal{D}_{X/R} \to \mathrm{End}_R(\mathcal{O}_X)$ need not be injective. Indeed, with $X = \mathbb{A}^1_{\mathbb{F}_p}$ over $\mathbb{F}_p$, we see that the iterated derivation ∂_t^p kills all polynomials (because $p! = 0$).

Remark 1.51. The various relations make it so that a $\mathcal{D}$-module is exactly a vector bundle with a flat connection $\nabla\colon M \to M \otimes \Omega^1_{X/R}$.

Remark 1.52. The algebra $\mathcal{D}_{X/R}$ admits a filtration by the subgroups $\mathcal{D}^{\leq n}_{X/R}$ of differentials at most n. It turns out that $\mathrm{gr}\,\mathcal{D}_{X/R} = \mathrm{Sym}^\bullet T_X$, which is probably a variant of the PBW theorem.

Here is why $\mathcal{D}$-modules are relevant to us.

Lemma 1.53. Fix a smooth scheme X over a ring R. Then the de Rham complex $\Omega^\bullet_{X/R}$ is quasi-isomorphic to $\mathrm{RHom}_{\mathcal{D}_{X/R}}(\mathcal{O}_X, \mathcal{O}_X)$ in $D(\mathrm{Sh}(X))$.

Proof. We are going to find a locally projective resolution for $\mathcal{O}_X$ as a $\mathcal{D}_{X/R}$-module. To start, note that there is a projection

$$\mathcal{D}_{X/R} \to \mathcal{O}_X$$

given by sending $\delta \mapsto \delta(1)$. This is surjective because functions act by left multiplication. By definition, all the derivations ∂_v are in the kernel, and these are exactly what goes to zero, so we continue our sequence by writing

$$\mathcal{D}_{X/R} \otimes_{\mathcal{O}_X} T_{X/R} \to \mathcal{D}_{X/R} \to \mathcal{O}_X \to 0,$$

where the left map sends $\delta \otimes v \mapsto \delta\partial_v$. Then $\partial_v \otimes w - \partial_w \otimes v$ is in the kernel, and we can find that we should extend our complex by

$$\mathcal{D}_{X/R} \otimes \mathcal{O}_X \wedge^2 T_{X/R} \to \mathcal{D}_{X/R} \otimes_{\mathcal{O}_X} T_{X/R} \to \mathcal{D}_{X/R} \to \mathcal{O}_X \to 0,$$

where the left map sends $\delta \otimes (v_1 \wedge v_2)$ to $\delta\partial_{v_1} \otimes v_2 - \delta\partial_{v_2} \otimes \partial_{v_1} - \delta \otimes [v_1, v_2]$. This suggests that we are looking at a Koszul complex

$$\cdots \to \mathcal{D}_{X/R} \otimes \mathcal{O}_X \wedge^2 T_{X/R} \to \mathcal{D}_{X/R} \otimes_{\mathcal{O}_X} T_{X/R} \to \mathcal{D}_{X/R} \to \mathcal{O}_X \to 0,$$

and one can check that it is acyclic by passing to the associated graded everywhere.

We now compute RHom by hitting our complex with $\mathrm{Hom}_{\mathcal{D}_X}(-, \mathcal{O}_X)$. Then the point is that there is an isomorphism $\mathrm{Hom}_{\mathcal{D}_X}(\mathcal{D}_X \otimes E, \mathcal{O}_X) = E^\vee$ in the obvious way, so taking $\mathrm{Hom}_{\mathcal{D}_X}(-, \mathcal{O}_X)$ produces the complex

$$\mathcal{O}_X \xrightarrow{d} \Omega^1_{X/R} \xrightarrow{d} \Omega^2_{X/R} \xrightarrow{d} \cdots,$$

which is exactly the de Rham complex. ∎

Remark 1.54. One can replace the target $\mathcal{O}_X$ with any quasicoherent sheaf to obtain a similar result, which allows us to compute de Rham cohomology of an arbitrary quasicoherent sheaf.

1.3.2 The Category of Crystals

We are now ready to construct crystalline cohomology.

Theorem 1.55. Suppose $p > 2$. Fix any smooth variety X over a perfect field k of characteristic p. For each $n \geq 1$, then there is an abelian tensor category $\mathrm{Cris}(X/W_n(k))$ of crystals with unit object $\mathcal{O}_X^{\mathrm{cris}}$, with the following property: for any lift X_n of X to $W_n(k)$, there is an isomorphism

$$\mathrm{Cris}(X/W_n(k)) \cong \mathrm{Mod}(\mathcal{D}_{X_n/W_n})$$

sending $\mathcal{O}_X^{\mathrm{cris}}$ to $\mathcal{O}_X$.

Remark 1.56. This certainly implies Theorem 1.32 simply by defining

$$\mathrm{R}\Gamma_{\mathrm{cris}}(X/W_n(k)) := \mathrm{RHom}_{\mathrm{Cris}}(X/W_n(k))(\mathcal{O}_X^{\mathrm{cris}}, \mathcal{O}_X^{\mathrm{cris}}).$$

Namely, we may view the target as living in $\mathcal{D}_{X_n/W_n}$-modules for any lift X_n, which is then related to de Rham cohomology via Lemma 1.53. One can construct $\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k))$ by taking a suitable limit over n.

Remark 1.57. The category quasicoherent sheaves on X_n (as a tensor category) recovers X_n as a tensor category, so it is remarkable that the category of $\mathcal{D}$-modules does not!

Remark 1.58. The sheaf of algebras $\mathcal{D}_{X_n/W_n}$ does depend on the lift to X_n, so it is remarkable that we are able to construct $\mathrm{Cris}(X/W_n(k))$ at all!

We have been using a bit of deformation theory (for motivation) throughout, but let's state what we need more explicitly.

Theorem 1.59 (Deformation)**.** Fix a commutative ring A and a nilpotent thickening $A \to A/I$ where $I^2 = 0$, and choose a smooth scheme Y over A/I.

(a) There is a (natural) class $\mathrm{ob}_{Y,A} \in \mathrm{H}^2(Y; T_{Y/(A/I)} \otimes_{A/I} I)$ which vanishes if and only if there is a smooth lift $\widetilde{Y}$ over A for which $\widetilde{Y}_{A/I} \cong Y$.

(b) Given a choice of lift $\widetilde{Y}$ with a choice of isomorphism $\rho\colon \widetilde{Y}_{A/I} \to Y$, then $\mathrm{H}^1(Y; T_{Y/(A/I)} \otimes I)$ parameterizes all lifts.

(c) Given a choice of lift $\widetilde{Y}$ with a choice of isomorphism $\rho\colon \widetilde{Y}_{A/I} \to Y$, then the group of automorphisms $f\colon \widetilde{Y} \to \widetilde{Y}$ which reduce to the identity modulo I is isomorphic to $\mathrm{H}^0(Y; T_{Y/(A/I)} \otimes_{A/I} I)$.

Example 1.60. By (a), we see that one can always lift smooth curves.

Remark 1.61. One can directly prove the commutativity of the group of automorphisms (roughly) as follows: f amounts to a deformation which looks like $\mathrm{id} + \varepsilon$ (e.g., on the sheaf of rings), which all commute with each other because $I^2 = 0$.

Let's explain how we use Theorem 1.59: if X is a smooth variety over a field k, then X is separated, so it admits an affine open cover $\{U_i\}_{i \in \Lambda}$ so that each intersection $U_i \cap U_j$ is still affine. By the first two parts of Theorem 1.59 (applied iteratively), we are granted a unique smooth lift $\widetilde{U}_\alpha$ over $W_n(k)$ (because higher cohomology of affine schemes will vanish). By the uniqueness, we are granted isomorphisms

$$f_{ij}\colon \mathcal{O}_{\widetilde{U}_i}|_{U_{ij}} \cong \mathcal{O}_{\widetilde{U}_j}|_{U_{ij}},$$

but the third part of Theorem 1.59 tells us that there are many choices for such an automorphism. Nonetheless, for any triple (i, i', i''), we see that the composite $f_{i''i} f_{i'i''} f_{ii'}$ is the identity $\pmod p$. But this does not mean that our sheaves glue! This explains why we cannot lift X.

However, we will be able to lift $\mathcal{D}$-modules to a category of $\mathcal{D}$-modules with W_n-coefficients without ever having to lift X to X_n. The key is the following, which asserts that such "automorphisms which are the identity $\pmod p$" are inner for $\mathcal{D}$-modules.

Lemma 1.62. Suppose $p > 2$, and fix a smooth affine scheme Y_n over W_n. Then for any automorphism f of Y_n which is the identity $\pmod p$, there is a $D_f \in \mathcal{D}_{Y_n/W_n}(Y_n)^\times$ which is $1 \pmod p$, and

$$f^*\delta = D_f \delta D_f^{-1}$$

for any $\delta \in \mathcal{D}_{Y_n/W_n}(Y_n)$.

Proof. We may find Δ_f to satisfy

$$p\Delta_f = \log f^*,$$

where $\log f^*$ refers to the power series $\sum_{r\geq 1}(-1)^{r+1}(f-\mathrm{id})^{\circ r}/r$. (This exists because $p > 2$.) This logarithm is some endomorphism on $\mathcal{O}_{Y_n}$, and it is even a derivation of $\mathcal{O}_{Y_{n-1}}$, which one can check directly. We now define $D_f := \exp(p\partial_{\Delta_f})$, and one can check the identity directly from the identity $f^* = \exp(p\Delta_f)$. ■

Remark 1.63. The moral of Lemma 1.62 is that any $\mathcal{D}_{Y_n}$-module M admits a natural isomorphism $M \cong (f^*)_* M$, where the right-hand module means that $\mathcal{D}_{Y_n}$ acts on M through the automorphism f^*. Indeed, simply send m to $D_f m$.

Remark 1.64. Lemma 1.62 even has a formulation in characteristic zero, which admits basically the same proof.

We are now ready to prove Theorem 1.55.

Proof of Theorem 1.55. Fix an affine open cover $\{U_i\}_{i\in\Lambda}$ on X with f_{ij}s as before. The category of $\mathcal{D}_{X/k}$-modules can be thought of as the category of big tuples $\{M_i\}_i$, where M_i is a module for $\mathcal{D}_{X/k}(U_i)$, and we are equipped with isomorphisms between $M_i|_{U_{ij}}$ and $M_j|_{U_{ij}}$ that satisfy some cocycle condition. In light of the discussion preceding Lemma 1.62, our problem is that we do not know how to fix the cocycle condition.

We now imitate this definition. Define our category $\mathrm{Cris}(X/W_n(k))$ of crystals as being the category of tuples $\{M_i\}_i$ of modules, where M_i is a module over $\mathcal{D}_{\widetilde{U}_i/W_n}(\widetilde{U}_i)$, equipped with isomorphisms

$$\alpha_{ij} \colon f_{ij}^* M_i|_{U_{ij}} \to M_j|_{U_{ij}}$$

such that the composites $\alpha_{i''i}\alpha_{i'i''}\alpha_{ii'}$ produces an automorphism $f_{i''i}^* f_{i'i''}^* f_{ii'}^* M_i \to M_i$, where this is an automorphism where we have identified the left module with M_i via Lemma 1.62.

It may appear that our construction depends rather poorly on the choice of cover. However, this is not the case: one can simply show that the uniqueness of our constructions means that our construction does not depend on refinements of open covers, so it does not actually depend on the choice of open cover. (It may look like this is so because the identifications depend on the choice of D_f in Lemma 1.62, but this construction was in fact canonical in the proof.) Professor Petrov seems to think that one will need to check some cocycle condition on intersections of four open subsets.

We now run our checks.

- For example, we note that the $\mathcal{O}_{\widetilde{U}_i}$s glue to an object $\mathcal{O}_X^{\mathrm{cris}}$ because all data is only every constructed $\pmod p$, and the cocycle condition is satisfied by the particular construction of D_f.
- Given a choice of lift X_n, we may actually lift everything to X_n, which shows that our crystals agree with $\mathcal{D}$-modules on X_n. ■

Remark 1.65. One can relax the hypothesis that X is separated by further covering the U_{ij}s by affines.

Remark 1.66. These are called "crystals" because they are rigid and they grow. Namely, we constructed them locally, but they were rigid enough to be able to be glued to a global category.

Remark 1.67. Additionally, morphisms of varieties are finite type and separated, so it is not too hard to construct functoriality for our category of crystals. Explicitly, a morphism $f\colon X \to Y$ produces a morphism

$$f^*\colon \mathrm{R}\Gamma_{\mathrm{cris}}(Y/W(k)) \to \mathrm{R}\Gamma(X/W(k)).$$

1.4 February 11

Today, we give some applications of crystalline cohomology.

1.4.1 The Ax–Katz Theorem

Crystalline cohomology comes with a natural Frobenius.

Example 1.68. For any smooth scheme X over k, there is a Frobenius $F\colon X \to X^{(1)}$, which produces an endomorphism

$$F^*\colon \mathrm{R}\Gamma_{\mathrm{cris}}\left(X^{(1)}/W(k)\right) \to \mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k)).$$

For example, if X is defined over $\mathbb{F}_{q^r}$, then $X = X^{(r)}$, then we have an endomorphism $(F^*)^r$ of the complex $\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(k))$.

Remark 1.69. If X is smooth and proper over $\mathbb{F}_q$ where $q = p^r$, then $\mathrm{H}^n_{\mathrm{cris}}(X/W(\mathbb{F}_q))$ is finitely generated over $W(\mathbb{F}_q)$, and it turns out that

$$\#X(\mathbb{F}_q) = \sum_{i\geq 0}(-1)^i \operatorname{tr}\left((F^*)^r; \mathrm{H}^i_{\mathrm{cris}}(X/W(\mathbb{F}_q))\left[\frac{1}{p}\right]\right).$$

One has to check that crystalline cohomology is a Weil cohomology theory, which is true. It is not too difficult to check this formula directly when X is zero-dimensional, meaning that it is a finite disjoint union of points.

Theorem 1.70 (Ax–Katz)**.** Let X be a smooth proper geometrically connected variety over $\mathbb{F}_q$ for which $\mathrm{H}^i(X;\mathcal{O}_X) = 0$ for all $i > 0$. Then

$$\#X(\mathbb{F}_q) \equiv 1 \pmod{q}.$$

Proof. The idea is to use Remark 1.69. Note that $\mathrm{H}^0_{\mathrm{cris}}(X/W(\mathbb{F}_q))$ is the one-dimensional space with trivial Frobenius action, so it is enough to show that

$$\operatorname{tr}\left((F^*)^r; \mathrm{H}^i_{\mathrm{cris}}(X/W(\mathbb{F}_q))\left[\frac{1}{p}\right]\right) \equiv 0 \pmod{q}$$

for all $i \geq 1$. With an inversion of p, one will not be able to do better than $\pmod{p}$ information, so we see that we have to use the integral structure.

For a given module M over $W(\mathbb{F}_q)$, we let M_{tf} denotes the quotient by torsion so that M_{tf} is torsion-free. It is now enough to show that the map

$$F^* \colon \mathrm{H}^i_{\mathrm{cris}}(X^{(1)}/W(\mathbb{F}_q))_{\mathrm{tf}} \to \mathrm{H}^i_{\mathrm{cris}}(X/W(\mathbb{F}_q))_{\mathrm{tf}}$$

is divisible by p. Indeed, the trace after $(-)[1/p]$ is the same as the trace on the torsion-free part (because then inverting p does nothing), so we can just iterate this divisibility r times to prove the claim. We now reduce $\pmod p$, for which we recall that

$$\mathrm{R}\Gamma_{\mathrm{cris}}(X/W(\mathbb{F}_q)) \otimes^{\mathbb{L}}_{W(\mathbb{F}_q)} \mathbb{F}_q \cong \mathrm{R}\Gamma_{\mathrm{dR}}(X/\mathbb{F}_q).$$

Now, the same argument as in Universal coefficient theorem yields an exact sequence

$$0 \to \mathrm{H}^n_{\mathrm{cris}}(X/W(\mathbb{F}_q))/p \to \mathrm{H}^n_{\mathrm{dR}}(X/\mathbb{F}_q) \to \mathrm{H}^{n+1}_{\mathrm{cris}}(X/W(\mathbb{F}_q))[p] \to 0.$$

By functoriality of the short exact sequence, if we could show that Frobenius was zero on de Rham cohomology (i.e., the map $F^* \colon \mathrm{H}^n_{\mathrm{dR}}(X^{(1)}/\mathbb{F}_q) \to \mathrm{H}^n_{\mathrm{dR}}(X/\mathbb{F}_q)$ vanishes), then it would vanish on $\mathrm{H}^n_{\mathrm{cris}}(X/W(\mathbb{F}_q))/p$ and thus vanish on $\mathrm{H}^n_{\mathrm{cris}}(X/W(\mathbb{F}_q))_{\mathrm{tf}}/p$, which was the required claim.

We are thus reduced to computing some Frobenius action on the de Rham complex. Let's recall how this Frobenius action is defined: by pulling back along F^*, there are natural maps

$$\begin{tikzcd}
F^{-1}\mathcal{O}_{X^{(1)}} \arrow[r] \arrow[d, "F^*"'] & F^{-1}\Omega^1_{X^{(1)}} \arrow[r] \arrow[d, "F^*"'] & F^{-1}\Omega^2_{X^{(1)}} \arrow[r] \arrow[d, "F^*"'] & \cdots \\
\mathcal{O}_X \arrow[r] & \Omega^1_X \arrow[r] & \Omega^2_X \arrow[r] & \cdots
\end{tikzcd}$$

which then descend to cohomology. But note that $F^*(f\,dg) = f^p\,d(g^p) = 0$, so F^* vanishes on all the higher terms. Thus, the above morphism F^* of complexes factors through the complex $\mathcal{O}_X \to 0 \to 0 \to \cdots$. Thus, after taking hypercohomology, we see that F^* factors through $\mathrm{H}^n(X; \mathcal{O}_X)$, which vanishes by assumption! ■

Remark 1.71. We can remove the properness assumption, and one still finds that F^* finds on de Rham cohomology.

Remark 1.72 (Chevalley–Warning)**.** If $X \subseteq \mathbb{P}^N_{\mathbb{F}_q}$ is a hypersurface of degree at most N, then the hypothesis is satisfied. It is a classical result of Chevalley–Warning that $\#X(\mathbb{F}_q) \equiv 1 \pmod p$. A cohomological argument could proceed by using the Atiyah–Bott formula

$$\#X(\mathbb{F}_q) = \sum_{i \geq 0} (-1)^i \operatorname{tr}\left((F^*)^r; \mathrm{H}^n(X; \mathcal{O}_X)\right) \pmod p$$

and then argue as above. To access $\pmod q$ information, we need a different cohomology theory.

Remark 1.73. If one wants to prove that the Frobenius acting on crystalline cohomology was instead divisible by p^2, then there starts to be contributions of the cohomology of Ω_X, via some discussion of Hodge–Tate weights.

1.4.2 A Weak Hodge Decomposition

Here is another application.

Notation 1.74. Given a formal scheme $\widetilde{X}$ over $W(k)$, we define

$$\widetilde{X}^{(1)} := \widetilde{X} \times_{W(k)} W(k),$$

where the internal map $W(k) \to W(k)$ is the natural Frobenius on $W(k)$.

Proposition 1.75. Fix a smooth scheme X over a perfect field k of characteristic p. Suppose that we can lift X to a formal scheme $\widetilde{X}$ over $W(k)$ as well as a lift $\widetilde{F}\colon \widetilde{X} \to \widetilde{X}^{(1)}$ of F. Then the de Rham complex

$$\mathcal{O}_{\widetilde{X}} \xrightarrow{d} \Omega^1_{\widetilde{X}} \xrightarrow{d} \Omega^2_{\widetilde{X}} \xrightarrow{d} \cdots$$

is quasi-isomorphic to

$$\mathcal{O}_{\widetilde{X}^{(1)}} \xrightarrow{pd} \Omega^1_{\widetilde{X}^{(1)}} \xrightarrow{pd} \Omega^2_{\widetilde{X}^{(1)}} \xrightarrow{pd} \cdots .$$

Remark 1.76. If X is affine, then such lifts $\widetilde{X}$ and $\widetilde{F}$ always exist.

As an application, we get a version of the Hodge decomposition.

Corollary 1.77. Fix a smooth scheme X over a perfect field k of characteristic p. Suppose that we can lift X to a formal scheme $\widetilde{X}$ over $W(k)$ as well as a lift $\widetilde{F}\colon \widetilde{X} \to \widetilde{X}^{(1)}$ of F. Then the de Rham complex

$$\mathcal{O}_X \xrightarrow{d} \Omega^1_X \xrightarrow{d} \Omega^2_X \to \cdots$$

is quasi-isomorphic to

$$\mathcal{O}_{X^{(1)}} \xrightarrow{0} \Omega^1_{X^{(1)}} \to \Omega^2_{X^{(2)}} \to \cdots .$$

Thus,

$$\mathrm{H}^n_{\mathrm{dR}}(X/k) = \bigoplus_{i+j=n}^{n} \mathrm{H}^i\left(X^{(1)}; \Omega^j_{X^{(1)}}\right).$$

Proof. To show the quasi-isomorphism, merely reduce Proposition 1.75 modulo p, and then we know that crystalline cohomology becomes de Rham cohomology. To prove the last claim, note that the hypercohomology of the first complex is the de Rham cohomology, and hypercohomology of the second complex produces the right-hand side. Indeed, to take hypercohomology, one needs to resolve the second complex, but the "columns" in our resolution do not need to communicate with each other, so

$$\mathbb{H}^n\left(\mathcal{O}_{X^{(1)}} \xrightarrow{0} \Omega^1_{X^{(1)}} \to \Omega^2_{X^{(2)}} \to \cdots\right) = \bigoplus_{i+j=n}^{n} \mathrm{H}^i\left(X^{(1)}; \Omega^j_{X^{(1)}}\right)$$

follows after we keep track of our degrees. ■

Proof of Proposition 1.75. We are interested in filling in the following diagram.

$$\begin{tikzcd}
\mathcal{O}_{\widetilde{X}} \arrow[r] & \Omega^1_{\widetilde{X}} \arrow[r] & \Omega^2_{\widetilde{X}} \arrow[r] & \cdots \\
\mathcal{O}_{\widetilde{X}^{(1)}} \arrow[r] \arrow[u, "\widetilde{F}^*"] & \Omega^1_{\widetilde{X}^{(1)}} \arrow[r] \arrow[u, dashed] & \Omega^2_{\widetilde{X}^{(1)}} \arrow[r] \arrow[u, dashed] & \cdots
\end{tikzcd}$$

Staring at the diagram, we see that the map $\Omega^i_{\widetilde{X}^{(1)}} \to \Omega^i_{\widetilde{X}}$ had better be $p^{-i}F^*$. The proof of Theorem 1.70 checked that $p^{-1}F^*$ makes sense for $i > 0$, and by discussing Frobenius on higher wedges of differentials finds that even $p^{-i}F^*$ will always make sense.

We have thus provided our morphism of complexes, which we want to be a quasi-isomorphism. Because the terms are p-adically complete, it turns out that it is enough to check that it is a quasi-isomorphism $\pmod p$.[1] It turns out that one can proceed as in the proof of Theorem 1.22. ■

This decomposition remains true with weaker assumptions, which will be our next goal.

Theorem 1.78 (Berthelot–Ogus)**.** Fix a smooth scheme X over a perfect field k which can be lifted to a formal scheme $\widetilde{X}$ over $W(k)$. Suppose further that $\dim X < p$. Then

$$\mathcal{O}_{\widetilde{X}} \xrightarrow{d} \Omega^1_{\widetilde{X}} \xrightarrow{d} \Omega^2_{\widetilde{X}} \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{\dim X}_{\widetilde{X}}$$

is quasi-isomorphic to

$$\mathcal{O}_{\widetilde{X}^{(1)}} \xrightarrow{pd} \Omega^1_{\widetilde{X}^{(1)}} \xrightarrow{pd} \Omega^2_{\widetilde{X}^{(1)}} \xrightarrow{pd} \cdots \xrightarrow{pd} \Omega^{\dim X}_{\widetilde{X}^{(1)}}.$$

Remark 1.79. The same argument as in Corollary 1.77 is able to show that

$$\mathrm{H}^n_{\mathrm{dR}}(X/k) \approx \bigoplus_{i+j=n} \mathrm{H}^i(X^{(1)}; \Omega^j_{X^{(1)}}).$$

(We will not be precise about what $\approx$ means now.) It turns out that this decomposition remains true if X can merely be lifted to $W_2(k)$, and it even depends on this choice of lift.

More precisely, our goal will be to prove the above remark.

1.4.3 $\mathcal{D}$-modules by Crystals

In order to access our Hodge decomposition, we are going to realize $\mathcal{D}_X$-modules as quasicoherent sheaves on some stack X^{dR}; if X is affine, then X^{dR} will be able to be realized as the quotient of some formal scheme by a formal group scheme. Notably, quasicoherent sheaves can then just be thought of as certain equivariant sheaves on a formal scheme.

To understand where X^{dR} may come from, we return to characteristic 0.

Lemma 1.80. Fix a smooth scheme F over a field F of characteristic zero. For a $\mathcal{D}_X$-module M and a point $x \in X(F)$, there is a natural isomorphism

$$M|_{\widehat{\mathcal{O}}_{X,x}} \cong M_x \widehat{\otimes}_F \widehat{\mathcal{O}}_{X,x}.$$

Here, the left-hand side is $\lim M \otimes_{\mathcal{O}_X} \mathcal{O}_{X,x}/\mathfrak{m}_x^\bullet$, and $M_x = M \otimes_{\mathcal{O}_X} \mathcal{O}_{X,x}/\mathfrak{m}_x$.

Remark 1.81. The moral is that there is a canonical trivialization "M_x" of M in a formal neighborhood of a point.

Proof. We will produce an isomorphism $M_x \to \left(M|_{\widehat{\mathcal{O}}_{X,x}}\right)^{\nabla=0}$, where the target means that we are looking for sections with flat connection. Accordingly, choose $s \in M_x$, which can be lifted to some $\widetilde{s} \in M(U)$. Now, $\widetilde{s}$ has no reason to be flat, so we have to fix it. Accordingly, we use Taylor's formula to expand out U in local coordinates $(t_1, \ldots, t_d)$ and then define

$$\sum_{i_1,\ldots,i_d \geq 0} (-1)^{i_1+\cdots+i_d} \frac{t_1^{i_1} \cdots t_d^{i_d}}{i_1! \cdots i_d!} \nabla^{i_1}_{\partial_{t_1}} \circ \cdots \circ \nabla^{i_d}_{\partial_{t_d}}(\widetilde{s}),$$

[1] This is a variant of Nakayama's lemma. For example, a sample statement is that a morphism $f\colon M \to N$ of torsion-free p-adically complete $\mathbb{Z}_p$-modules (meaning that $M \cong \lim M/p^\bullet$) can be checked to be an isomorphism $\pmod p$. After deriving, it turns out that we don't have to check that it is torsion-free.

and one can hit this with ∇ to check that it is a flat section. Further, note that $i_1 = \cdots = i_d = 0$ reproduces s, so this section continues to lift s. We have thus defined our map, and one can check that it produces the required isomorphism by Nakayama's lemma. ■

Example 1.82. Let's write this out for $d = 1$. Then in the first degree, we have replaced $\widetilde{s}$ with $t - \nabla_{\partial_t}\widetilde{s}$; hitting this with ∇_{∂_t}, we receive $-t\nabla^2_{\partial_t}(\widetilde{s})$. This is still nonzero, and we then further correct by $\frac{1}{2}t^2\nabla^2_{\partial_t}(\widetilde{s})$ and continue.

We may want to extend these canonical trivializations into a full trivialization of M, but then we have to explain how to relate the various points.

Notation 1.83. Fix a smooth scheme X over a field F of characteristic zero. Then we define $(X \times_F X)^\wedge_\Delta$ to be the union of the closed subschemes of $X \times X$ cut out by the powers of the quasicoherent ideal sheaf $\mathcal{I}_\Delta$ (of the diagonal). We can define other powers similarly.

Remark 1.84. Intuitively, $(X \times_F X)^\wedge_\Delta$ is the union of all nilpotent thickenings of Δ.

Theorem 1.85. Fix a smooth scheme X over a field F of characteristic zero. Define the formal scheme $(X \times_F X)^\wedge_\Delta$ to be the formal completion at the diagonal, which is the union of the nilpotent thickenings of the diagonal Δ. Then the data of a $\mathcal{D}_X$-module M is equivalent to the data of a quasicoherent sheaf M on X and an isomorphism $\alpha\colon \operatorname{pr}_1^* M \to \operatorname{pr}_2^* M$ on $(X \times_F X)^\wedge_\Delta$ making the diagram of sheaves on $(X \times X \times X)^\wedge_\Delta$

$$\begin{tikzcd} \operatorname{pr}_1^* M \arrow[r, "\operatorname{pr}_{12}^*\alpha"] \arrow[dr, "\operatorname{pr}_{13}^*\alpha"'] & \operatorname{pr}_2^* M \arrow[d, "\operatorname{pr}_{23}^*\alpha"] \\ & \operatorname{pr}_3^* M \end{tikzcd}$$

commute.

Sketch. We describe how to turn a $\mathcal{D}_X$-module M into the desired data. Namely, we need to provide the data of α. For example, to define α along the fiber $\operatorname{pr}_1^{-1}(x) \cong \operatorname{Spf} \widehat{\mathcal{O}}_{X,x}$ needs to provide an isomorphism

$$\operatorname{pr}_1^* M|_{\operatorname{pr}_1^{-1} x} \to \operatorname{pr}_2^* M|_{\operatorname{pr}_1^{-1} x}.$$

The left-hand side is just $M_x \otimes_F \widehat{\mathcal{O}}_{X,x}$ because we start at the point x (which gives M_x) which is then expanded along the formal neighborhood. On the other hand, the right-hand side is $M|_{\widehat{\mathcal{O}}_{X,x}}$ because the pullback is able to remember some horizontal information.[2] It turns out that the maps of Lemma 1.80 now suitably glue over all points. ■

Remark 1.86. The moral of α is that it lets us identify infinitesimally nearby fibers. Once one has identified "nearby" fibers, then one can expect to be able to take derivatives, which gives the $\mathcal{D}_X$-module structure.

Remark 1.87. The data (M, α) is sometimes called a "crystal."

[2] This discussion is a little confusing. It becomes easier to think about in the case of $X = \mathbb{A}^1$, where we can think of Δ as the genuine diagonal of a square.

Example 1.88. Consider $X = \mathbb{A}^1_F$ (or any other algebraic group). Then we find that $\mathcal{D}_{\mathbb{A}^1}$-modules can be seen to be the same as quasicoherent sheaves on $\mathbb{A}^1$ with $\widehat{\mathbb{G}}_a$-equivariant structure. Well, we are asking for a suitable identification along the two projections

$$\left(\mathbb{A}^1 \times \mathbb{A}^1\right)^\wedge_\Delta \rightrightarrows \mathbb{A}^1.$$

However, the left-hand formal scheme is $\mathbb{A}^1 \times \widehat{\mathbb{G}}_a$ (given by sending (x, y) to $(x, x - y)$), and one can check that the cocycle condition corresponds to an equivariant structure. It follows that we are looking at quasicoherent sheaves on $\mathbb{A}^1/\widehat{\mathbb{G}}_a$. Unwinding, we remark that a $\mathcal{D}_{\mathbb{A}^1}$-module M goes to the same $R[t]$-module M with $\widehat{\mathbb{G}}_a$-(co)action $M \to M\widehat{\otimes}\mathcal{O}(\widehat{\mathbb{G}}_a)$ given by

$$m \mapsto \sum_{n \geq 0} \partial_t^n m \otimes \frac{t^n}{n!}.$$

1.5 February 17

Today, we discuss the Hodge decomposition.

1.5.1 Divided Powers

We are interested in sketching the following result, fulfilling our promise in Remark 1.79.

Theorem 1.89 (Drinfeld, Bhatt–Lurie)**.** Fix a smooth variety X over a perfect field k of positive characteristic p. Given a smooth lift $\widetilde{X}$ over $W_2(k)$, there is a natural $\mathbb{Z}/p\mathbb{Z}$-grading

$$\mathrm{H}^n_{\mathrm{dR}}(X/k) = \bigoplus_{i \in \mathbb{Z}/p\mathbb{Z}} \mathrm{H}^n_{\mathrm{dR}}(X/k)_i$$

such that $\mathrm{H}^n_{\mathrm{dR}}(X/k)_i \cong \mathrm{H}^{n-i}\left(X^{(1)}; \Omega^i_{X^{(1)}}\right)$ for $n < p$ and $i \in \{0, 1, \ldots, p-1\}$.

Remark 1.90. This is not exactly a Hodge decomposition because this is only a $\mathbb{Z}/p\mathbb{Z}$-grading. Additionally, the Hodge decomposition usually only exists when X is further proper.

Remark 1.91. It is possible to find a smooth proper surface without the lift to $W_2(k)$ and also so that

$$\dim \mathrm{H}^n_{\mathrm{dR}}(X/k) \neq \sum_{0 \leq i < p} \mathrm{H}^{n-i}\left(X^{(1)}; \Omega_{X^{(1)}}\right).$$

Such varieties even appear in nature.

It is difficult to find a grading directly on de Rham cohomology. Instead, we will pass through the de Rham stack. We may want to retell some of our story from last class with crystals, but there are problems with this construction in positive characteristic. For example, the power series defining the coaction in Example 1.88 have large denominators.

To fix this problem, we simply add in the required symbols by hand.

Definition 1.92. We define the affine group scheme $\mathbb{G}_a^\sharp$ over $\mathbb{F}_p$ to be given by the ring

$$\mathbb{Z}\left[t, t/1, t^2/2, t^3/3!, \ldots\right] \otimes_{\mathbb{Z}} \mathbb{F}_p.$$

The group structure $\mathbb{G}_a^\sharp \times \mathbb{G}_a^\sharp \to \mathbb{G}_a^\sharp$ is given by sending $t_1 \otimes t_2$ to $\frac{1}{n!}(t_1+t_2)^n = \sum_{i+j=n} \frac{t_1^i}{i!} \cdot \frac{t_2^j}{j!}$.

Remark 1.93. Explicitly, the ring is given by

$$\frac{\mathbb{F}_p[t, u_1, u_2, u_3 \ldots]}{(u_1 - t, 2tu_2 - u_1, 3tu_3 - u_2, \ldots)}.$$

It turns out to be enough to merely add in the divided powers coming from powers of p.

Remark 1.94. One can also take the formal group scheme $\widehat{\mathbb{G}}_a^\sharp$ over $\mathbb{F}_p$ to be given by the union of the affine group schemes given by the quotient rings

$$\frac{\mathbb{Z}\left[t, t/1, t^2/2, t^3/3!, \ldots\right]}{(t^m/m!)_{m \geq n}} \otimes_{\mathbb{Z}} \mathbb{F}_p.$$

Alternatively, this is the formal scheme attached to the completed ring $\mathbb{F}_p[t][[t^n/n!]]_{n \geq 0}$.

Remark 1.95. The previous remark provides a natural map $\mathbb{G}_a^\sharp \to \mathbb{G}_a$. This is some map of group schemes, and we see that its kernel is given by

$$\frac{\mathbb{F}_p\left[t, t^p/p!, t^{p^2}/(p^2)!, \ldots\right]}{(t)},$$

which continues to be some infinite-dimensional group scheme. Similarly, the kernel of the natural map $\widehat{\mathbb{G}}_a^\sharp \to \mathbb{G}_a$ is the formal scheme attached to the ring $\mathcal{O}(\widehat{\mathbb{G}}_a^\sharp)/(t)$.

Here is our analog of Example 1.88.

Lemma 1.96. Fix a ring R. Then the category of $\mathcal{D}_{\mathbb{A}^1_R/R}$-modules is equivalent to the category of quasi-coherent sheaves on $\mathbb{A}^1_R$ with an equivariant action by $\widehat{\mathbb{G}}_a^\sharp$.

Proof. The functor in the forward direction was already given by Example 1.88. We will not check that it is an equivalence. ■

Example 1.97. Because the natural map $\widehat{\mathbb{G}}_a^\sharp \to \mathbb{G}_a$ admits a kernel, we see that its kernel will act by endomorphisms on a $\mathcal{D}$-module M. (Indeed, this kernel does not interact with the rest of the $\mathbb{A}^1$-action.)

1.5.2 The p-curvature

It is remarkable that we have found so many additional endomorphisms. Let's explain this. We have already seen something like this before: recall that the natural map $\mathcal{D}_X \to \mathcal{E}nd_k(\mathcal{O}_X)$ admitted a kernel in positive characteristic.

Lemma 1.98. Fix a smooth scheme X over a field k of positive characteristic p. For any derivation $v\colon \mathcal{O}_X \to \mathcal{O}_X$, the p-fold composition $v^{\circ p}$ is also a derivation.

Proof. Direct calculation. ■

Notation 1.99. Fix a scheme X over a field k of positive characteristic p. Given a vector field v on some open subset $U \subseteq X$, we let $v^{[p]}$ denote the vector field corresponding to the p-fold derivation $v^{\circ p}$.

Example 1.100. With $X = \mathbb{A}^1$, we see that $(\partial_t)^{[p]} = 0$ because hitting any t^n with p derivatives will kill it. Similarly, we see that $(t\partial_t)^{[p]} = t\partial_t$ because hitting t^n with $t\partial_t$ a total of p times goes to $n^p t^n = nt^n$.

The point is that the element $\partial_v^{\circ p} - \partial_{v^{[p]}}$ acts by zero as a derivation on $\mathcal{O}_X$ because $\partial_v^{\circ p}$ and $\partial_{v^{[p]}}$ admit the same action on $\mathcal{O}_X$ (by construction of $v^{[p]}$). In particular, it lives in the kernel of the natural map $\mathcal{D}_X \to \mathcal{E}nd_k(\mathcal{O}_X)$! It turns out that these elements generate.

Proposition 1.101. Fix a smooth variety X over a perfect field k of positive characteristic p.

(a) Then $F_*\mathcal{D}_X$ is a sheaf of $\mathcal{O}_{X^{(1)}}$-algebras, and its center is the symmetric algebra generated by the elements
$$\partial_v^p - \partial_{v^{[p]}}$$
as v varies over $T_{X^{(1)}}$.

(b) The kernel of $\mathcal{D}_X \to \mathcal{E}nd_k(\mathcal{O}_X)$ is generated by the homogeneous elements generated by $\partial_v^p - \partial_{v^{[p]}}$ in positive degree.

Proof. Omitted. ■

Thus, we have found many endomorphisms.

Definition 1.102 (p-curvature)**.** Fix a smooth variety X over a perfect field k of positive characteristic p. For a vector field v and a $\mathcal{D}$-module M, the *p-curvature* of M is the action of
$$\psi_v := \partial_v^{\circ p} - \partial_v.$$

Remark 1.103. The action of ψ_v is an endomorphism as a $\mathcal{D}_X$-module, which more or less follows from part (a) of Proposition 1.101. (Note that the F_* does not actually do much to the commutativity condition we are trying to check.) This action turns out to be the same as the one found in Example 1.97.

1.5.3 Quasi-nilpotent Modules

We are now allowed to make the following definition, which will let us remove some formal schemes from view.

Definition 1.104 (quasi-nilpotent)**.** Fix a smooth variety X over a perfect field k of positive characteristic p. Then a $\mathcal{D}_X$-module M is *quasi-nilpotent* if and only if, for all vector fields v on $U \subseteq X$, the action of ψ_v on $M(U)$ is locally nilpotent: for any $m \in M(U)$, there is N for which $\psi_v^{\circ N}(m) = 0$. More generally, if X is a smooth scheme over a $\mathbb{Z}/p^n\mathbb{Z}$-algebra R, then a $\mathcal{D}$-module M is *quasi-nilpotent* if and only if $M_{R/p}$ is nilpotent as a $\mathcal{D}$-module over $X_{R/p}$.

Example 1.105. Because the action by the ψ_vs is zero on $\mathcal{O}_X$, we see that $\mathcal{O}_X$ is quasi-nilpotent.

Non-Example 1.106. The action of $\mathcal{D}_X$ on $\mathcal{D}_X$ is faithful, so $\mathcal{D}_X$ is not quasi-nilpotent.

Let's now upgrade Lemma 1.96.

Lemma 1.107. The category of quasi-nilpotent $\mathcal{D}$-modules on $\mathbb{A}^1_{\mathbb{F}_p}$ is equivalent to the category of quasicoherent sheaves on $\mathbb{A}^1_{\mathbb{F}_p}$ with an equivariant action by $\mathbb{G}_a^\sharp$.

Proof. Via Lemma 1.96, we only have to characterize the image of the constructed functor. Indeed, the functor defined by Example 1.88 finds that the sum

$$\sum_{n\geq 0} \partial_t^n m \otimes \frac{t^n}{n!}$$

is always finite because M is quasi-nilpotent. ■

Remark 1.108. For later use, it is worth noting that the de Rham complex continues to be computed by quasi-nilpotent sheaves: as in Lemma 1.53, we find that

$$\Omega^\bullet_{X/R} \cong \mathrm{RHom}_{\mathrm{Mod}(\mathcal{D}_X)^{\mathrm{quasi-nilpotent}}}(\mathcal{O}_X, \mathcal{O}_X).$$

We additionally have a notion of crystals!

Definition 1.109. Fix a closed, regular embedding $Y \hookrightarrow X$ of smooth varieties over a field k. Then we define the *de Rham stack* $X_Y^\sharp$ as the "divided power envelope"

$$X_Y^\sharp = \mathrm{Spec}_X \mathcal{O}_X \left[\frac{f^n}{n!}\right]_{n\geq 1, f\in\mathcal{I}_Y},$$

where $\mathcal{I}_Y$ is the ideal sheaf.

Remark 1.110. Explicitly, if Y is cut out by a regular sequence $\{f_1, \ldots, f_r\}$, then the ring can be described as the tensor product of $\mathcal{O}_X$ with the free divided power algebra on the letters $x_1, \ldots, x_r$, where x_i corresponds to f_i. It is not obvious that this ring does not depend on the choice of regular sequence!

As promised, here is our analog of Theorem 1.85.

Theorem 1.111. Fix a smooth variety X over a perfect field k of positive characteristic p. Then the category of quasi-nilpotent $\mathcal{D}_X$-modules is equivalent to the category of quasicoherent sheaves M on X equipped with an isomorphism $\mathrm{pr}_1^* M \cong \mathrm{pr}_2^* M$ (and satisfying a suitable cocycle condition), where pr_1 and pr_2 are the canonical projections $(X \times X)_\Delta^\sharp \to X$.

Proof. Omitted. ■

1.5.4 The de Rham Stack

We are now ready to define the de Rham stack.

Definition 1.112 (quotient stack)**.** A *quotient stack*, denoted $[X/G]$, is a pair (G, X), where G is a group scheme acting on a scheme X. Two quotient stacks $[X/G]$ and $[X'/G']$ are *strongly equivalent* if and only if there is a flat surjection $G' \to G$ of group schemes and a map $X' \to X$ so that $X' \to X$ is a torsor for $\ker f$. Two quotient stacks $[X/G]$ and $[X'/G']$ are *equivalent* if and only if they can be connected by a finite sequence of strong equivalences.

Notation 1.113. Given a quotient stack $[X/G]$, we define $\mathrm{QCoh}([X/G])$ to consist of the G-equivariant sheaves on X.

Remark 1.114. One can check that $\mathrm{QCoh}([X/G])$ does not depend on the equivalence class of $[X/G]$.

Here is a first definition of our de Rham stack, for affine schemes.

Definition 1.115 (de Rham stack)**.** Fix an affine scheme $X \subseteq \mathbb{A}^n_k$ over a perfect field k of positive characteristic p. Then we define the *de Rham stack* X^{dR} to be the quotient stack

$$\left[(\mathbb{A}^n)^\sharp_X \,/\, \left(\mathbb{G}^\sharp_a\right)^n\right].$$

It turns out that this construction does not depend on the choice of embedding to $\mathbb{A}^n_k$.

Remark 1.116. It turns out that the quasicoherent sheaves on X^{dR} correspond exactly to crystals and thus to quasi-nilpotent $\mathcal{D}_X$-modules. We have already seen this for $\mathbb{A}^1$ in Example 1.88.

To make a definition in general, note that one can locally fit X into a pullback square as follows.

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d] & \mathbb{A}^n \arrow[d] \\ \operatorname{Spec} k \arrow[r, hook] & \mathbb{A}^m \end{tikzcd}$$

(One may not be able to do this globally if X is not a complete intersection.) But now we may as well base-change by $(\mathbb{A}^m)^\sharp_0$, which produces the following pullback square.

$$\begin{tikzcd} X^{\mathrm{dR}} \arrow[r] \arrow[d] & (\mathbb{A}^n)^{\mathrm{dR}} \arrow[d] \\ \operatorname{Spec} k \arrow[r] & (\mathbb{A}^m)^{\mathrm{dR}} \end{tikzcd}$$

Here, note $(\operatorname{Spec} k)^{\mathrm{dR}} = \operatorname{Spec} k$ because $\mathcal{D}$-modules on a point are just vector spaces. But now we see that $(\mathbb{A}^n)^{\mathrm{dR}} = \left(\mathbb{A}^1\right)^{\mathrm{dR},n}$, so it becomes interesting to understand what the polynomial map $\mathbb{A}^n \to \mathbb{A}^m$ becomes after these isomorphisms.

Remark 1.117. Note that $(\mathbb{A}^1)^{\mathrm{dR}}$ is a ring stack: one can just base-change the addition and multiplication up from $\mathbb{A}^1$ to the de Rham stack. In fact, this is a k-algebra stack, meaning that each $a \in k$ produces a multiplication $\mu_a \colon (\mathbb{A}^1)^{\mathrm{dR}} \to (\mathbb{A}^1)^{\mathrm{dR}}$.

The moral is that the functor $X \mapsto X^{\mathrm{dR}}$ for every X is understood just from gluing and taking closed subschemes from the k-algebra stack $\left(\mathbb{A}^1\right)^{\mathrm{dR}}$. Indeed, one can make sense of the action of "polynomials," which is enough to cut out X. Thus, we could recover X^{dR} for affine schemes by only understanding the k-algebra stack structure on $\left(\mathbb{A}^1\right)^{\mathrm{dR}}$.

We end our class by giving a coordinate-free definition.

Definition 1.118 (de Rham stack)**.** Fix a smooth variety X over a perfect field k. Then we define the *de Rham stack* to be the quotient $X^{\mathrm{dR}} := X/(X \times X)^{\sharp}_{\Delta}$.

Remark 1.119. It turns out that $\mathrm{QCoh}\, X^{\mathrm{dR}}$ is the category of quasi-nilpotent $\mathcal{D}_X$-module.

1.6 February 18

Here we go.

1.6.1 How to Grade

We are going to prove Theorem 1.89 by finding new structure on X^{dR}.

Notation 1.120. We define $\mathbb{G}_m^{\sharp}$ to be the group scheme with ring given by

$$\operatorname{Spec} k\left[t, 1/t, \{(t-1)^n/n!\}_n\right].$$

Remark 1.121. There is a canonical map $k[t, 1/t] \to k[t, 1/t, \{(t-1)^n/n!\}_n]$, but note that it factors through

$$\frac{k[t, 1/t]}{(t-1)^p}$$

because $(t-1)^p$ is divisible by p in the target. Thus, the canonical map $\mathbb{G}_m^{\sharp} \to \mathbb{G}_m$ factors through μ_p.

Remark 1.122. There is a map $\log\colon \mathbb{G}_m^{\sharp} \to \mathbb{G}_a$ given by

$$\log t := \sum_{n \geq 1} (-1)^{n+1} \frac{(t-1)^n}{n},$$

and one can calculate that $\ker \log = \mu_p$. (Note that this sum is finite because $(t-1)^n/n = (t-1)^n/n! \cdot (n-1)!$, and $(n-1)!$ is eventually divisible by p.)

Theorem 1.123. Fix a smooth variety X over a perfect field k of positive characteristic p. Given a smooth lift $\widetilde{X}$ over $W_2(k)$, there is a natural action by $\mathbb{G}_m^{\sharp}$ on X^{dR}.

Proof that Theorem 1.123 implies Theorem 1.89. We will only explain how to produce the grading; the requirement in low degrees follows from the construction of the action. Now, an action of $\mathbb{G}_m^{\sharp}$ on X^{dR} amounts to an action of $\mathbb{G}_m^{\sharp}$ on

$$\mathrm{H}^{\bullet}_{\mathrm{dR}}(X/k) = \mathrm{Ext}^{\bullet}_{\mathrm{QCoh}(X^{\mathrm{dR}})}(\mathcal{O}_{X^{\mathrm{dR}}}, \mathcal{O}_{X^{\mathrm{dR}}})$$

because $\mathrm{QCoh}(X^{\mathrm{dR}})$ is the category of quasi-nilpotent $\mathcal{D}$-modules by Remark 1.108.

Now, a $\mathbb{G}_m$-action on a vector space is the data of a $\mathbb{Z}$-grading (by taking the eigenvalues of the $\mathbb{G}_m$-action). Thus, we may hope that our $\mathbb{G}_m^{\sharp}$-action can produce a similar grading. Thus, there is a canonical copy of μ_p in $\mathbb{G}_m^{\sharp}$, which then acts on $\mathrm{H}^{\bullet}_{\mathrm{dR}}(X/k)$. But now an action of μ_p on V corresponds to a $\mathbb{Z}/p\mathbb{Z}$-grading given by taking the eigenspaces $V_i := \{v \in V : tv = t^v \text{ for } t \in \mu_p\}$. We have thus exhibited the grading! ■

Remark 1.124. It turns out that $\mathbb{G}_m^\sharp \cong \mu_p \times \mathbb{G}_a^\sharp$. Thus, there is an interesting action by $\mathbb{G}_a^\sharp$ on the degree pieces.

For motivation, note that Theorem 1.123 can be refined if we give ourselves more lifting.

Proposition 1.125. Fix a smooth variety X over a perfect field k of positive characteristic p. Given a smooth lift $\widetilde{X}$ over $W_2(k)$ and a lift $\widetilde{F}\colon \widetilde{X} \to \widetilde{X}^{(1)}$ of the Frobenius, there is a natural action of $\mathbb{G}_m$ on X^{dR}.

Remark 1.126. As before, a natural action of $\mathbb{G}_m$ on X^{dR} induces a $\mathbb{Z}$-grading on the de Rham cohomology.

One may hope to use Proposition 1.125 by working with $\mathbb{A}^1$ and then passing to X using the k-algebra stack structure. This is not possible because there is no Frobenius lift of $\mathbb{A}^1$ on $W_2(k)$ which commutes with the structure as an additive group scheme: indeed, even for $k = \mathbb{F}_p$, there is no polynomial $f(x) \in \mathbb{Z}/p^2\mathbb{Z}[x]$ such that $f(x+y) = f(x) + f(y)$ and $f(x) \equiv x^p \pmod{p}$. Thus, we cannot hope to use some "canonical" Frobenius lifts and also reduce everything to the ring stack structure of $\mathbb{A}^1$: one needs to use the data of the lift to $W_2(k)$ to produce the Frobenius lift.

Anyway, let's prove Theorem 1.123 in the case of $\mathbb{A}^1$.

Lemma 1.127. The kernel of the projection $\mathbb{G}_a^\sharp \to \mathbb{G}_a$ is given by $\mathbb{G}_a^{\sharp,(1)}$.

Proof. The idea is that the map $\mathbb{G}_a^\sharp \to \mathbb{G}_a$ factors through α_p because $t^p \in k[t]$ is divisible by p in the divided power ring, which is the kernel of $F\colon \mathbb{G}_a \to \mathbb{G}_a^{(1)}$. ■

Example 1.128. We exhibit a $\mathbb{G}_m^\sharp$-action on $\mathbb{A}^{1,\mathrm{dR}} = \mathbb{A}^1/\mathbb{G}_a^\sharp$. Indeed, $\mathbb{G}_m^\sharp$ acts on $\mathbb{G}_a^{\sharp,(1)}$ simply by scaling $\lambda\colon t^p/p \mapsto \lambda t^p/p$. This then embeds into $\mathbb{G}_a^\sharp$ as the kernel down to $\mathbb{G}_a$. The action follows.

1.6.2 Witt Vectors

We now move to the general case. Motivated by Proposition 1.125, we try to find a "universal" algebra with a Frobenius lift, which is the Witt vectors.

Lemma 1.129. There is a unique flat $\mathbb{Z}_p$-algebra $\mathcal{O}(W)$ with a Frobenius lift $F\colon \mathcal{O}(W) \to \mathcal{O}(W)$ such that $F(y) \equiv y^p \pmod{p}$ for all y satisfying the following universal property: for every map $\mathbb{Z}_p[x] \to R$ with R flat and admitting a Frobenius lift factors uniquely through $\mathcal{O}(W)$.

Proof. As one does in commutative algebra, we simply define

$$\mathcal{O}(W) := \mathbb{Z}_p\left[x, F(x), \frac{F(x)-x^p}{p}, F\left(\frac{F(x)-x^p}{p}\right), \ldots\right],$$

where $F(x)$ is viewed as a formal variable, and then one explicitly finds some generators and relations. There is some work to show that W is flat, and it satisfies the universal property. ■

Definition 1.130 (Witt vectors)**.** We define the *Witt vector scheme* W to be the scheme $\operatorname{Spec}\mathcal{O}(W)$.

Proposition 1.131. There is a unique ring scheme structure on W such that the map $W \to \mathbb{A}^1$ is a ring map. In fact, $\mathcal{O}(W)$ admits a presentation

$$\mathcal{O}(W) = \mathbb{Z}_p[a_0, a_1, \ldots]$$

so that $F^n(a_0) = a_0^{p^n} + pa_1^{p^{n-1}} + \cdots + p^n a_n$.

Proof. Omitted. ■

Example 1.132. One has that $W(\mathbb{F}_p) = \mathbb{Z}_p$.

Remark 1.133. Here is how to work formally with $W(R)$: as a set, we know that $W(R) = \prod_{m \geq 0} R$, and we must have that the "ghost component" map

$$\prod_{m \geq 0} R \to R$$

given by $\mapsto a_0^{p^n} + \cdots + p^n a_n$ is a ring map. For example, $(a+b)_0 = a_0 + b_0$, and $(a+b)_1 = a_1 + b_1 + \frac{a_0^p + b_0^p - (a_0+b_0)^p}{p}$.

Remark 1.134. The Witt vectors W admit a Frobenius lift, which continues to be a ring map.

We are now ready to give some kind of Frobenius lift to $\mathbb{A}^{1,\mathrm{dR}}$.

Lemma 1.135. The de Rham stack $\mathbb{A}^{1,\mathrm{dR}}$ is isomorphic to $[W/pW]$, where pW means that W is acting additively on W by multiplication by p.

Proof. It turns out that the kernel of $F\colon W \to W$ is $\mathbb{G}_a^\sharp$. For example, $(Fa)_0 = a_0^p + pa_1$, so a_0 has pth power divisible by p, which explains why divided powers may appear (and allows us to define a map $\mathbb{G}_a^\sharp \to \ker F$). ■

Proof of Theorem 1.123. We would like a ring stack action of $\mathbb{G}_m^\sharp$ on $\mathbb{A}^{1,\mathrm{dR}}$; this then formally goes to all X^{dR}s. By Lemma 1.135, we know that $\mathbb{A}^{1,\mathrm{dR}} = [W/pW]$. Taking units, we know that $\mathbb{G}_m^\sharp$ is isomorphic to the kernel of $F\colon W^\times \to W^\times$. Thus, $\mathbb{G}_m^\sharp$ is able to rescale W, and it intertwines the multiplication-by-p map, where the target W has trivial action by $\mathbb{G}_m^\sharp$.

Let's quickly explain why one may need a smooth lift of X. Suppose in fact that X admits a lift $\widetilde{X}$ fully to $W(k)$. Then, working locally on X, we may assume that $\widetilde{X}$ is a locally complete intersection (e.g., the fiber of some $f\colon \mathbb{A}^n \to \mathbb{A}^m$ even over $W(k)$), and then the action on X^{dR} is found by pullback on ring stacks $f^{\mathrm{dR}}\colon \mathbb{A}^{n,\mathrm{dR}} \to \mathbb{A}^{m,\mathrm{dR}}$. ■

Remark 1.136. If we merely have a smooth lift to $W_2(k)$, then one has to redo the argument replacing Ws with W_2s everywhere.

Minicourse 2
Hitchin Systems

These talks were given by Pavel Etingof.

2.1 February 18

Here we go.

2.1.1 Principal Bundles

Let X be a smooth irreducible projective curve over a field k, and let G be a split, connected reductive group over k.

Example 2.1. We will usually take $k = \mathbb{C}$ and G to be some kind of linear group.

Definition 2.2 (principal bundle)**.** A *principal G-bundle* is a variety P over k equipped with a map $\pi\colon P \to X$ and a right G-action which preserves π and locally on X is isomorphic to $G \times U \to U$.

Remark 2.3. There was some disagreement about what "locally" means. In general, we should work étale-locally, but it is enough to work Zariski-locally if $k = \mathbb{C}$.

Remark 2.4. Concretely, we may define a G-bundle by choosing an (étale) atlas $\{U_i\}$ on X and some "clutching maps" $g_{ij}\colon U_{ij} \to G$ satisfying the cocycle conditions $g_{ii} = 1$ and $g_{ij}g_{ji} = 1$ and $g_{ij}g_{jk}g_{ki} = 1$. Indeed, given the variety P, we take the atlas $\{U_i\}$ to be a trivializing cover for P, and g_{ij} is induced by the composite

$$G \times U_i \cong \pi^{-1}(U_i) \supseteq \pi^{-1}(U_i j) \subseteq \pi^{-1}(U_j) \cong G \times U_j.$$

Conversely, given the atlas $\{U_i\}$, we define P by taking a disjoint union of the varieties $\{G \times U_i\}_i$ and then gluing along the g_{ij}s; the resulting P is in fact a variety by some faithfully flat descent.

Remark 2.5. Let $\mathcal{O}_{X,G}$ be the sheaf of functions on X valued in G. The previous remark explains that the data of a principal G-bundle comes from some Čech 1-cocycle valued in $\mathcal{O}_{X,G}$, and one can check adjusting such a cocycle by a coboundary does nothing to the isomorphism class of the actual bundle. In fact, one finds that cocycle classes are in bijection with isomorphism classes of bundles, so $\mathrm{H}^1_{\text{ét}}(X; \mathcal{O}_{X,G})$ classifies our bundles. Note that this $\mathrm{H}^1_{\text{ét}}(X; \mathcal{O}_{X,G})$ is not even a group in general!

Our bundles can be classified by stacks.

Definition 2.6. There is a stack $\mathrm{Bun}_G(X)$ which classifies G-bundles on X: namely, given a test scheme T, maps $T \to \mathrm{Bun}_G(X)$ are exactly given by G-bundles on $X \times T$.

Example 2.7. If $G = \mathrm{GL}_1$, then principal bundles are line bundles, so $\mathrm{Bun}_{\mathrm{GL}_1}(X) = \mathrm{Pic}\, X$.

Remark 2.8. It turns out that $\mathrm{Bun}_G(X)$ is a smooth algebraic stack.

Example 2.9. Here is an example of a stack: given an algebraic group H acting on a scheme Y (not necessarily freely), the quotient stack $[Y/H]$ is defined by its functor of points. Indeed, maps from a test scheme T to $[Y/H]$ are given by a pair (E, φ) of a principal H-bundle on S and a map $\varphi\colon E \to Y$ which commutes with the H-action. The point is that a test scheme T should fit into a commutative diagram as follows.

$$\begin{tikzcd} E \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d] \\ S \arrow[r] & {[Y/H]} \end{tikzcd}$$

However, it is possible for $\mathrm{Bun}_G(X)$ to have stabilizers of arbitrary dimension, so it is not a quotient stack. Instead, it turns out to be a nested union of quotient stacks.

2.2 February 23

Here we go.

2.2.1 Bundles on $\mathbb{P}^1$

We continue letting X be a curve over $\mathbb{C}$.

Example 2.10. The simplest stack is $\mathrm{B}G$ is the quotient stack $[\mathrm{pt}/G]$. Thus, maps $S \to \mathrm{B}G$ are the principal G-bundles on S.

Example 2.11. The stack $\mathrm{Pic}\, X := \mathrm{Bun}_{\mathrm{GL}_1}(X)$ classifies line bundles on X. (Indeed, one is simply trying to glue scalars.) For example, when X is a curve, then $\mathrm{Pic}\, X$ admits a surjective degree map, so it has infinitely many automorphisms. Even the degree-zero component $\mathrm{Pic}^0 X$ may not quite be a variety because line bundles may have automorphisms.

Example 2.12. With $G = \mathrm{GL}_n$, we find that $\mathrm{Bun}_G(X)$ classifies vector bundles on X of rank n.

Example 2.13. With $G = \mathrm{SL}_n$, we find that $\mathrm{Bun}_G(X)$ classifies vector bundles with trivial determinant. Similarly, if $G = \mathrm{PGL}_n$, then we are classifying vector bundles up to equivalence by tensoring with line bundles.

Example 2.14. We classify GL_2-bundles on the complex curve $\mathbb{P}^1$ as direct sums of two line bundles.

Proof. Note that any bundle E admits some meromorphic section on $X = \mathbb{P}^1$, so $\mathrm{Hom}(\mathcal{O}(m), E) \neq 0$ for sufficiently small m, and this map is an inclusion for generic such maps. Thus, we get a short exact sequence

$$0 \to \mathcal{O}(m) \to E \to \mathcal{O}(n) \to 0,$$

where m is small, and $\mathcal{O}(n)$ is the line bundle representing the quotient. (The quotient is a line bundle for generic inclusions $\mathcal{O}(m) \to E$.) Recall that the transition function for $\mathcal{O}(m)$ is given by z^m (used to transition between charts). Thus, the short exact sequence given above shows that the transition function for E is given by

$$\begin{bmatrix} z^m & f(z) \\ 0 & z^n \end{bmatrix},$$

where $f(z) \in k[z, 1/z]$. By changing the choice of trivialization, one can multiply this matrix on the left or right by some upper-triangular matrices, and one can use such moves to force $f(z)$ to have monomials concentrated in degrees between m and n.

One can now study the actual isomorphism class of taking each affine open for E. In particular, letting r be the maximal integer which admits an inclusion $\mathcal{O}(r) \to E$, which we see lives between m and n. By massaging with these transition maps, one is able to show that $r \leq (m+n)/2$ with some degree arguments. Eventually, one finds that the corresponding short exact sequence

$$0 \to \mathcal{O}(r) \to E \to \mathcal{O}(r') \to 0,$$

with $r' := m+n-r$, which by using a similar argument to before forces this short exact sequence to genuinely split. While we're here, we remark that these r and r' have been chosen canonically from E, and so the decomposition of E into $\mathcal{O}(r) \oplus \mathcal{O}(r')$ has unique r and r'. ■

Remark 2.15. Here is another way to see the class of short exact sequence of E: the short exact sequence is some class in

$$\mathrm{Ext}^1(\mathcal{O}(n), \mathcal{O}(m)) \cong \mathrm{H}^1\left(X; \mathcal{O}(m-n)\right),$$

which by Serre duality is

$$\mathrm{H}^0(X; \mathcal{O}(n-m-2))$$

because the canonical bundle is $\mathcal{O}(-2)$. But this space gives the homogeneous polynomials of degree $n-m-2$, which has dimension $\max\{n-m-1, 0\}$.

Remark 2.16. One can use this to find connected components of $\mathrm{Bun}_{\mathrm{GL}_2}(\mathbb{P}^1)$, which are labeled by their degree $m+n$ where the bundles are $\mathcal{O}(m) \oplus \mathcal{O}(n)$. Further, the points inside a connected component can be labeled by the invariant $n-m$ when $n \geq m$, so the points are $P_0, P_1, \ldots, P_{n-m}$.

It turns out that P_i specializes to P_j whenever $j \geq i$, which one sees by suitably choosing degenerations of families of bundles. For example, consider the family of bundles E_t with the clutching map $\begin{bmatrix} z^{-1} & t \\ 0 & z \end{bmatrix}$. Then $E_t \cong \mathcal{O} \oplus \mathcal{O}$ for $t \neq 0$ (by explicitly finding a global trivialization), but $E_0 \cong \mathcal{O}(1) \oplus \mathcal{O}(-1)$

One can upgrade this discussion as follows.

Theorem 2.17 (Grothendieck)**.** A vector bundle of rank N on $\mathbb{P}^1$ can be uniquely written as

$$\bigoplus_{i=1}^{N} \mathcal{O}(m_i),$$

where $\{m_i\}$ is a decreasing sequence of integers.

Proof. We proceed by induction. The case of $N = 1$ is classical, so we focus on the inductive step. Let E be a vector bundle of rank $N + 1$. As before, choose maximal m_0 so that there is an embedding $\mathcal{O}(m_0) \to E$, and generic such choices makes the quotient into another vector bundle E'. Thus, by the induction, we have a short exact sequence

$$0 \to \mathcal{O}(m_0) \to E \to \mathcal{O}(m_1) \oplus \cdots \oplus \mathcal{O}(m_N) \to 0.$$

Now, for each i, we can let E' be the kernel of E projecting onto the sum of the $\mathcal{O}(m_j)$s with $j \neq i$ and $j > 0$, so there is a short exact sequence

$$0 \to \mathcal{O}(m_0) \to E' \to \mathcal{O}(m_i) \to 0.$$

The argument on transition functions from before on $N = 2$ shows that $m_0 \geq (m_0 + m_i)/2$, so $m_0 \geq m_i$ for all i follows. A calculation with Ext groups as before shows that the short exact sequence for E must split, so we are done! (As before, to show uniqueness of the sequence $\{m_i\}$, we simply note that the maximal term m_0 is uniquely constructed above.) ■

Remark 2.18. This argument is more or less due to Birkhoff, though reinterpreted by Grothendieck.

More generally, one has the following.

Theorem 2.19. Fix a reductive group G with maximal torus T. Then a G-bundle on $\mathbb{P}^1$ admits a T-structure, and there is a bijection given by induction

$$[\mathrm{Bun}_T/W] \to \mathrm{Bun}_G$$

Remark 2.20. The moral is that bundles can be labeled by dominant integral weights for the dual group.

2.2.2 Double Quotients

We will want the following motivational tool.

Theorem 2.21 (Harder)**.** Fix a reductive group G over an algebraically closed field k. Then any G-bundle on an affine curve is trivial.

Proof. Omitted. This is pretty hard! ■

The moral is that one expects to be able to glue bundles over just two charts on a projective curve. For us, we will use the affine chart $X \setminus \{x\}$ and the formal disk D_x around x, which have intersection $D_x^\times := D_x \setminus \{x\}$.

Notably, the transition function on $D_x^\times$ is some element $g \in G(K)$, where $K = k[D_x^\times]$; note that K is isomorphic to $k((t))$, where t is a choice of uniformizer.[1] Thus, elements $g \in G(K)$ parameterize bundles; note that there is no required cocycle condition because there are only two charts.

However, two G-bundles will be isomorphic if we decide to take a quotient on the right side by $G(\mathcal{O})$ and on the left side by $G(R)$, where $R = k[X \setminus \{x\}]$. It follows that

$$\mathrm{Bun}_G(X) = G(R) \backslash G(K) / G(\mathcal{O}).$$

We are now motivated to make the following definition.

[1] Technically, any such g spreads out to an open neighborhood of x, which then allows us to glue.

Definition 2.22 (affine Grassmannian)**.** Fix an affine algebraic group G. Then we define the *affine Grassmannian* to be

$$\mathrm{Gr}_G \coloneqq G(K)/G(\mathcal{O}),$$

which we note is just $G(k((t)))/G(k[[t]])$. In particular, Gr_G does not depend on the choice of curve.

We can tell the same story for any finite set of points $S \subseteq X$, from which we find that

$$\mathrm{Bun}_G(X) = G(R)\backslash \prod_{x\in S} G(K_x)/\prod_{x\in S} G(\mathcal{O}_x).$$

This does not quite work when G is no longer assumed to be semisimple because G does not trivialize on $X \setminus \{x\}$. In fact, G has no reason to trivialize on any given set of points: even line bundles have no reason to trivialize away from a given set of points.

The right thing to do is to simply take the colimit over all points in $X(k)$. Then we still know that any G-bundle can be trivialized away from finite set of points in $X(k)$, and we get the following.

Proposition 2.23. Fix a curve X over an algebraically closed field k, and let G be an affine algebraic group over k. Then

$$\mathrm{Bun}_G(X)(k) = G(F)\backslash G(\mathbb{A}_F)/G(\mathcal{O}_{\mathbb{A}}),$$

where $F = k(X)$, $\mathbb{A} = \prod_{x\in X(k)}(K_x, \mathcal{O}_x)$, and $\mathcal{O}_{\mathbb{A}} = \prod_{x\in X(k)} \mathcal{O}_x$.

Remark 2.24. The restricted direct product appears as a colimit of those "meromorphic power series" which are regular almost everywhere.

Remark 2.25. The same discussion works even if k is not algebraically closed. In this case, the restricted product should take place over the Galois orbits of $X(\overline{k})$.

Remark 2.26. This realizes $\mathrm{Bun}_G(X)(k)$ as an arithmetic quotient. Namely, let F be a number field or field of functions of a curve. Then let $V(F)$ be its set of valuations (constant on the base field when F is over a curve), and let F_v be the completion for each $v \in V(F)$, and let $\mathcal{O}_v$ be the ring of integers in F_v when v is nonarchimedean. Then the arithmetic quotient is

$$G(F)\backslash G(\mathbb{A})/G(\mathcal{O}_{\mathbb{A}}).$$

The case where F is a number field is of interest to number theorists. When $F = \mathbb{Q}$, one finds that this double quotient is $G(\mathbb{Z})\backslash G(\mathbb{R})$ when one has weak approximation, meaning that $\prod_p (G(\mathbb{Q}_p), G(\mathbb{Z}_p)) = G(\mathbb{Q}) \prod_p G(\mathbb{Z}_p)$. For example, modular forms arise from the case of $G = \mathrm{SL}_2$.

2.2.3 Higgs Fields

Let's use the double quotient to write down some tangent spaces. Let E_γ be the line bundle defined by $\gamma \in G(k((t)))$. If we are in the case that $k = \mathbb{C}$ so that G is a Lie group with Lie algebra $\mathfrak{g}$, then taking the quotient shows that

$$T_{E_\gamma} \mathrm{Bun}_G(X) = \frac{\mathfrak{g}((t))}{\left(\mathrm{Ad}_\gamma^{-1}\, \mathfrak{g}(X \setminus \{x\}) + \mathfrak{g}[[t]]\right)}.$$

This is the space of Čech 1-cocycles for the "adjoint" vector bundle $\mathrm{ad}\, E_\gamma$ modulo coboundaries for our given open cover, so we see that $E \coloneqq E_\gamma$ ought to have

$$T_E \mathrm{Bun}_G(X) \cong \mathrm{H}^1(X; \mathrm{ad}\, E),$$

which by Serre duality is $\mathrm{H}^0(X; K_X \otimes (\operatorname{ad} E)^*)^*$. If one fixes an invariant inner product on $\mathfrak{g}$, then we may identify $\operatorname{ad} E$ with its dual. Removing the dual on the outside produces Higgs fields.

Definition 2.27 (Higgs field)**.** Fix a complex curve X, and let G be an affine algebraic group. Then the space of *Higgs fields* is

$$\mathrm{H}^0(X; K_X \otimes \operatorname{ad} E).$$

We let $\mathcal{M}_G(X)$ be the *Hitchin stack* classifying pairs (E, φ), where E is a principal G-bundle, and φ is a Higgs field.

Remark 2.28. This argument also works over an algebraically closed field k of sufficiently large characteristic.

The above discussion motivates us to study 1-forms on X valued in $\operatorname{ad} E$. For simplicity, we assume that a generic bundle has automorphism group $Z(G)$, which is true as long as G is semisimple and the genus of X exceeds 1.

Notation 2.29. Fix a complex curve X, and let G be an affine algebraic group. Let $\operatorname{Bun}_G^0(X)$ be the open subset of regularly stable G-bundles.

Remark 2.30. Any regularly stable bundle has automorphism group $Z(G)$.

Example 2.31. If $G = \mathrm{GL}_n$ (or PGL_n or SL_n), then "stable" means that for all sub-bundles $F \subseteq E$, we have

$$\frac{\deg F}{\operatorname{rank} F} < \frac{\deg E}{\operatorname{rank} E}.$$

Remark 2.32. Take G semisimple. For a regular stable E, one finds that $\mathrm{H}^0(X; \operatorname{ad} E) = \operatorname{Lie} \operatorname{Aut} E$, which vanishes because $Z(G)$ is finite. Thus, Serre duality implies $\mathrm{H}^1(X; K_X \otimes \operatorname{ad} E) = 0$, so

$$\dim \mathrm{H}^0(X; K_X \otimes \operatorname{ad} E) = \chi(K_X \otimes \operatorname{ad} E).$$

Because $\operatorname{ad} E$ is some degree-zero bundle, one finds that the right-hand side is $\chi(K_X) \dim \mathfrak{g}$. (Indeed, one can reduce to the case of line bundles by producing a filtration of $\operatorname{ad} E$ by degree-zero vector bundles.) Thus, our dimension is $\dim \mathfrak{g}(g(X) - 1)$.

Thus, we see that $\operatorname{Bun}_G^0(X)$ is a smooth variety of dimension $\dim \mathfrak{g}(g(X) - 1)$.

Remark 2.33. If G is reductive, then a similar calculation shows that

$$\dim \operatorname{Bun}_G^0(X) = \dim \mathfrak{g}(g(X) - 1) + \dim Z(\mathfrak{g}).$$

For example, $G = \mathrm{GL}_n$ gives $n^2(g - 1) + 1$.

This has application to our Hitchin stack $\mathcal{M}_G(X)$ because $T^* \operatorname{Bun}_G^0(X)$ is an open subscheme of the Hitchin stack $\mathcal{M}_G(X)$. In fact, $T^* \operatorname{Bun}_G^0(X)$ is a symplectic variety because cotangent spaces have a canonical symplectic structure.

2.2.4 Hamiltonian Reduction

Let's recall something about Hamiltonian reduction. If a group H with Lie algebra $\mathfrak{h}$ acts on a variety Y, then H acts no T^*Y in a Hamiltonian way. There is also a moment map $\mu\colon T^*Y \to \langle^*$ which is dual to the canonical

map $\mathfrak{h} \to \operatorname{Vect} Y$ given by the infinitesimal action. (Namely, the Lie algebra should act infinitesimally by vector fields.)

Now, if H acts freely, then

$$\mu^{-1}(0)/H \cong T^*(Y/H).$$

In fact, H-stable functions f on T^*Y automatically descend to $T^*(Y/H)$. The left-hand object is called the Hamiltonian reduction. In this way, one sees that $T^* \operatorname{Bun}_G(X)$ is the Hamiltonian reduction of $T^*G(K)$ by $G(R) \times G(\mathcal{O})$.

Remark 2.34. Here is something called the Poisson bracket: given a nondegenerate symplectic form ω on M, meaning that $d\omega = 0$, then ω^{-1} defines a non-degenerate symplectic form on $\wedge^2 TM$. Accordingly, we may define

$$\{f, g\} := \left\langle \omega^{-1}, df \otimes dg \right\rangle,$$

where f and g live on $\mathcal{O}(M)$. For example, on $M = k^{2n}$ with coordinates $\{x_i, p_i\}$, there is an explicit way to write this out in terms of some derivatives.

2.3 February 25

Today we talk about the classical Hitchin integrable system.

2.3.1 The Classical Hitchin Integrable Systems

For today, we continue to let X be an irreducible smooth projective curve over $\mathbb{C}$ of genus $g \geq 2$, and G is a connected reductive group over $\mathbb{C}$.

Definition 2.35 (Hitchin base)**.** The *Hitchin base* of X for PGL_n is the vector space

$$\mathcal{B} := \bigoplus_{i=1}^{n-1} \mathrm{H}^0\left(X; K_X^{\otimes(i+1)}\right).$$

Remark 2.36. If $m \geq 2$, then

$$\dim \mathrm{H}^0\left(X; K_X^{\otimes m}\right) = (2m-1)(g-1)$$

by the Riemann–Roch theorem, so

$$\dim \mathcal{B} = \sum_{m=2}^{n} (2m-1)(g-1) = \left(n^2 - 1\right)(g-1).$$

Thus, $\mathcal{B}$ has the same dimension as $\operatorname{Bun}_{\mathrm{PGL}_n}(X)$.

Definition 2.37 (Hithin map)**.** Suppose $G = \mathrm{PGL}_n$. The *Hitchin map* is the map $p\colon T^* \operatorname{Bun}_G^0(X) \to \mathcal{B}$ is the map

$$\rho(E, \varphi) := \left(\operatorname{tr} \wedge^2 \varphi, -\operatorname{tr} \wedge^3 \varphi, \ldots, (-1)^n \operatorname{tr} \wedge^n \varphi\right).$$

To make sense of this, we note that $\varphi \in \mathrm{H}^0(X; K_X \otimes \mathfrak{sl}(E))$ has trace zero. Notably, taking each further wedge power increases the level of the differentials.

Theorem 2.38 (Hitchin)**.** Suppose $G = \mathrm{PGL}_n$. Then the map p is generically a Lagrangian fibration, meaning that the generic fibers are Lagrangian.

Proof. We will do this later! ■

Remark 2.39. For example, a dimension argument is able to show that p is dominant.

Remark 2.40. Concretely, this means that coordinate functions on $\mathcal{B}$ will pull back to algebraically independent functions, and their Poisson brackets vanish. Explicitly, by choosing a basis $\{b_j\}$ of $\mathcal{B}$ (compatible with the direct sum decomposition of $\mathcal{B}$), we see that

$$p(E,\varphi) = \sum_j H_j(E,\varphi) b_j,$$

where $H_j \colon T^* \operatorname{Bun}_G^0(X) \to \mathbb{C}$ are some algebraically independent coordinate functions, and $\{H_i, H_j\} = 0$ for all i and j. This is exactly the data of an integrable system.

Remark 2.41. For example, we can see that the fiber over 0 is the nilpotent cone. Thus, p is not smooth! However, p turns out to be flat.

One can define the Hitchin map in other situations. For example, GL_n should add in the trace.

Definition 2.42 (Hitchin map)**.** The *Hitchin base* of X for GL_n is the vector space

$$\mathcal{B} := \bigoplus_{i=0}^{n-1} \mathrm{H}^0\left(X; K_X^{\otimes(i+1)}\right).$$

In this case, the *Hitchin map* $p \colon T^* \operatorname{Bun}_G^0(X) \to \mathcal{B}$ is the map

$$\rho(E,\varphi) := \left(-\operatorname{tr}\varphi, \operatorname{tr}\wedge^2\varphi, -\operatorname{tr}\wedge^3\varphi, \ldots, (-1)^n \operatorname{tr}\wedge^n\varphi\right).$$

Remark 2.43. Here, we find that $\dim \mathcal{B} = \left(n^2-1\right)(g-1) = n^2(g-1)+1$, which is $\dim \operatorname{Bun}_G^0(X)$.

We turn to general groups. To define our Hitchin base, we must generalize our functions $\operatorname{tr}\wedge^\bullet\varphi$. The importance of these functions is that they are functions on $\mathfrak{g}$ which are G-invariant (with the adjoint action). This is desirable because we want to define a function on pairs (E,φ), but changing charts may adjust the local coordinates of E by some conjugation by elements of G. Thus, whatever function on φ we choose needs to be invariant!

We are thus motivated to study $\mathbb{C}[\mathfrak{g}]^G$, for which we recall the following result.

Theorem 2.44 (Chevalley)**.** Fix a connected reductive group G over $\mathbb{C}$ with Lie algebra $\mathfrak{g}$ and rank r. Then there are homogeneous polynomials $\{_1, \ldots, Q_r\}$ for which

$$\mathbb{C}[\mathfrak{g}]^G = \mathbb{C}[Q_1, \ldots, Q_r].$$

In fact, the tuple of degrees $\{\deg Q_i\}_i$ are independent of the precise choices of $Q_\bullet$s.

Example 2.45. For $G = \mathrm{GL}_n$, the degrees are $\{1, \ldots, n\}$, and one can take $Q_i := (-1)^i \operatorname{tr}\wedge^i\varphi$ for each $\varphi \in \mathfrak{g}$.

Example 2.46. For $G = \mathrm{SL}_n$, the degrees are $\{2, \ldots, n\}$. One can take $Q_i := (-1)^i \operatorname{tr}\wedge^i\varphi$ again, where $i \in \{2, \ldots, n\}$.

Remark 2.47. It turns out that the degrees $\{d_i\}$ of the group G arise via the decomposition

$$\mathfrak{g} = \bigoplus_{i=1}^{r} L_{2(d_i-1)}$$

as a representation of the principal $\mathfrak{sl}_2$-triple. (The principal $\mathfrak{sl}_2$-triple is given by setting e to be the sum of the simple e_is, which is a regular nilpotent element.)

Definition 2.48 (Hitchin map)**.** Let the degrees of G be $\{d_i\}_{i=1}^r$. The *Hitchin base* for G is

$$\mathcal{B}_{G,X} := \bigoplus_{i=1}^{r} \mathrm{H}^0\left(X; K_X^{\otimes d_i}\right).$$

When no confusion is possible, we write $\mathcal{B}$ for $\mathcal{B}_{G,X}$. We define the *Hitchin map* $p\colon T^*\operatorname{Bun}_G^0(X) \to \mathcal{B}$ given by

$$p(E,\varphi) := (Q_1(\varphi), \ldots, Q_r(\varphi)).$$

Remark 2.49. The space $\mathcal{B}_{G,X}$ admits a "canonical" definition as the space of sections of the weighted vector bundle $\mathfrak{h}/W \otimes K_X$.

Remark 2.50. By the same Riemann–Roch argument, one finds that

$$\dim \mathcal{B} = \sum_{i=1}^{r} (2d_i - 1)(g-1) + \#\{i : d_i = 1\}.$$

The latter term is the rank $\dim Z(G)$, so the entire dimension is $\dim \operatorname{Bun}_G^0(X)$.

We are now ready to restate Theorem 2.38.

Theorem 2.51. Fix an irreducible smooth projective curve X over $\mathbb{C}$ of genus $g \geq 2$, and let G be a connected reductive group over $\mathbb{C}$. Then the Hitchin map $p\colon T^*\operatorname{Bun}_G^0(X) \to \mathcal{B}$ is generically a Lagrangian fibration.

We will prove that the relevant $H_\bullet$s are Poisson-commutative for all semisimple G, but we will only show the independence for $G = \mathrm{GL}_n$.

Proof of commutativity. For the Poisson commutativity, we use Hamiltonian reduction. Choose a local coordinate t at some $x \in X$. Then our discussion of double quotients granted

$$\operatorname{Bun}_G(X) = G(R)\backslash G(\mathbb{C}((t)))/G(\mathbb{C}[[t]]),$$

where $R = \mathbb{C}[X \setminus \{x\}]$. (We have identified $\mathbb{C}[[t]]$ with $\widehat{\mathcal{O}}_x$.) Then the pre-image of $\operatorname{Bun}_G^0(X)$ is some $G(\mathbb{C}((t)))^0$.

Now, the group $H := G(R) \times G(\mathbb{C}[[t]])/Z(G)$ acts freely on $Y := G(\mathbb{C}[[t]])^0$, and the quotient is $\operatorname{Bun}_G^0(X)$. Because this action is free, we conclude that

$$T^*(Y/H) = T^*Y/H,$$

where the latter is the Hamiltonian reduction $\mu^{-1}(\{0\})$, where $\mu\colon T^* \to \mathfrak{h}^*$ is the moment map. Now, if $\{F_i\}$ are H-invariant Poisson-commutative functions on T^*Y, then they descend to $\mu^{-1}(\{0\})$ by restriction and then to $\mu^{-1}(\{0\})/H$ by the invariance, so they give functions in $T^*(Y/H)$. In fact, these descended functions

continue to be Poisson-commutative. The moral of this reduction is that it is comparatively easier to find functions on T^*Y.

We are thus on the hunt for Poisson-commutative functions on

$$T^*G(\mathbb{C}((t)))^0 = \mathfrak{g}((t))^* \times G((t))^0.$$

The idea is to use left translation to produce invariant functions. Let's start by computing this dual. Note that $\mathfrak{g}((t)) = \mathfrak{g} \otimes \mathbb{C}((t))$, and there is a canonical invariant pairing $\mathbb{C}((t)) \times \mathbb{C}((t))\, dt \to \mathbb{C}$ given by the residue, so $\mathfrak{g}((t))^* = \mathfrak{g} \otimes \mathbb{C}((t))\, dt$.

Let's now define our functions. By definition, a point in $T^*G(\mathbb{C}((t)))$ is represented by a pair (g, φ), where $g \in G((t))$, and $\varphi \in \mathfrak{g} \otimes \mathbb{C}((t))\, dt$. But this φ is a Higgs field in the punctured formal disk $D_x^\times$, so we can use our prior discussion to get well-defined maps

$$\varphi \mapsto (Q_1(\varphi), \ldots, Q_r(\varphi)),$$

where $Q_i(\varphi) \in \mathbb{C}((t))\,(dt)^{\otimes d_i}$. Taking Laurent coefficients of Q_i produces many functions $Q_{i,n}$ on $T^*G((t))$. They are conjugation-invariant (because Q_i is), and they are invariant by left translation (because of how we parallelized our cotangent space), so they are invariant under the two-sided action by $G(R) \times G(\mathbb{C}[[t]])$.

Let's check that the $Q_{i,n}$s are Poisson-commutative. In general, for any Lie group A with Lie algebra $\mathfrak{a}$, we are finding a Poisson structure on $\mathfrak{a}^*$ given by the Lie bracket: choosing a basis $\{a_i\}$ of $\mathfrak{a}$ so that $[a_i, a_j] = \sum_k c_{ij}^k a_k$, one defines

$$\{f, g\} := \sum_{i,j,k} c_{ij}^k \frac{\partial f}{\partial a_i} \frac{\partial g}{\partial a_j} a_k$$

for any $f, g \colon \mathfrak{a}^* \to \mathbb{C}$. In particular, if f is invariant, then $\{a_j, f\} = 0$ for all j by a direct calculation, so $\{-, f\}$ vanishes in general. The same sort of argument works in our situation because the functions $Q_{i,n}$ are suitable invariant.

Thus, we receive functions which descend to Y/H on $\operatorname{Bun}_G^0(X)$. Poisson-commutativity will now follow as soon as we show that these functions descend to the required Hitchin map p. Well, Hamiltonian reduction produces a diagram

$$\begin{tikzcd}
\mu^{-1}(\{0\}) \arrow[r] \arrow[d] & T^*G(\mathbb{C}((t)))^0 \arrow[r, "Q_1,\ldots,Q_r"] & \bigoplus_{i=1}^r \mathbb{C}((t))\, dt^{\otimes d_i} \\
T^* \operatorname{Bun}_G^0(X) \arrow[rr] & & \bigoplus_{i=1}^r \mathrm{H}^0\left(X; K_X^{\otimes d_i}\right) \arrow[u]
\end{tikzcd}$$

where the right vertical map is given by expanding the global differentials at x. It is not too hard to check that this diagram commutes, so the p^*b_js are linear combinations of the $Q_{i,n}$s, so they are Poisson commutative. ■

Remark 2.52. The expansion of p^*b_j in terms of the $Q_{i,n}$s is far from unique!

2.3.2 Spectral Curve

It remains to show the algebraic independence in Theorem 2.51, which we only do for $G = \mathrm{GL}_n$. For this, we will use the notion of the "spectral curve."

Definition 2.53 (spectral curve)**.** Take $G = \mathrm{GL}_n$. For $b \in \mathcal{B}$, write $b = (b_1, \ldots, b_n)$, where we have $b_i \in \mathrm{H}^0(X; K_X^{\otimes i})$ for each i, and consider the factorized polynomial

$$\lambda^n + b_1\lambda^{n-1} + \cdots + b_n = \prod_{i=1}^{n}(\lambda - \lambda_i).$$

Thus, for each $x \in X$, we receive a set of roots $\{\lambda_1, \ldots, \lambda_n\}$in $T_x^*X = K_{X,x}$. As $x \in X$ varies, this cuts out the *spectral curve* C of X. Equivalently, $C \subseteq T^*X$ cut out by the above polynomial (which notably outputs to $(T^*X)^{\otimes n}$).

Remark 2.54. The curve C has no reason to be connected or regular, but it is projective.

Remark 2.55. This is called the spectral curve because if $b = p(E, \varphi)$, then it turns out that C gives the spectrum of φ.

Here are some facts about this spectral curve.

Theorem 2.56 (Hitchin)**.** For generic $b \in \mathcal{B}$, the fiber C_b is smooth and irreducible.

Proof. The fiber being smooth and irreducible is an open condition, so it is enough to exhibit a single $b \in \mathcal{B}$. We will want the following result.

Lemma 2.57. Fix a smooth irreducible curve X over $\mathbb{C}$ of genus $g \geq 2$.

(a) If $\mathcal{L}$ is a line bundle on X of degree at least $2g$, hen a generic sectino of $\mathcal{L}$ has only simple zeroes.

(b) A generic section of $\mathcal{K}_X^{\otimes n}$ has only simple zeroes for $n \geq 2$.

Proof. Quickly, note that (a) implies (b) by a degree calculation. It remains to show (a). Well, let $Y \subseteq \mathrm{H}^0(X; \mathcal{L})$ be the subset of sections which have a double root, so we want to show that Y is positive codimension. By the Riemann–Roch theorem, we know that any line bundle $\mathcal{M}$ of degree $m \geq 2g - 2$ has

$$\dim \mathrm{H}^0(X; \mathcal{M}) = \begin{cases} m - g + 1 & \text{if } \mathcal{M} \neq K_X, \\ g & \text{if } \mathcal{M} = K_X. \end{cases}$$

Now, for each $s \in Y$ with a double root at $z \in X$, we can view s as a section of $\mathcal{L} \otimes \mathcal{O}(-2z)$. But $\deg(\mathcal{L} \otimes \mathcal{O}(-2z)) \geq 2g-2$. Thus, if $\mathcal{L} \otimes \mathcal{O}(-2z) \neq \mathcal{K}_X$ (which happens for all but finitely many z because this requires $\mathcal{O}(2z)$ to equal a fixed line bundle, which is impossible for divisor reasons), then the space of global sections has dimension $\deg \mathcal{L} - g - 1$. Looping over all zs shows that

$$\dim Y \leq d - g < \dim \mathrm{H}^0(X; \mathcal{L}),$$

so we are done. ■

We work with b of the form $(0, \ldots, 0, s)$, where $s \in \mathrm{H}^0\left(X; \mathcal{K}_X^{\otimes n}\right)$ is chosen generically to have no simple zeroes. Thus, our spectral curve is defined by the equation

$$\lambda^n = s(x),$$

which one can check is smooth and irreducible. Indeed, it is irreducible because this polynomial is Eisenstein, and smoothness can be checked by some Jacobian calculation. ■

Remark 2.58. In fact, this argument can also show that the fibers C_b generically have genus $n^2(g-1)+1$. Indeed, one can calculate this genus via the Hurwitz formula with the projection $C_b \to X$. Indeed, the section s admits $2n(g-1)$ simple zeroes, so

$$\chi(C_b) = n\chi(X) - (n-1) \cdot 2n(g-1),$$

which rearranges into the required claim.

2.4 March 2

Here we go.

2.4.1 More on the Higgs Field

We continue with $G = \mathrm{GL}_n$. It is remarkable that the genus of C_b is (generically) $n^2(g-1)+1$, which agrees with $\dim \mathrm{Bun}_G^0(X)$. We are going to attempt to embed $\mathrm{Bun}_G^0(X)$ into the Jacobian of C_b.

Remark 2.59. Suppose that $b \in \mathcal{B}$ is of the form $p(E, \varphi)$ and so that C_b is smooth and irreducible. Then C_b admits a line bundle $\mathcal{L}$ which is an eigenbundle for φ. Indeed, for some $\lambda \in C$ lying over $x \in X$, simply let $\mathcal{L}_\lambda$ be the eigenspace of φ acting on the space E_x; generically, φ has distinct eigenvalues, so we do in fact produce a well-defined line bundle.

Remark 2.60. Note that E and φ can both be reconstructed from the data of C and $\mathcal{L}$: one has $\mathcal{E} = \pi_*\mathcal{L}$, where $\pi\colon C \to X$ is the canonical projection (indeed, this simply glues together the eigenspaces of φ from $\mathcal{L}$ together into $\mathcal{E}$). Further, $\pi_*\mathcal{L}$ admits a canonical Higgs field $\pi_*\mathcal{L} \to \pi_*\mathcal{L} \otimes K_X$, where each summand of $\pi_*\mathcal{L}$ receives the corresponding eigenvalue; this Higgs field goes back to φ.

We are now ready to complete the proof of Theorem 2.51 in this case.

Proof. To start, we restrict p to a subset of $T^* \mathrm{Bun}_G^0(X)$ of a fixed degree ℓ so that it is connected. We see that Higgs fields (E, φ) are in bijection with our pairs $(C, \mathcal{L})$. One can calculate that $\deg \mathcal{L}$ is some fixed degree d, so $p^{-1}(\{b\})$ embeds in $\mathrm{Pic}^d C$, which is isomorphic to $\mathrm{Jac}\, C$. It follows that

$$\dim p^{-1}(\{b\}) \leq n^2(g-1)+1.$$

However, p is of relative dimension at least $n^2(g-1)+1$ by calculating the difference of the dimensions of the codomain and domain. We conclude that all the fibers of p must be the same dimension, and p must be dominant, completing our proof. ■

Remark 2.61. For $G = \mathrm{SL}_n$, it turns out that $p^{-1}(\{b\})$ embeds into the kernel of $\mathrm{Jac}\, C \to \mathrm{Jac}\, X$, which is some kind of Prym variety. For other classical groups, a similar argument works, but the story is more complicated in full generality.

2.4.2 Parabolic Structures

We would like to say something for curves X of smaller genus because curves of high genus are rather complicated to do calculations with.

Definition 2.62 (structure)**.** Choose reductive groups $H \subseteq G$, and let $\mathcal{E}$ be a G-bundle on X. Then an *H-structure* on $\mathcal{E}$ at some fixed $x \in X$ is an H-orbit in the fiber $\mathcal{E}_x$.

Notation 2.63. Choose several parabolic subgroups $P_1, \ldots, P_N$ and points $t_1, \ldots, t_N \in X$. Then we let

$$\mathrm{Bun}_G(X, t, P)$$

be the moduli stack of G-bundles on X with parabolic P_i-structure at t_i for each i.

Example 2.64. For $G = \mathrm{GL}_N$ with P given by the block upper-triangular matrices with sizes $N_1 + \cdots + N_s = N$, then a P-structure on a vector bundle $\mathcal{E}$ of rank N means that we are fixing a partial flag (whose quotient dimensions are given by the $N_\bullet$s) at a point.

Remark 2.65. There is a natural map $\mathrm{Bun}_G(X, t, P) \to \mathrm{Bun}_G(X)$ which simply forgets the parabolic structures. Thus, this is a fibration whose fibers are $G/P_1 \times \cdots \times G/P_N$. It follows that

$$\dim \mathrm{Bun}_G(X, t, P) = \dim \mathrm{Bun}_G(X) + \sum_i (\dim G - \dim P_i).$$

To retell our story of Hitchin systems, our next goal is to introduce some open substack of "stable" bundles with minimal stabilizers (namely, it would be nice to have stabilizer $Z(G)$). This is doable for large N, even for our curves of low genus.

Example 2.66. Take $X = \mathbb{P}^1$, and work with trivial G-bundles with some parabolic structures. It follows that the stack of these bundles is

$$\left(\prod_{i=1}^N G/P_i \right) \Big/ G.$$

For example, $G = \mathrm{PGL}_2$ and P is the Borel subgroup, then we take $N \geq 3$, and we are looking at the quotient $\left(\mathbb{P}^1\right)^N / \mathrm{PGL}_2$. In the open locus where the last three points of $\mathbb{P}^1$ are distinct, there is a unique way to move them to $\{0, 1, \infty\}$ with the PGL_2-action; thus, this locus is isomorphic to the variety $\left(\mathbb{P}^1\right)^{N-3}$.

From now on, we will assume that we have a suitable definition of stable bundle, whose moduli we (unsurprisingly) denote $\mathrm{Bun}_G^0(X, t, P)$.

2.4.3 Parabolic Hitchin Systems

Let's now attempt to define our Hitchin systems.

Remark 2.67. For a given $\mathcal{E}$, the tangent space

$$T^*_{\mathcal{E}} \mathrm{Bun}_G^0(X, t, P)$$

should consist of those Higgs fields $\varphi \colon \mathcal{E} \to \mathcal{E} \otimes K_X$ which have at worst simple poles at the t_is and with residues α that strictly preserve the flag $\{V_\bullet\}$ at t_i, meaning that $\alpha V_i \subseteq V_{i-1}$.

Definition 2.68 (Hitchin map)**.** The *Hitchin map* is now defined exactly as before as

$$p(E, \varphi) := (Q_1(\varphi), \ldots, Q_r(\varphi)).$$

Example 2.69. We continue with the case where $G = \mathrm{PGL}_2$ and the P_is are all B. A parabolic structure at t_i is an element $s_i \in \mathbb{P}\mathcal{E}_{t_i}$. Thus, we are asking for our Higgs fields $\varphi\colon \mathcal{E} \to \mathcal{E} \otimes K_X$ (which must have vanishing trace to descend to PGL_2) to have simple poles at t_i with residue A_i such that $A_i s_i = 0$. Now, we know that

$$p(E, \varphi) = \operatorname{tr} \wedge^2 \varphi.$$

Note that φ is trace-zero and admits a kernel at each t_i, so it is nilpotent. It follows that $p(\mathcal{E}, \varphi)$ is $-\frac{1}{2} \operatorname{tr} \varphi^2$, and staring at our poles shows that this lands in $\mathrm{H}^0(X; K_X^{\otimes 2} \otimes \mathcal{O}(t_1) \otimes \cdots \otimes \mathcal{O}(t_N))$.

Let's continue with the above example, where $X = \mathbb{P}^1$, and we will work with trivial bundles. (Note that all bundles are trivial up to some twist.) Suppose are asking for parabolic structures at the points $t_1, \ldots, t_N \in X$, which we may as well move away from ∞. Now, after doing a partial fraction decomposition, having the required poles means that φ looks like

$$\varphi = \sum_{i=1}^{N} \frac{A_i}{z - t_i},$$

where A_i are the residues; recall that they must satisfy $A_i^2 = 0$ to strictly preserve the flag while having vanishing trace. Furthermore, in order to be regular at ∞, a coordinate change ($z \mapsto 1/z$) has no pole at ∞. It follows that $\sum_i A_i = 0$.

Let's put some coordinates down. Write

$$A_i = \begin{bmatrix} a_i & b_i \\ c_i & -d_i \end{bmatrix}$$

for each i. To strictly preserve our flag, it must send some vector $(y_i, 1)$ to zero. A short calculation shows that this matrix must actually take the form

$$A_i = p_i \begin{bmatrix} y_i & -y_i^2 \\ 1 & -y_i \end{bmatrix}.$$

Thus, we see that $\frac{1}{2} \operatorname{tr} \varphi^2$ is

$$\frac{1}{2} \operatorname{tr} \sum_{i,j} \frac{A_i A_j}{(z - t_i)(z - t_j)} (dz)^2.$$

When $i = j$, the term goes away because $A_i^2 = 0$. Thus, we are left with

$$\frac{1}{2} \operatorname{tr} \varphi^2 = \sum_{i<j} \frac{\operatorname{tr} A_i A_j}{(z - t_i)(z - t_j)} (dz)^2.$$

Another calculation shows that each summand is $p_i p_j (y_i - y_j)^2$. Doing a partial fraction decomposition then shows

$$\frac{1}{2} \operatorname{tr} \varphi^2 = \sum_i \frac{G_i}{z - t_i} (dz)^2,$$

where

$$G_i \coloneqq \sum_{j \neq i} \frac{p_i p_j (y_i - y_j)^2}{t_j - t_i}.$$

These expressions G_i are "Garnier" Hamiltonians. One can calculate that $\{G_i, G_j\} = 0$, and there are N of them, so it may look like we have produced an integrable system. However, these G_is are not independent! Recall $\sum_i A_i = 0$, so it follows that $\sum_i p_i = \sum_i p_i y_i = \sum_i p_i y_i^2 = 0$. (These are "moment map" conditions coming from $\mu = 0$, where $\mu\colon T^*\mathbb{C}^N \to \mathfrak{sl}_2^*$ is the moment map.) On the other hand, $\sum_i G_i = \sum_i t_i G_i = \sum_i t_i^2 G_i = 0$.

Remark 2.70. What is going on here is that $T^*\left(\mathbb{P}^1\right)^{N-3}$ is being found inside $T^*\left(\mathbb{P}^1\right)^N/\operatorname{PGL}_2$. Thus, we cannot hope to have N independent functions because they all come from a smaller space!

Nonetheless, after passing to the quotient, we obtain an integrable system. Formally speaking, our "phase space" $T^*\left(\mathbb{P}^1\right)^N/\operatorname{PGL}_2$ should be given by pairs $(y,p)\in\mathbb{C}^{2N}$ given by $\sum_i p_i=\sum_i p_iy_i=\sum_i p_iy_i^2$, and PGL_2 acts on the $y_\bullet$s by fractional linear transformations and on the $p_\bullet$s dually.

2.4.4 Twisted Structures

Suppose that all of our parabolic subgroups are a fixed Borel subgroup B. Using the double quotient argument from before, we find that

$$\operatorname{Bun}_G(X,t,P)=G(R)\backslash G(\mathbb{C}((z_1)))\times\cdots\times G(\mathbb{C}((z_n)))/(I_1\times\cdots\times I_N),$$

where $R=\mathbb{C}[X\setminus\{t_1,\ldots,t_N\}]$, and $I\subseteq G[[z]]$ is the subalgebra of elements which at $z=0$ evaluate to an element of B at $z=0$; in particular, $G(\mathbb{C}[[z]])/I$ is just G/B. Now, we can choose characters $\lambda_i\colon\operatorname{Lie}I\to\mathbb{C}$ coming from $\mathfrak{h}^*$, and we let $\lambda=(\lambda_1,\ldots,\lambda_N)$ be the tuple of characters. There is now a twisted Hamiltonian reduction $\mu^{-1}(\lambda)/(G(R)\times I_1\times\cdots\times I_N)$, which is some twisted cotangent bundle of Bun_G^0.

Remark 2.71. Let's say something about this twisted Hamiltonian reduction. For some group H acting on Y, we have $T^*(Y/H)=T^*Y/H$ by usual Hamiltonian reduction. Then at a point $y\in Y$, one finds that $T_y^*(Y/H)$ is the kernel of $T_y^*Y\to\mathfrak{h}^*$, where $\mathfrak{h}=\operatorname{Lie}H$. But we can instead take a pre-image by any choice of vector in $\mathfrak{h}^*$, thereby "twisting" our reduction.

Example 2.72. We return to the case of $G=\operatorname{PGL}_2$, where $X=\mathbb{P}^1$. In this case, the cotangent space at $(\mathcal{E},s_1,\ldots,s_N)$ consists of those Higgs bundles $\varphi\colon\mathcal{E}\to\mathcal{E}\otimes K_X$ with simple poles at t_i with residues A_i at t_i for which $A_i|_{s_i}$ is exactly given by the character λ_i. Redoing the calculation of the previous subsection, one finds that A_i satisfies $A_i^2=\lambda_i^2\operatorname{id}$ and so takes the form

$$\begin{bmatrix}-\lambda_i+p_iy_i & 2\lambda_iy_i-p_iy_i^2\\ p_i & \lambda_i-p_iy_i\end{bmatrix}.$$

One then can calculate $\varphi_i=\sum_i\frac{A_i}{z-z_i}$ has an explicit $\frac{1}{2}\operatorname{tr}\varphi^2$ to receive Hamiltonians

$$G_i:=\sum_{j\neq i}\frac{(y_i-y_j)^2p_ip_j-2(\lambda_ip_j-\lambda_jp_i)(y_i-y_j)-2\lambda_i\lambda_j}{t_j-t_i}.$$

This gives an integrable system on the twisted cotangent bundle of $\left(\mathbb{P}^1\right)^{N-3}$, where our coordinate is $\omega=\sum_i dy_i\wedge dp_i$. One can calculate (as before) that $\sum_i G_i=0$, and both $\sum_i t_iG_i$ and $\sum_i t_i^2G_i$ do not depend on λ. Additionally, we have moment map conditions $\sum_i p_i=\sum_i(y_ip_i-\lambda_i)=\sum_i\left(y_i^2p_i-2\lambda_iy_i\right)=0$.

The moral is that we have a family of integrable systems living on some family of cotangent bundles.

2.5 March 4

Today, we discuss a little about quantization.

2.5.1 The Elliptic Calogero–Moser Hamiltonians

Let's do another example of a twisted parabolic Hitchin system. Let X be an elliptic curve with origin 0, and let $P\subseteq\operatorname{PGL}_n$ be the maximal parabolic subgroup of block upper-triangular matrices with sizes $(1,n-1)$. For the twisting, we are asking for a choice of scalar $c\in\mathbb{C}$.

It turns out that a generic degree-0 rank n vector bundle on X is a direct sum of degree-0 line bundles. But line bundles on X of degree zero are parameterized by points: for each $q \in X$, there is a line bundle

$$\mathcal{L}_q \coloneqq \mathcal{O}(q) \otimes \mathcal{O}(0)^{-1}.$$

This line bundle consists of those functions which look like $\theta(z-q)/\theta(z)$, where

$$\theta(z) \coloneqq \sum_{m,n} (-1)^n e^{2\pi i n z + \pi n^2 i \tau}$$

is the theta function on $X = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ (which is entire on $\mathbb{C}$ and admitting only simple zeroes at $m + n\tau$, satisfying $\theta(z) = z + 1$ and $\theta(z+\tau) = -e^{-z}\theta(z)$). In other words, we are looking at meromorphic functions with a single pole at 0 and vanishes at q. Thus, a generic degree-0 rank n vector bundle on X looks like

$$\mathcal{E} = \bigoplus_{i=1}^{n} \mathcal{L}_{q_i}.$$

With $\mathcal{E}$ generic, we may even assume that the $q_\bullet$s are distinct. Now, K_X is the trivial bundle (because X has genus one), so a Higgs field $\mathcal{E} \to \mathcal{E} \otimes K_X$ is just the data of an endomorphism of $\mathcal{E}$. Such an endomorphism φ corresponds to a matrix of endomorphisms $\varphi_{ij} \colon \mathcal{L}_{q_i} \to \mathcal{L}_{q_j}$, meaning that φ_{ij} should be a global section of $\mathcal{L}_{q_j} \otimes \mathcal{L}_{q_i}^{-1}$. Keeping track of the parabolic structure, we are allowing φ_{ij} to admit a simple pole at 0 with some controlled residue. Accordingly, write

$$\varphi_{ij} = a_{ij} \cdot \frac{\theta'(0)\theta(z + q_j - q_i)}{\theta(z)\theta(q_j - q_i)}$$

for $i \neq j$, and $\varphi_{ii} = p_i$ is some momentum variable. Notably, φ_{ij} has residue a_{ij} at 0, and φ_{ii} has residue 0 at 0.

On the other hand, let's talk about what the parabolic structure gives us: a parabolic structure for our given parabolic amounts to the choice of a line in $\mathbb{C}^n$. Now, for the twisting, $\operatorname{Res}_0 \varphi$ acts by $(n-1)c$ on this vector and on $-c$ on the quotient; by some normalization process using actions of the torus, we may even assume that this vector is $(1, \ldots, 1)$. Explicitly, our twisting is given by the character χ on $\operatorname{Lie} P$ given by

$$\begin{bmatrix} a & v \\ 0 & A \end{bmatrix} \mapsto (n-1)ca - c \operatorname{tr} A.$$

Anyway, the point is that $\operatorname{Res}_0 \varphi + c \operatorname{id}$ should have rank 1. But the φ_{ii}s have vanishing residue at 0, so in the basis of the previous paragraph, we need the residue a_{ij} to be c everywhere. Thus,

$$\varphi_{ij} = c \cdot \frac{\theta'(0)\theta(z + q_j - q_i)}{\theta(z)\theta(q_j - q_i)}.$$

We are now ready to compute some polynomials in our Hitchin system. Doing a little calculation, we find that $H_2(z) = \frac{1}{2} \operatorname{tr} \varphi^2$ is

$$\frac{1}{2} \sum_i p_i^2 + c^2 \sum_{i \neq j} \frac{\theta'(0)^2 \theta(z + q_i - q_j)\theta(z + q_j - q_i)}{\theta(z)^2}.$$

The latter sum turns out to be $\wp(z) - \wp(q_i - q_j)$. We are thus left with

$$c^2 \cdot \frac{n(n-1)}{2} \wp(z) + \frac{1}{2} \left(\sum_i p_i^2 - c^2 \sum_{i \neq j} \wp(q_i - q_j) \right).$$

The second sum is known as the elliptic Calogero–Moser Hamiltonian. The higher terms $\operatorname{tr} \varphi^m$ look like $\sum_i p_i^m$ plus some lower-order terms, which give the other coordinates. For example, one can calculate that H_3 looks like

$$\sum_i p_i^2 + \sum_i (a_i(q) p_i + b(q)).$$

2.5.2 Desiderata of Quantization

We should start by explaining what our goal is when quantizing. A classical Hitchin system is an integrable system on T^*Y. In particular, we are trying to live in the ring $\mathcal{O}(T^*Y)$, which is locally generated by $\mathcal{O}(T)$ and some differential coordinates $p_i := dy_i$. Quantization asks us to replace the variables p_i by $\hbar\partial_i$, where $\partial_i := \partial/\partial x_i$. Thus, $\mathcal{O}(T^*Y)$ is quantized to quasiclassical differential operators $\mathcal{D}(Y)$, which is a Rees alegbra

$$\mathcal{D}(Y) = \sum_{j=0}^{\infty} \hbar^j \mathcal{D}_{\leq j}(Y).$$

The point is that each derivative should have a formal factor of $\hbar$; sending $\hbar\partial_i \to p_i$ and then send $\hbar \to 0$ recovers the original.

To quantize our Hitchin system, we are thus interested in taking $H_\bullet$s from the original Hitchin system, and lifting them to $\widehat{H}_\bullet$s in $\mathcal{D}_\hbar(Y)$ which commute in the sense that $[\widehat{H}_i, \widehat{H}_j] = 0$. Notably, we want $\widehat{H}_i$ to specialize to H_i by the limiting process we just described.

Here is a basic example which tells us that such a thing might be possible.

Example 2.73 (Gaudin system)**.** For the Garnier system, we saw

$$G_i = \sum_{j \neq i} \frac{(x_i - x_j)^2 p_i p_j - 2(x_i - x_j)(\lambda_i p_j - \lambda_j p_i)}{t_j - t_i}.$$

Naïvely, we may try to quantize this by setting

$$\widehat{G}_i := \sum_{j \neq i} \frac{\hbar^2 (x_i - x_j)^2 \partial_i \partial_j - (x_i - x_j)(\hbar\lambda_i \partial_j - \hbar\lambda_j \partial_i)}{t_i - t_j}.$$

By some miracle, it turns out that these operators commute, which is not at all expected because $\partial_i \partial_j$ do not commute. In fact, the resulting system is independent of the ordering of the variables (up to some constants): one could even have used (for example) have put all ∂s before the variables.

Let's explain where this construction comes from. Choose representations $V_1, \ldots, V_N$ of a simple Lie algebra $\mathfrak{g}$. Set $\mathcal{H} := V_1 \otimes \cdots \otimes V_N$. There is a Casimir tensor $\Omega \in \left(S^2\mathfrak{g}\right)^G$. One can then divide Ω into terms Ω_{ij} with $i \neq j$, meaning that the first component of the pure tensor Ω_{ij} acts on V_i, and the second acts on V_j. Then we set

$$\widehat{G}_i := \sum_{j \neq i} \frac{\Omega_{ij}}{t_i - t_j},$$

which are operators on $\mathcal{H}$. It then turns out that having $[\Omega_{ij}, \Omega_{k\ell}] = 0$ and $[\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0$ for all i and j and k implies that $[\widehat{G}_i, \widehat{G}_j] = 0$ and descend to an action on $\mathcal{H}^G$.

Example 2.74. With $\mathfrak{g} = \mathfrak{sl}_2$, take the representations V_i to be $\mathbb{C}[x]$, where f acts by $-\partial_x$, e acts by $x^2\partial_x - 2\Lambda_i x$, and h acts by $2x\partial_x - \Lambda_i$.

Remark 2.75. This process can only produce at most quadratic differential operators. In general, quantization must do something more complicated.

Here is another example of quantization.

Example 2.76. We can quantize the Calogero–Moser Hamiltonian

$$H_2 = \frac{1}{2}\sum_i p_i^2 - c^2 \sum_{i \neq j} \wp(q_i - q_j)$$

to

$$\widehat{H}_2 = \frac{1}{2}\sum_i \hbar^2 \partial_i^2 - c^2 \sum_{i \neq j} \wp(q_i - q_j).$$

After factoring out $\hbar^2$, one receives the "quantum Calogero–Moser Hamiltonian" L_2.

Remark 2.77. It further turns out that there are operators $L_3, \ldots, L_n$ which commute and have L_k is equal to $\sum_i \partial_i^k$ up to lower-order terms. The commutativity of the classical system does not directly imply the commutativity of the higher-order terms, as one can see already from our choice of quantization of H_3.

2.5.3 Beilinson–Drinfeld Quantization

Let's sketch the main ideas of how to quantize our classical Hitchin systems. For the time being, set $\hbar = 1$. This requires us to redo much of the construction of the classical Hitchin system. Recall that the classical Hitchin system arose from Hamiltonian reduction: we wrote $\operatorname{Bun}_G(X)$ as $G(R)\backslash G(K_x)/G(\mathcal{O}_x)$, so Hamiltonian reduction granted us an identification

$$T^* \operatorname{Bun}_G(X) = T^* G(K_x)/(G(R) \times G(\mathcal{O}_x)).$$

Thus, we could find a Hitchin system by finding two-sided invariant functions on $T^*G(K_x)$ and then descend to the quotient.

Thus, we must replace $T^* \operatorname{Bun}_G(X)$ by $\mathcal{D}_\hbar(\operatorname{Bun}_G(X))$, and then we need to take 2-sided invariant elements in $\mathcal{D}_\hbar(G(K_x))$ and descend to the quotient. This will not quite work, but it is a reasonable first attempt.

Example 2.78. To get our bearings, we work in the case where X is a Lie group K. Then left-invariant differential operators in $\mathcal{D}(K)$ are given by $U\mathfrak{k}$, where $\mathfrak{k} := \operatorname{Lie} K$, so the two-sided invariant differential operators in $\mathcal{D}(K)$ are given be the center $Z\mathfrak{k}$. Now, our $H_2 = \frac{1}{2}\operatorname{tr}\varphi^2$ looks like $\varphi = \sum_n \varphi z^{-n-1}\,dz$, where $\varphi_n \in \mathfrak{g}$. Now, letting $\{a_i\}$ be an orthonormal basis of $\mathfrak{g}$ for the Killing form so that $\varphi = \sum_i \varphi^i a_i$, we find that

$$\frac{1}{2}\operatorname{tr}\varphi^2 = \sum_n T_n z^{-n-2}\,(dz)^2,$$

where $T_n = \sum_{m,i} \varphi_m^i \varphi_{n-m}^i$. (To make sense of this, we can view the T_ns as some quadratic elements of $U\mathfrak{g}((t))$.) For commutativity, we require $[\varphi_\ell^j, T_n] = 0$; classically, the Poisson bracket does vanish, but we no longer achieve commutativity because there are ordering issues. After a calculation, one finds $[\varphi_\ell^j, T_i] = \ell h^\vee \varphi_{\ell+n}^j$, where $h^\vee$ is the dual Coxeter number.

There is no obvious fix for this, and indeed, there shouldn't be: Beilinson and Drinfeld showed that $\operatorname{Bun}_G(X)$ admits no global nonconstant differential operators. The solution is to consider differential operators on $\frac{1}{2}$-densities.

Remark 2.79. In fact, this is more natural from the perspective of quantum mechanics: the quantization of T^*M should be a Hilbert space. But to have a canonical pairing, we need to use half-densities instead of functions.

Accordingly, we should replace $\mathcal{D}(G(K))$ should be replaced by differential operators on some line bundle. (Here, $G(K)$ is the loop group.) Luckily, there is a suggestion of such a line bundle: the Kac–Moody group $\widehat{G}$ fits into a central extension

$$1 \to \mathbb{C}^\times \to \widehat{G} \to G((t)) \to 1,$$

and the Lie algebra of $\widehat{G}$ is some Kac–Moody algebra. It turns out that the canonical bundle of $\mathrm{Bun}_G(X)$ is $\mathcal{L}^{-2h^\vee}$, where $\mathcal{L}$ is the associated line bundle on the loop group (which notably descends to Bun_G). The moral is that we can replace $U\mathfrak{g}((t))$ with the enveloping algebra of this Kac–Moody algebra above up to identifying K with $-h^\vee$. Doing this quotient expands the center significantly (called the Feigen–Frenkel center), so we get our integrable system by redoing the whole system.

Minicourse 3
Conjugacy Classes

From March 9th to March 18th, these talks were given by George Lusztig. The talk on March 30th was given by Zhiwei Yun.

3.1 March 9

We will assume that we know something about reductive groups already.

3.1.1 Historical Remarks

Let's discuss the history of the following theorem.

Theorem 3.1. For a reductive group G, there are only finitely many unipotent conjugacy classes.

Reductive groups go back to Killing, who apparently did not know that symplectic groups exist. Eventually, he managed to construct G_2. Conjugacy classes (in the form of the Jordan normal theorem) go back to Weierstrass, though often attributed to Jordan. In particular, by the Jordan normal theorem, conjugacy classes are parameterized by a pair of a unipotent conjugacy class and a semisimple conjugacy class. Over $\mathbb{C}$, semisimple conjugacy classes is not so hard (they all diagonalize). For unipotent conjugacy classes, the classification is due to Weierstrass for $\mathfrak{gl}_n$ (via partitions) and due to Williamson for orthogonal and symplectic groups.

Let's discuss the finiteness of unipotent classes. The first important step was by Morozov, who showed that unipotent classes in a reductive group G are parameterized by algebraic maps from SL_2. Then Malcev showed that there are only finitely many such maps up to conjugacy, which together shows that there are only finitely many unipotent conjugacy classes. This result is frequently attributed to Dynkin, who shortly fully classifies all such maps up to conjugacy. Later, it was shown that the parameterization given by Morozov was in fact a bijection, therefore enumerating conjugacy classes.

Chevalley showed that the theory of reductive groups works well over any algebraically closed field, but the discussion of the previous paragraph does not generalize to positive characteristic (even for good primes): Morozov's theorem is no longer true.

Example 3.2. In E_8, there are 70 conjugacy classes in characteristic zero, 74 conjugacy classes in characteristic two, and 71 conjugacy classes in characteristic three.

Remark 3.3. In general, there is a natural embedding from conjugacy classes in characteristic zero to conjugacy classes in positive characteristic. However, there may be more conjugacy classes in the latter!

Wall would classify conjugacy classes for classical groups in characteristic 2. For good primes, the finiteness was proved for exceptional groups by Richardson (though without classification). Students of Iwahori would classify these conjugacy classes for types G_2 and F_4. Lusztig (based on work with Deligne) showed finiteness in full generality shortly thereafter, which was new for the exceptional types E, and classification followed in types E by Mizuno.

3.1.2 Characteristic Zero to Positive Characteristic

Let's begin doing math.

Notation 3.4. Fix a connected reductive group G over an algebraically closed field k. Then we define $\mathcal{D}_G$ to be the set of algebraic group homomorphisms $\omega\colon \mathbb{G}_m \to G$ such that ω lifts to an algebraic homomorphism $\widetilde{\omega}\colon \mathrm{SL}_2 \to G$ for which

$$\widetilde{\omega}\left(\mathrm{diag}(t, 1/t)\right) = \omega(t).$$

Given some $\omega \in \mathcal{D}_G$, we define a descending filtration $\{G_n^\omega\}$ of G so that $\mathrm{Lie}\, G_n^\omega = \mathfrak{g}_n \oplus \mathfrak{g}_{n+1} \oplus \mathfrak{g}_{n+2} \oplus \cdots$, where $\mathfrak{g}_n := \{X \in \mathfrak{g} : \omega(t) = t^n X\}$.

Remark 3.5. It turns out that G_0^ω is parabolic, and G_1^ω is its unipotent radical. (These claims are checked by passing to Lie algebras. For example, for the former claim, one needs to show that $\mathfrak{g}_{\geq n}^\omega$ contains a Borel subgroup.)

Remark 3.6. We work over $k = \mathbb{C}$. The natural map from $\mathrm{Hom}(\mathrm{SL}_2, G)$ to unipotent classes (which descends to a bijection up to conjugacy) factors through $\mathcal{D}_G$ simply by restricting the map $\mathrm{SL}_2 \to G$ to the diagonal.

Notation 3.7. Fix a connected reductive group G over an algebraically closed field k. For a G-orbit $\overline{\omega}$ of some $\omega \in \mathcal{D}_G$, we define

$$\widetilde{H}^{\overline{\omega}} := \bigcup_{\omega \in \overline{\omega}} G_2^\omega,$$

and

$$H^{\overline{\omega}} := \widetilde{H}^{\overline{\omega}} \setminus \bigcup_{\substack{\overline{\omega}' \in \mathcal{D}_G/G \\ \overline{\omega}' \neq \overline{\omega}}} \widetilde{H}^{\overline{\omega}'}.$$

These $H^{\overline{\omega}}$s turn out to partition the unipotent variety.

Remark 3.8. The set $\mathcal{D}_G/G$ is independent of the field (it only depends on root datum). The $H^{\overline{\omega}}$ contains a unique dense unipotent class, so the moral is that there is an embedding from unipotent classes of G over $\mathbb{C}$ to unipotent classes of G over k.

3.1.3 The Springer Resolution

We will want to know a little bit about the Springer resolution.

Notation 3.9. Fix a connected reductive group G over an algebraically closed field k. Then $\mathcal{B}$ is the flag variety of Borel subgroups, and we define W as the set of G-orbits on $\mathcal{B} \times \mathcal{B}$.

Definition 3.10 (Springer resolution)**.** Fix a connected reductive group G over an algebraically closed field k. Then we have a *Springer–Grothendieck resolution*

$$\pi\colon \widetilde{\mathcal{B}} \to \mathcal{B},$$

where $\widetilde{\mathcal{B}} := \{(g, B) \in G \times \mathcal{B} : g \in B\}$. We let $\widetilde{\mathcal{B}}_{\mathrm{rs}}$ denote the pre-image of the regular semisimple locus of G.

Remark 3.11. The Springer resolution is generically W-to-1. Indeed, $\widetilde{\mathcal{B}}_{\mathrm{rs}} \to \mathcal{B}$ is an étale covering with fibers given by W. We then define an embedding $W \to \mathrm{Aut}(\mathcal{B}_{\mathrm{rs}})$ by sending $w \in W$ to the automorphism $B \mapsto B'$, where $(B, B') \in w$. This defines the group structure of W (as a subgroup of $\mathrm{Aut}(\mathcal{B}_{\mathrm{rs}})$.

This map π has some special properties.

Theorem 3.12 (Lusztig)**.** The map $\pi\colon \widetilde{\mathcal{B}} \to \mathcal{B}$ is small, meaning that the dimensions of the fibers are small in some technical sense.

Remark 3.13. Formal nonsense takes the smallness of π and produces an action of W on $\mathrm{H}^*(\mathcal{B}_g; \overline{\mathbb{Q}}_\ell)$ using some intersection cohomology. (Here, $\mathcal{B}_g$ is the fiber of $\mathcal{B}$ over g.) Such an action was expected over $\mathbb{C}$, but this definition now works in any characteristic.

We are now ready to produce our Springer representation.

Notation 3.14. Fix a connected reductive group G over an algebraically closed field. For $g \in G$, we let V_g be the co-image of the map

$$\mathrm{H}^*(\mathcal{B}; \overline{\mathbb{Q}}_\ell) \to \mathrm{H}^*(\mathcal{B}_g; \overline{\mathbb{Q}}_\ell)$$

in degree $2 \dim \mathcal{B}_g$.

Remark 3.15 (Springer correspondence)**.** The action of W on $\mathcal{B}$ produces an action on $\mathrm{H}^*(\mathcal{B}; \overline{\mathbb{Q}}_\ell)$. One can show that V_g is a representation of W, and it turns out to be irreducible. In terms of homology, it is equivalent to ask for the image of this map (in top-degree of the right-hand homology group) to be irreducible.

3.1.4 Our Strata

Here is the main definition of our course.

Notation 3.16. Fix a connected reductive group G over an algebraically closed field. Then $g, g' \in G$ are *equivalent*, written $g \sim g'$, if and only if $V_g \cong V_{g'}$ as representations of W. This stratifies G.

We will find these strata very interesting. Later, we will show the following.

- We will show that each stratum is a union of conjugacy classes of fixed dimension.
- Each stratum contains at most one unipotent conjugacy class.
- The strata are in bijection with the irreducible representations of W.
- Any stratum contains a unipotent element for some choice of characteristic.
- The strata turn out to be the same (roughly speaking) for a group and its Langlands dual. Notably, this is not true for unipotent classes!

Example 3.17. There are three unipotent classes for $G := \mathrm{GL}_3$ over $\mathbb{C}$, and it turns out that there are three strata. Let $G_d \subseteq G$ be the subset of those $g \in G$ for which $\dim Z_G(g) = d$. It turns out that $d \in \{0, 4, 6\}$, and these sets turn out to be the strata. (On the homework, we will check that $\dim Z_G(g) \in \{0, 4, 6\}$, and we will check that the G_ds are smooth irreducible varieties.)

Example 3.18. We work with $G := \mathrm{Sp}_4$ over $\mathbb{C}$, and again let $G_d \subseteq G$ be the subset of those $g \in G$ for which $\dim Z_G(g) = d$. The G_ds turn out to not be the strata: indeed, G_4 splits into $G_4' \sqcup G_4''$, where the former is the conjugacy class of $\mathrm{diag}(1, 1, -1, -1)$, and the latter corresponds to a unipotent conjugacy class in characteristic 2. But up to this splitting, we have indeed recovered our strata. Notably, G_4' contains no unipotent class in characteristic zero. We are asked to show that these strata have smooth irreducible components on the homework, but it may be too difficult.

Remark 3.19. In general, the strata have smooth irreducible components for classical groups, but Professor Lusztig does not know if this is true in general.

3.2 March 11

Here we go.

3.2.1 Characteristic Zero to Positive Characteristic, Again

For today, for primes p, we set $k_p := \overline{\mathbb{F}}_p$ and $k_0 := \mathbb{C}$. We then choose some characteristic p, and we let G_p be a connected reductive group over k_p, we let $U_p \subseteq G_p(k_p)$ is the set of unipotent elements, and $\mathrm{Conj}(U_p)$ is the set of unipotent conjugacy classes. Last time, we showed that there is an embedding $\mathrm{Conj}(U_0) \to \mathrm{Conj}(U_p)$; note that these are finite sets.

Remark 3.20. One has that U_p can be stratified into a disjoint union of locally closed subsets (of U_p) given by the $H^{\overline{\omega}}$s, and these locally closed subsets have the same indexing set as for $\mathrm{Conj}(U_0)$.

Now, for any of our $\mathcal{P} \subseteq U_p$ and finite field $\mathbb{F}_q \subseteq k_p$, one has that $|\mathcal{P}(\mathbb{F}_q)|$ is independent of p (as a function of q).

Example 3.21. Consider $\mathrm{Sp}_4(\mathbb{F}_q)$. This has four pieces for odd q.

- There is the identity (with 1 element).
- There is a class with $q^4 - 1$ elements.
- The subregular unipotent elements split into two classes, which have $\frac{1}{2}q(q-1)\left(q^4-1\right)$ elements and $\frac{1}{2}q(q+1)\left(q^4-1\right)$ elements. (This rejoins as a single conjugacy class over $\overline{\mathbb{F}}_q$.)
- The regular unipotent elements split into two classes, each with $\frac{1}{2}q^2\left(q^2-1\right)\left(q^4+1\right)$ elements.

For $p = 2$, the subregular elements split into two conjugacy classes (even over $\overline{\mathbb{F}}_2$) which have a different number of elements than above (namely, $\left(q^2-1\right)\left(q^4-1\right)$ and q^4-1), but the union of all the subregular unipotent elements is the same!

Remark 3.22. Steinberg has shown that the total number of unipotent elements is a power of q.

The point of this discussion is that we may define the following.

Notation 3.23. We define $\mathrm{Conj}(U_*)$ to be the quotient of $\bigsqcup_{p>0} \mathrm{Conj}(U_p)$ by the equivalence relation $\sim$ given by $x_0 \sim x_p$ for any $x_0 \in \mathrm{Conj}(U_0)$.

Example 3.24. For classical groups, it turns out that $\mathrm{Conj}(U_*) = \mathrm{Conj}(U_2)$.

Example 3.25. For E_8, one has that $\mathrm{Conj}(U_*) = (\mathrm{Conj}(U_2) \sqcup \mathrm{Conj}(U_3))/\mathrm{Conj}(U_0)$. Here, $\#\,\mathrm{Conj}(U_2) = 74$ and $\#\,\mathrm{Conj}(U_3) = 71$, so $\#\,\mathrm{Conj}(U_*) = 75$.

We are going to use $\mathrm{Conj}(U_*)$ to index the strata of G discussed last time.

Let's use the Springer representation to give another embedding $\mathrm{Conj}(U_0) \to \mathrm{Conj}(U_p)$. Let's start with symplectic types.

Notation 3.26. Recall that the irreducible representations of the Weyl group in type C_n is given by a sequence (a_i) with $a_0 \le a_2 \le \cdots \le a_{2m}$ and $a_1 \le a_3 \le \cdots \le a_{2m-1}$ and total sum $\sum_i a_i = n$; two such sequences are the same if we prepend by two 0s. These representations are found by inducing from the two copies of GL_m in a parabolic. We might label this set by $\mathrm{Irr}\, C_n$.

Remark 3.27. Professor Lusztig labeled this set as $\mathrm{Irr}\, B_n$ because the Weyl group in types B_n and C_n are the same. We will continue to use C_n to avoid confusion.

Here are some alternative parameterizations.

Notation 3.28. We define $\mathrm{Sym}\, C_n$ to be given by sequences (b_i), where we replace the $\le$s requirements with $<$ and replace the sum condition to summing to $n + m^2$. Note that there is a bijection $\mathrm{Irr}\, C_n \to \mathrm{Sym}\, C_n$ by taking (a_i) to $(a_i + \lfloor i/2 \rfloor)_i$. The equivalence classes of increasing sequences (b_i) parameterize the "special" representations of the Weyl group.

The notion of "special" representation was defined in a paper of Lusztig in 1978.

Notation 3.29. We define $\mathrm{Sym}'\, C_n$ to be given by sequences (c_i), where we replace the $\le$s requirements with $\le\ 2+$ and replace the sum condition to summing to $n + 2m^2 + m$. One should also adjust the equivalence to be $(c_i) \sim (0, 1, c_0 + 2, c_1 + 2, \ldots)$. Note that there is a bijection $\mathrm{Sym}\, C_n \to \mathrm{Sym}'\, C_n$ by taking (b_i) to $(b_i + \lfloor (i+1)/2 \rfloor)_i$. The equivalence classes of increasing sequences (c_i) parameterizes the representations of W which correspond to unipotent classes under the Springer correspondence for $p \ne 2$.

Notation 3.30. We define $\mathrm{Sym}''\, C_n$ to be given by sequences (d_i), where we replace the $\le$s requirements with $\le 4+$ and replace the sum condition to summing to $n + 4m^2 + 2m$; we also need $d_1 \ge 2$. One should also adjust the equivalence to be $(d_i) \sim (0, 2, c_0 + 4, c_1 + 4, \ldots)$. Note that there is a bijection $\mathrm{Sym}'\, C_n \to \mathrm{Sym}''\, C_n$ by taking (c_i) to $(c_i + i)_i$. The equivalence classes of increasing sequences (d_i) parameterizes the representations of W which correspond to unipotent classes under the Springer correspondence for $p = 2$.

The statements about the unipotent classes were worked out by Lusztig (and Spaltenstein for $p = 2$) in the 1980s. We remark that these parameterizations automatically provide embeddings $\mathrm{Conj}(U_p) \to \mathrm{Conj}(U_2)$ for $p \ne 2$.

Remark 3.31. One can do something similar in other types. In particular, basically everything is the same in the odd orthogonal types B_n, except Sym' changes slightly. In type D_n, all the definitions are the same, except one removes a_0s from $\mathrm{Irr}\, D_n$, removes $b_0 s$ from $\mathrm{Sym}\, D_n$ (and sums to $n+m^2-m$), removes c_0s from $\mathrm{Sym}'\, D_n$ (and sums to $n+2m^2-2m$), and removes d_0s from $\mathrm{Sym}''\, D_n$ (and sums to $n+4m^2-4m$). Importantly, one also needs to adjust our equivalence so that $(a_1, a_2, a_3, a_4, \ldots) \sim (a_2, a_1, a_4, a_3, \ldots)$, and when this movement does nothing, our sets should have two copies of this sequence. The statements about the parameterizations are all the same.

Here is some more notation.

Notation 3.32. Let $\mathrm{Irr}_*(W)$ be the set of Weyl group representations $\{V_g\}_{g \in G_p}$. One can show that $\mathrm{Irr}_*(W)$ is independent of p.

Remark 3.33. It turns out that

$$\mathrm{Irr}_* W = \bigcup_{\text{prime } p} \mathrm{Irr}_p W.$$

Remark 3.34. The Springer correspondence provides a bijection $\mathrm{Conj}(U_*) \to \mathrm{Irr}_* W$ which restricts to isomorphisms $\mathrm{Conj}(U_p) \to \mathrm{Irr}_p W$ for each prime p.

Example 3.35. For E_8, one finds that $\mathrm{Irr}\, W$ has 112 elements, but $\mathrm{Irr}_* W$ has 75 elements.

The moral is that the indexing of our strata will be independent of the characteristic.

3.2.2 2-Special Representations

Let V and V^* be dual rational vector spaces, and let $\langle -, - \rangle$ be the induced pairing. We also let $\Phi \subseteq V$ and $\Phi^\vee \subseteq V^*$ be dual root systems, and we let W be the corresponding Weyl group generated by the (simple) reflections.

Notation 3.36. For $e \in V$, we define

$$\Phi_e := \{\alpha \in \Phi : \langle e, \alpha^\vee \rangle \in \mathbb{Z}\},$$

and

$$\Phi_e^\vee := \{\alpha^\vee \in \Phi : \alpha \in \Phi e\}.$$

Dually, for $e \in V^*$, we may also define Φ_e and $\Phi_e^\vee$ as above (replacing Φ with $\Phi^\vee$ and adjusting equations as needed).

Remark 3.37. Now, for a pair (e, e'), it turns out that $(V, V^*, \Phi_e \cap \Phi_{e'}, \Phi_e^\vee \cap \Phi_{e'}^\vee)$ is a root system. We let its Weyl group be $W_{ee'}$.

Example 3.38. Here are the possible subgroups $W_{ee'}$ of the same rank.

- In type A_n, the Weyl subgroups $W_{ee'}$ are A_n.
- In types B_n and C_n, the Weyl subgroups $W_{ee'}$ take the form $B_? \times B_? \times D_? \times D_?$.
- For type D_n, the Weyl subgroups look like $D_? \times D_? \times D_? \times D_?$.
- There are also lists for the exceptional types.

Let's explain why this relates to $\operatorname{Irr}_* W$.

Notation 3.39. For an irreducible representation E of $W_{ee'}$, we let n_E be the smallest nonnegative integer for which $E \subseteq V^{\otimes n_E}$, where V is the standard (reflection) representation of $W_{ee'}$. We have a similar definition for irreducible representations of W.

Notation 3.40. Let E be the special representation of $W_{ee'}$. Then its induction to W has a unique irreducible component $\widetilde{E}$ for which $n_{\widetilde{E}} = n_E$, which we denote by $j_{W_{ee'}}^W(E)$.

Definition 3.41 (special)**.** An irreducible representation $\widetilde{E}$ is 2-*special* if and only if it takes the form $j_{W_{ee'}}^W(E)$ for some $(e, e') \in V \times V^*$ and some special representation E of $W_{ee'}$.

It turns out that the 2-special representations of W are in bijection with $\operatorname{Irr}_* W$.

Remark 3.42. It turns out that there is another definition of a 1-special representation, where one deals with W_e instead of $W_{ee'}$.

3.3 March 16

Here we go.

3.3.1 Weyl Conjugacy Classes

Our next goal is to redefine our strata in terms of conjugacy classes of the Weyl group W of the group G.

Definition 3.43. Fix $w \in W$ and a conjugacy class γ of a reductive group G_p in characteristic p. Then we define

$$\mathcal{B}_{\gamma,w} := \left\{(g, B) \in \gamma \times \mathcal{B}_p : \left(B, gBg^{-1}\right) \text{ are in relative position } w\right\}.$$

Remark 3.44. One can similarly fix g and only look at the Borel subgroups in $\mathcal{B}_{\gamma,w}$, which are refinements of the Springer fibers.

Theorem 3.45 (Lusztig 2011)**.** Fix $w \in W$ and a conjugacy class γ of a reductive group G_p in characteristic p. Suppose that w has minimal length its in conjugacy class. Then there is a unique unipotent conjugacy class γ in $\operatorname{Conj}(U_p)$ so that $\mathcal{B}_{\gamma,w}$ is nonempty and has minimal dimension. In fact, γ depends only on the conjugacy class of w.

Theorem 3.45 was settled by a complicated argument for classical types and using a computer for exceptional types.

Remark 3.46. Professor Zhiwei Yun has another construction of this map in characteristic zero.

Theorem 3.45 produces the following definition.

Definition 3.47. Fix a reductive group G_p in characteristic p. Then we define a map $\varphi_p \colon \operatorname{Conj} W \to \operatorname{Conj}(U_p)$ by sending the conjugacy class of w (of minimal length in its conjugacy class) to the class γ for which $\mathcal{B}_{\gamma,w}$ is nonempty of minimal dimension.

We now extend this to $\mathrm{Conj}(U_*)$.

Definition 3.48. For a split reductive group G defined over $\mathbb{Z}$, we extend the φ_ps to a map $\varphi_* \colon \mathrm{Conj}\, W \to \mathrm{Conj}\, U_*$ as follows.

- If $\varphi_p(w)$ is independent of p, then we define $\varphi_*(w)$ as the value at any p.
- If $\varphi_p(w)$ is not independent of p, then it turns out to only be different at a single prime p_0, and we define $\varphi_*(w) = \varphi_{p_0}(w)$.

The fibers in W are called *strata*.

Example 3.49. Composing with the Springer correspondence, we see that there is thus a surjective map $\mathrm{Conj}\, W \to \mathrm{Irr}_*\, W$. Here are the fibers of this map for E_8; throughout, A_B denotes the representation of dimension A which appears for the first time in the Bth symmetric power of the reflection representation of the Weyl group.

- The representation 15_{56} has five elements in its fiber. This kind of looks like an A_5.
- The representation 700_{28} has four elements in its fiber. It kind of looks like an A_4.
- The representations 1400_{29} or 2016_{19} have fibers of size four, which look like $A_2 \times A_2$.
- There are also five representations with fiber of size three, fourteen representations with fiber of size two, and fifty-two representations with fiber of size one.

One makes the fibers into the graphs described above by finding paths from the most elliptic to the least elliptic conjugacy classes; it turns out that each class contains a unique most elliptic class.

Remark 3.50. In fact, in characteristic two, one finds that the reductive part of the centralizer of a unipotent element corresponding to an elliptic Weyl group element is a product of symplectic groups, and it kind of explains how one finds the sizes of the fibers. This feature works in general, not just for E_8.

The moral is that we can parameterize conjugacy classes by representations, which we understand better.

Example 3.51. Let's describe the map φ_* in type C. This type is classical, so $\mathrm{Conj}(U_*) = \mathrm{Conj}(U_2)$, and it turns out that $\varphi_* = \varphi_2$. The classes admit a combinatorial description as partitions $\nu \vdash 2n$ for which an odd part appears an even number of times, and any even part which appears an even number of times has an associated number $\varepsilon \in \{0, 1\}$. (The invariant ε disappear in characteristic zero.) The Weyl group W can be described as the permutations in S_{2n} which commute with the involution $k \mapsto (2n + 1 - k)$. It turns out that the map $\mathrm{Conj}(W) \to \mathrm{Conj}(U_2)$ sends $w \in W$ to the partition given by the lengths of its cycles; for cycles of even length, it may not commute with the involution, in which case we give it $\varepsilon = 1$ (and we give it $\varepsilon = 0$ otherwise).

Example 3.52. In type C, the sequences (d_i) in $\mathrm{Sym}''\, C_n$ with

$$d_0 \le d_1 + 2 \le d_2 + 4 \le \cdots$$

parameterize those elements in $\mathrm{Irr}_*\, W$ corresponding to an elliptic conjugacy class in W.

3.4 March 18

Here we go.

3.4.1 More on Strata

Last time, we explained how to take a Weyl conjugacy class and produce an element of $\operatorname{Conj} U_*$, which of course index the strata.

Remark 3.53. If we restrict to the "elliptic" Weyl elements, then the conjugacy classes inject into the strata. We may call the image of this map the "elliptic" strata.

Remark 3.54. If G is semisimple, then for any w of minimal length in an elliptic conjugacy class, the length is equal to the dimension of the centralizer of the corresponding stratum. Professor Lusztig has only checked this in a case-by-case manner.

Remark 3.55. Except for one stratum in type E_8, each elliptic stratum contains a semisimple element.

Example 3.56. In type A_n, the unique elliptic conjugacy class is given by the n-cycle. It goes to the regular semisimple stratum.

Let's discuss the proof of the following conjecture of Lusztig.

Theorem 3.57 (Carnovale)**.** Let G be a reductive group. Any stratum of G is locally closed.

Proof. Let G_m be the collection of $g \in G$ of dimension m. This is a union of strata, and it is known to be locally closed.

The proof will involved "pieces." Let's review this definition.

Notation 3.58. Let G be a reductive group. For each $g \in G$, let g_s be the semisimple element, and we define $H_G(g) := Z_G(g_s)^\circ$.

Remark 3.59. Note that $H_G(g)$ is the Levi subgroup of some parabolic because it is the centralizer of a torus.

Definition 3.60 (piece)**.** Let G be a reductive group. For any Levi subgroup L of a parabolic, we may let S_1 be the pre-image of an isolated class in $L/Z^\circ(L)$. Here, an element of a connected reductive group is said to be isolated if and only if the connected components of the centralizer of its semisimple part is not contained in a Levi subgroup of a proper parabolic subgroup. We can then define Y_{L,S_1} to be the union of all conjugacy classes in G which meet those $g \in S_1$ and having $H_G(g) = L$. It turns out that there are only finitely many pairs (L, S_1) up to conjugacy, and these Y_{L,S_1}s are the pieces of G.

Remark 3.61. Lusztig has shown that the closure of a piece is a union of pieces.

Now, Carnovale shows that a stratum Σ is a union of pieces. In fact, for any piece $J \subseteq \Sigma$, one can find m with $\Sigma \subseteq G_m$, and then one can show that $\overline{J} \cap G_m \subseteq \Sigma$. It follows that Σ is the union of these closed subsets $\overline{J} \cap G_m$, so Σ is closed in G_m and hence locally closed in G. ■

Remark 3.62. In fact, the proof has shown that each stratum Σ is closed in G_m. One can show further that a stratum is a union of irreducible components.

Remark 3.63. There is a map ε_p from irreducible representations to W to $\operatorname{Conj} U_p$: any irreducible representation of W is parameterized by the Springer correspondence by a unipotent element and a local system, so we can forget about the local system. We can then return another irreducible representation by adding back in the trivial local system. For an irreducible representation E of W, we may write $\varepsilon_*(E)$ for the value of $\varepsilon_p(E)$ if this irreducible representation is independent of p; if it is not independent of p, then it only changes at a single prime p_0, and we define $\varepsilon_*(E)$ to be this exceptional value.

Remark 3.64. There is a notion of a "superstratum." One can decompose $\operatorname{Irr} W$ into sets $\operatorname{Irr}_c W$, where c is a two-sided cell. For example, in type C, one has the symbols $(0,1,2)$ and $(0,2,1)$ and $(1,0,2)$ in the same two-sided cell. A superstratum is then the union of the strata Σ which fix a two-sided cell, meaning that there is a map $\delta\colon G \to \operatorname{Irr}_* W$ (maybe given by the Springer correspondence), and the superstratum is the union of the Σ for which $\delta(\Sigma) \in c$, for fixed two-sided cell c. It is conjectured that each superstratum is locally closed and that its irreducible components are rationally smooth.

Example 3.65. In type A, the strata agree with the superstrata.

Example 3.66. For Sp_4, the superstrata are given as follows.

- There is a superstratum of the zero-dimensional conjugacy classes.
- There is a superstratum of the eight-dimensional conjugacy classes.
- Everything else belongs to a third superstratum.

3.5 March 30

This talk was given by Zhiwei Yun.

3.5.1 Review

Let's recall what was covered. Fix a reductive group G over an algebraically closed field, and let W be the Weyl group. For the past few weeks, we have been discussing the strata of a reductive group G, which today we will denote $\mathrm{St}(G)$. It was related to both conjugacy classes in the Weyl group and the irreducible representations of the Weyl group W. In particular, we had the following two maps.

- Taking some inductions produces an embedding map

$$\mathrm{Str}(G) \to \mathrm{Irr}(G).$$

  The image is known as the 2-special representations and is denoted $\operatorname{Irr}_* W$. This map admits a section given by the Springer correspondence, which maps to $\mathrm{Conj}(U_p)$ (where p is the characteristic), which then embeds into $\operatorname{Irr}_* W$.

- There was a complicated map $\mathrm{cl}(W) \to \mathrm{Irr}(G)$. We will give a construction in characteristic zero shortly.

- The complicated map $\mathrm{cl}(W) \to \mathrm{Irr}(G)$ admits two canonical sections, given by choosing either the most or least elliptic elements of G.

Let's specialize to characteristic zero. Then let $\mathcal{N}$ be the nilpotent orbits of the Lie algebra $\mathfrak{g}$, and we are able to gives maps to and from $\mathcal{N}$ as above.

- By the Springer correspondence, there is a map $\mathcal{N} \to \operatorname{Irr}(W)$ given by sending a nilpotent orbit $\mathcal{O}$ to $\mathrm{H}^{\mathrm{top}}(\mathcal{B}_e)$. Conversely, the Springer correspondence associates an orbit and a local system to each irreducible representation in $\operatorname{Irr} W$, so one can forget the local system to recover a nilpotent orbit.
- There is also a surjection $\operatorname{cl}(W) \to \mathcal{N}$. Namely, given a class in W, let w be a representative of minimal length. Then it is a highly nontrivial theorem of Lusztig that $BwB \cap \mathcal{U}$ admits intersection with a unique minimal nilpotent orbit. (Here, $\mathcal{U}$ is the unipotent variety.) There are once again sections $\mathcal{N} \to \operatorname{cl}(W)$ given by choosing the most or least elliptic elements.

Today, we will focus on the maps to and from $\operatorname{cl}(W)$.

3.5.2 The Flag Variety

For the rest of today, we fix a reductive group G over $\mathbb{C}$. Here is some notation for today.

Definition 3.67 (loop group)**.** Fix a reductive group G over $\mathbb{C}$. Then the *loop group* LG is $LG(\mathbb{C}) := G(\mathbb{C}((t)))$. We may also write $L^+G := G(\mathbb{C}[[t]])$, and then there is a Grassmannian $\mathrm{Gr} := LG/L^+G$.

Remark 3.68. The group LG is an infinite-dimensional Lie group. Notably, it may admit different connected components if G is not simply connected.

Definition 3.69 (flag variety)**.** Fix a reductive group G over $\mathbb{C}$. Then the *Iwahori subgroup* I consists of those $g \in L^+G$ such that specializing $t \to 0$ places g in B. We then define the *flag variety* as

$$\mathrm{Fl} := LG/I.$$

Remark 3.70. This flag variety turns out to admit a stratification $\bigcup_{w \in \widetilde{W}} \mathrm{Fl}_w$, where $\widetilde{W} := X_*(T) \rtimes W$ is the affine Weyl group.

Notation 3.71. There are affine Springer fibers defined as follows: for $\gamma \in \mathfrak{g}((t))$, we define

$$\mathrm{Fl}_\gamma := \{g \in LG/I : \operatorname{Ad}_{g^{-1}} \gamma \in \{X \in \mathfrak{g}[[t]] : X_{t=0} \in \mathfrak{b}\}\}.$$

This is exactly analogous to the usual Springer fiber $\mathcal{B}_e = \{g \in G/B : \operatorname{Ad}_{g^{-1}} e \in \mathfrak{b}\}$. While we're here, we set

$$\mathrm{Gr}_\gamma := \{g \in LG/L^+G : \operatorname{Ad}_{g^{-1}} \gamma \in \mathfrak{g}[[t]]\}.$$

Remark 3.72. It turns out that Fl_γ and Gr_γ are nonempty if and only if γ is integral, which means that γ can be conjugated into $\mathfrak{g}[[t]]$. (This has no content for Gr_γ.) For GL_n, being integral amounts to saying that the characteristic polynomial lives in $\mathbb{C}[[t]]$.

Remark 3.73. The fiber of the map $\mathrm{Fl}_\gamma \to \mathrm{Gr}_\gamma$ looks like $\mathcal{B}_e$, which varies over the base Gr_γ.

We will not be interested in all γs, even those which are integral.

Remark 3.74. If γ is regular and semisimple, then Fl_γ and Gr_γ are finite-dimensional, but they may have infinitely many irreducible components. In this case, Kazhdan–Lusztig (modifying an idea of Spaltenstein) showed that the Fl_γs are equidimensional with a conjectural formula for the dimension, which was proven by Bezrukavnikov; Ngô showed that the Gr_γs are equidimensional.

Definition 3.75 (topologically nilpotent)**.** Fix a Lie algebra $\mathfrak{g}$ over $\mathbb{C}$, and choose $\gamma \in \mathfrak{g}[[t]]$. Then γ is *topologically nilpotent* (for the t-adic topology) if and only if it can be conjugated into $\{X \in \mathfrak{g}[[t]] : X_{t=0} \in \mathfrak{n}\}$.

Example 3.76. For GL_n, being topologically nilpotent amounts to saying that the characteristic polynomial has vanishing constant term.

3.5.3 Kazhdan–Lusztig Maps

We are now able to define a map in one direction.

Definition 3.77 (Kazhdan–Lusztig map)**.** There is a Kazhdan–Lusztig map $\mathrm{KL}\colon \mathcal{N} \to \mathrm{cl}(W)$ defined as follows. Given $e \in \mathcal{N}$, choose a generic power series

$$\gamma = e + X_1 t + X_2 t^2 + \cdots \in \mathfrak{g}[[t]].$$

(The genericity is rather non-explicit, even once we assume that γ is topologically nilpotent—which is automatic here—and regular semisimple.) Then we define the class $[w]_\gamma \in \mathrm{cl}(W)$ to be given by the splitting type of γ. For generic γ, this $[w]_\gamma$ does not depend on the choice of γ.

Example 3.78. For $G = \mathrm{GL}_n$, the characteristic polynomial of γ can be written as a product of irreducible polynomials in $\mathbb{C}((t))[x]$. This factorization gives a partition of n, which corresponds to a conjugacy class in S_n.

Remark 3.79. For general G, the centralizers $C_{LG}(\gamma)$ are some tori over $\mathbb{C}((t))$, which is classified by $\mathrm{H}^1(\mathbb{C}((t)); W)$, but the latter is $\mathrm{cl}(W)$ because $\mathrm{Gal}(\overline{\mathbb{C}((t))}/\mathbb{C}((t))) = \widehat{\mathbb{Z}}$. This explains how to recover a class from the "splitting type" of γ.

Remark 3.80. Kazhdan–Lusztig conjectured that KL is injective. This was proven in many cases by Spaltenstein and proven in full by Yun.

To upgrade the Kazhdan–Lusztig map, we will also need some analog of parabolic subgroups for our situation.

Definition 3.81 (parahoric)**.** Fix a reductive group G over $\mathbb{C}$. Then we define the *parahoric subgroups* P as those between I and LG°. We then define the *pseudo-Levi subgroup* $L_P := P/P^u$, where P^u is the unipotent radical.

Example 3.82. One may take P to be I or L^+G.

Example 3.83. For $G = \mathrm{SL}_2$, one can also take P to be the subgroup

$$\begin{bmatrix} t & \\ & 1 \end{bmatrix} L^+\mathrm{SL}_2 \begin{bmatrix} t & \\ & 1 \end{bmatrix}.$$

Remark 3.84. It turns out that P/P^u is a finite-dimensional reductive group. Its Lie algebra $\mathfrak{l}_\mathfrak{p}$ sits in a split short exact sequence

$$0 \to \mathfrak{p}^u \to \mathfrak{p} \to \mathfrak{l}_\mathfrak{p} \to 0.$$

We can now upgrade our Kazhdan–Lusztig map.

Definition 3.85 (Kazhdan–Lusztig map)**.** Fix a parahoric subgroup P. There is a Kazhdan–Lusztig map $\mathrm{KL}_P \colon \mathcal{N}_{L_P} \to \mathrm{cl}(W)$ which sends $e \in \mathcal{N}_{L_P}$ to the splitting type of some generic γ with constant term γ.

Remark 3.86. These Kazhdan–Lusztig maps can be assembled together. Choose parahoric subgroups P and Q, and suppose $P \subseteq Q$. Then P/Q^u is a parabolic in L_Q, and it admits a natural surjection onto L_P. Now, Lusztig–Spaltenstein defined a map $\mathrm{ind} \colon \mathcal{N}_{L_P} \to \mathcal{N}_{L_Q}$ given by sending e to e plus a generic element in the unipotent radical. It now turns out that

$$\begin{tikzcd} \mathcal{N}_{L_P} \arrow[r, "\mathrm{ind}"] \arrow[dr, "\mathrm{KL}_P"'] & \mathcal{N}_{L_Q} \arrow[d, "\mathrm{KL}_Q"] \\ & \mathrm{cl}(W) \end{tikzcd}$$

commutes. Thus, there is a map $\operatorname{colim}_P \mathcal{N}_{L_P} \to \mathrm{cl}(W)$.

Example 3.87. Take $G = \mathrm{GL}_n$, and suppose that P is maximal, and $Q = LG$. Then the induction $\mathrm{ind} \colon \mathcal{N}_{L_P} \to \mathcal{N}_{L_Q}$ basically takes a pair of partitions (λ, μ) to the partition $(\lambda_1 + \mu_1, \lambda_2 + \mu_2, \ldots)$.

Remark 3.88. If P and P' are parahoric subgroups with the same Levi (i.e., $L_P = L_{P'}$), then $\mathrm{KL}_P = \mathrm{KL}_{P'}$. Thus, we may write KL_{L_P} instead of KL_P.

Conjecturally, the Kazhdan–Lusztig map knows about strata.

Conjecture 3.89 (Lusztig)**.** Fix a reductive group G over $\mathbb{C}$. Then the image of the map $\operatorname{colim}_P \mathcal{N}_{L_P} \to \mathrm{cl}(W)$ agrees with the image of the map $\mathrm{Str}(G) \to \mathrm{cl}(W)$ given by choosing the most elliptic map.

Remark 3.90. It is also conjectured that the image of KL agrees with the image of the nilpotent orbits $\mathcal{N}$, which is a subset of $\mathrm{Str}(G)$, in $\mathrm{cl}(W)$.

Remark 3.91. Roughly speaking, we are hoping to define a map $\operatorname{colim} \mathcal{N}_{L_P} \to \operatorname{Str} G$, which we then hope recovers the Kazhdan–Lusztig map once we compose with the map $\mathrm{Str}(G) \to \mathrm{cl}(W)$. Here is a proposal for this map: send $e \in \mathcal{N}_{L_P}$ to a stratum containing a generic element of the form $s \exp(e)$, where s is semisimple.

3.5.4 The Minimal Reduction Type

We now hunt for a map $\mathrm{cl}(W) \to \mathcal{N}$, and then we will conjecturally construct a map $\mathrm{cl}(W) \to \mathrm{Str}(G)$.

Definition 3.92 (minimal reducton type)**.** We define the *minimal reduction type* $\mathrm{RT}_{\min}$ to send a topolotically nilpotent element $\gamma \in \mathfrak{g}[[t]]$ to the minimal nilpotent orbits $\mathcal{O}$ of $\mathfrak{g}$ such that γ can be conjugated (by LG) into $\mathcal{O} + t\mathfrak{g}[[t]]$. We let $\mathrm{RT}_{\min}(\gamma)$ be the collection of these orbits. Then for $[w] \in \mathrm{cl}(W)$, we let $\mathrm{RT}_{\min}([w])$ be $\mathrm{RT}_{\min}(\gamma)$, where γ is chosen generically to have splitting type $[w]$.

Remark 3.93. In many cases, Yun has shown that $\mathrm{RT}_{\min}(\gamma)$ is a unique nilpotent orbit, and he has computed this in classical types. This is conjectured to be true in all cases. In particular, for generic γ with splitting type $[w]$, one finds that $\mathrm{RT}_{\min}(\gamma)$ is a singleton, so $\mathrm{RT}_{\min} \colon \mathrm{cl}(W) \to \mathcal{N}$ can be seen to be well-defined.

Example 3.94. For $G = \mathrm{GL}_n$, the singleton $\mathrm{RT}_{\min}(\gamma)$ is described using Newton polygons.

Remark 3.95. These constructions were motivated by affine Springer fibers, which are the finite-dimensional representations $M_\gamma := \mathrm{H}^{\mathrm{top}}(\mathrm{Fl}_\gamma)^{A_\gamma}$. (Lusztig has shown that these admit action from $\widetilde{W}$.) Now, for $g \in \mathrm{Gr}_\gamma$, one has that $\mathrm{Ad}_{g^{-1}}\gamma \in \mathfrak{g}[[t]]$ with $\mathrm{Ad}_{g^{-1}}\gamma \equiv e \pmod{t}$ for some nilpotent e, and it turns out that the fiber over $g \in \mathrm{Gr}_\gamma$ in Fl_γ is exactly given by $\mathcal{B}_e$. In other words, we have a pullback square as follows.

$$\begin{tikzcd} \mathcal{B}_e \arrow[r, hook] \arrow[d] & \mathrm{Fl}_\gamma \arrow[d] \\ \{g\} \arrow[r, hook] & \mathrm{Gr}_\gamma \end{tikzcd}$$

Roughly speaking, the uniqueness claim of $\mathrm{RT}_{\min}(\gamma)$ shows that there is a unique Springer representation $E_{\mathcal{O}}$ in M_γ with longest "fake degree" $\dim \mathcal{B}_e$.

The above discussion assembles a map $\mathrm{cl}(W) \to \mathcal{N}$.

Notation 3.96. We define a multivalued map $\mathrm{cl}(W) \to \mathrm{Irr}_*(G)$. We start with $[w] \in \mathrm{cl}(W)$, and then we choose generic topologically nilpotent γ of splitting type $[w]$. Then among the 2-special irreducible representation $E \in \mathrm{Irr}_*(W)$, we let $\widetilde{\mathrm{RT}}_{\min}([w])$ consist of those E appearing in M_γ with largest "fake degree."

Conjecture 3.97 (Yun)**.** The above construction $\widetilde{\mathrm{RT}}_{\min}([w])$ is a singleton.

Remark 3.98. The above conjecture induces a map $\mathrm{cl}(W) \to \mathrm{Irr}_*(W)$, which is in bijection with strata.

Remark 3.99. The above conjecture implies Conjecture 3.89. The main point is that the diagram

$$\begin{tikzcd} \mathrm{colim}_P \mathcal{N}_{L_P} \arrow[r, "\widetilde{\mathrm{KL}}"] \arrow[dr, "(e \mapsto s\exp(e))"'] & \mathrm{cl}(W) \arrow[d, "\widetilde{\mathrm{RT}}_{\min}"] \\ & \mathrm{Str}(G) \end{tikzcd}$$

commutes.

Remark 3.100. For $\mathcal{O} \in \mathcal{N}$, Chua has shown that $\widetilde{\mathrm{RT}}_{\min}([w])$ is $E_{\mathcal{O}}$ if $[w] = \mathrm{KL}(\mathcal{O})$. In other words, Conjecture 3.97 is known if $[w]$ already comes from the Kazhdan–Lusztig map.

Remark 3.101. We close the lecture with a question: choose a maximal torus $T \subseteq G_{\mathbb{C}((t))}$, which we do not assume to be split. Then it is an interesting question to compute Gr^{LT°. In types A and C, these are points; in types B and D, these are given by quadratic spaces. However, this has not been computed for all exceptional types.

Minicourse 4
The Brauer–Manin Obstruction

These talks were given by Bjorn Poonen.

4.1 April 1

Here we go.

4.1.1 The Local-Global Principle

We are interested in finding rational points on varieties. Let's start with some notation for the present subsection. Fix a global field k, like $\mathbb{Q}$ or $\mathbb{F}_q(t)$, and for each place v, we let k_v be the corresponding complete local field. If v is finite, we let $\mathcal{O}_v$ be its ring of integers; we may as well define $\mathcal{O}_v = k_v$ when v is archimedean. As usual, we define the ring of adeles $\mathbb{A}_k$ to be the restricted product

$$\mathbb{A}_k := \prod_v (F_v, \mathcal{O}_v),$$

which we recall is locally compact.

Merely in terms of finding rational points, it turns out that one can reduce to the case where the variety is smooth, projective and geometrically irreducible.

Definition 4.1 (nice)**.** Fix a variety X over a field k. Then X is *nice* if and only if X is smooth, projective, and geometrically irreducible.

Remark 4.2. Given smoothness, one can weaken "geometrically irreducible" to "connected."

Remark 4.3. It turns out that nice varieties admit k_v-rational points for all but finitely many v, and this finite set can be made effective.

In general, there is no known algorithm to determine if a variety X over a global field k admits a rational point. However, there are some known cases.

Theorem 4.4 (Hasse–Minkowski)**.** Fix a global field k, and let X be a nice quadric hypersurface in $\mathbb{P}^n_k$. Then $X(k)$ is nonempty if and only if $X(k_v)$ is nonempty for all places v.

Remark 4.5. This was proven in the 1890s over $\mathbb{Q}$ by Minkowski and for general number fields by Hasse in 1924.

The moral is that one can test if $X(k_v)$ is nonempty for each place v fairly explicitly: there is no content over $\mathbb{C}$, one uses some inequalities over $\mathbb{R}$, and one can use Hensel's lemma for finite places.

Remark 4.6. It turns out that even in $\mathbb{P}^2$, being able to find the rational point is at least as hard as integer factorization.

The conclusion of Theorem 4.4 is nice enough to be worth a name.

Definition 4.7 (Hasse principle)**.** A variety X over a global field k satisfies the *Hasse principle* or the *local-global principle* if and only if $X(k)$ is nonempty is equivalent to $X(k_v)$ being nonempty for every v.

Remark 4.8. For projective X, one has that $X(\mathcal{O}_v) = X(k_v)$ by the valuative criterion for properness, so

$$X(\mathbb{A}_k) = \prod_v X(k_v).$$

Indeed, one just has to choose a model for X which is projective away from finitely many places. The moral is that the Hasse principle can be stated as having $X(k) \neq \varnothing$ equivalent to $X(\mathbb{A}_k) \neq \varnothing$.

The Hasse principle can fail.

Example 4.9. Hasse showed in 1931 that the variety of norms from $\mathbb{Q}(\sqrt{-3}, \sqrt{13})$ fails the Hasse principle.

Example 4.10. Lind and Reichardt in the 1940s produced some example of genus-1 curves where the Hasse principle fails, such as $3x^3+4y^2+5z^3 = 0$ (in $\mathbb{P}^2_{\mathbb{Q}}$) and a smooth projective model of $2y^2 = 1-17x^4$.

Remark 4.11. For elliptic curves E over a global field k, one has that E has a nonzero point over every local field (by some argument with the Implicit function theorem), but there is no reason for E to have a nonzero rational point.

Example 4.12. Swinnerton-Dyer constructed some cubic surfaces where the Hasse principle fails, such as $5x^3 + 9y^3 + 10z^3 + 12w^3 = 0$ in $\mathbb{P}^3_{\mathbb{Q}}$.

In 1970, Manin would explain what is going on in all these examples, which is the point of the present course.

4.1.2 The Brauer Group, from Algebra

Let's start with some algebra. Throughout, when k is a field, we let k^{sep} be its separable closure, and $\overline{k}$ is its algebraic closure. For convenience, we also set $G_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$.

Definition 4.13. An *algebra* over a field k is an associative unital ring with k-action.

One has the following structure theorem for so-called "central simple" algebras.

Proposition 4.14. Fix a finite-dimensional k-algebra A. Then the following are equivalent.

(i) One has that $A \otimes_k k^{\mathrm{sep}} \cong M_n(k^{\mathrm{sep}})$ for some $n \geq 1$.

(ii) One has that $A \otimes_k \overline{k} \cong M_n(\overline{k})$ for some $n \geq 1$.

(iii) The algebra A is central (i.e., $Z(A) = k$) and simple (i.e., A admits exactly two ideals).

(iv) One has that $A \cong M_r(D)$ for some $r \geq 1$ and some finite-dimensional central division k-algebra D.

Proof. Omitted. This is a standard result in non-commutative algebra. ■

Remark 4.15. In (iv), it turns out that r and D are uniquely determined by A.

Example 4.16. Over $\mathbb{R}$, the only central division algebras are $\mathbb{R}$ and $\mathbb{H}$, so the central simple algebras look like $M_r(\mathbb{R})$ and $M_r(\mathbb{H})$.

Example 4.17 (quaternion algebra)**.** For any field k of characteristic not 2, one can define a quaternion algebra as follows: choose $a, b \in k^\times$, and then

$$H_{a,b} := \frac{k\langle i, j\rangle}{(i^2 - a, j^2 - b, ij + ji)}.$$

By reducing words, one sees that $H_{a,b}$ is four-dimensional (spanned by $\{1, i, j, ij\}$), and one can check that this is central simple. Note that this does not have to be a division algebra: for example, if a is a square, then $H_{a,b}$ admits zero-divisors because $(i - \sqrt{a})(i + \sqrt{a}) = 0$.

Remark 4.18. One can check that all four-dimensional central simple algebras A are isomorphic to some quaternion algebra. As a sketch, one finds some $i \in A \setminus k$, which one can check must satisfy some quadratic equation. By completing the square, one may assume that $i^2 = a$ for some $a \in k^\times$. Then one can diagonalize the action of i in A (by conjugation), and choose j to be in the (-1)-eigenspace.

Our structure theory allows us to partition central simple algebras.

Definition 4.19 (Brauer group)**.** Fix a field k. Two central simple algebras A and B are *equivalent* if and only if there is a division algebra D and integers r and s such that $A \cong M_r(D)$ and $B \cong M_s(D)$. The *Brauer group* $\mathrm{Br}(k)$ is the collection of central simple k-algebras, considered up to equivalence.

Remark 4.20. We will not bother to check that equivalence is an actual equivalence relation. One can remove the division algebra D by instead asking for $M_s(A) \cong M_r(B)$. Indeed, the point is that

$$M_r(M_s(D)) = M_{rs}(D) = M_s(M_r(D)).$$

Remark 4.21. The collection $\mathrm{Br}(k)$ is a group under $\otimes$ (which turns out to send central simple algebras to central simple algebras). The unit is k, and the inverse of A is A^{op}; for example, one needs to check that $A \otimes A^{\mathrm{op}}$ is isomorphic to a matrix algebra over k (it's $\mathrm{End}_k(A)$).

Remark 4.22. A field extension $k \hookrightarrow \ell$ induces a group homomorphism $\mathrm{Br}(k) \to \mathrm{Br}(\ell)$ given by $- \otimes_k \ell$. Notably, $- \otimes_k \ell$ sends central simple algebras to central simple algebras by Proposition 4.14(b), and it preserves equivalence by the definition given in Remark 4.20.

4.1.3 The Brauer Group, from Cohomology

There is also a cohomological construction of the Brauer group.

Proposition 4.23. Fix a field k. Then $\mathrm{Br}(k) \cong \mathrm{H}^2(k; k^{\mathrm{sep}\times})$.

Proof. For each n, Proposition 4.14 central simple algebras of dimension n^2 are Galois twists of $M_n(k^{\mathrm{sep}})$, so they are classified by the pointed set $\mathrm{H}^1(k; \mathrm{Aut}\, M_n(k^{\mathrm{sep}}))$. By the Skolem–Noether theorem, all (algebra) automorphisms of $M_n(k^{\mathrm{sep}})$ are inner, so this automorphism group is $\mathrm{PGL}_n(k^{\mathrm{sep}})$. To compute this cohomology group, we note that there is a short exact sequence

$$1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$$

of group schemes, so taking the long exact sequence in cohomology, we receive an exact sequence

$$\mathrm{H}^1(k; \mathbb{G}_m) \to \mathrm{H}^1(k; \mathrm{GL}_n) \to \mathrm{H}^1(k; \mathrm{PGL}_n) \to \mathrm{H}^2(k; \mathbb{G}_m)$$

of pointed sets. The left two groups vanish by (a version of) Hilbert's theorem 90. (This is basically the statement that all GL_n-bundles on $\mathrm{Spec}\, k$ are trivial, which is true because all rank-n modules over k are free.) It turns out that this is enough to produce an injection $\mathrm{H}^1(k; \mathrm{PGL}_n) \to \mathrm{H}^2(k; \mathbb{G}_m)$.[1]

Now, taking the colimit over n, one shows that the collection of all central simple algebras to $\mathrm{H}^2(k; \mathbb{G}_m)$. It remains to check that this map descends to equivalence classes and then defines an isomorphism on $\mathrm{Br}(k)$. We will not run any of these checks. ■

Remark 4.24. In characteristic away from n, one sees that the map $\mathrm{SL}_n(k^{\mathrm{sep}}) \to \mathrm{PGL}_n(k^{\mathrm{sep}})$ is surjective with kernel μ_n. Thus, the image of $\mathrm{H}^1(k; \mathrm{PGL}_n)$ in $\mathrm{H}^2(k; \mathbb{G}_m)$ comes from $\mathrm{H}^2(k; \mu_n)$, which is n-torsion. The same argument works in characteristic dividing n if we use flat cohomology instead of étale (Galois) cohomology; namely, this allows us to pass from k^{sep} to $\overline{k}$ everywhere. It follows that $\mathrm{Br}(k)$ is a torsion group.

Now that we know that $\mathrm{Br}(k)$ is torsion, we can hope to compute its n-torsion. This is a variant of Kummer theory.

Proposition 4.25. Fix a field k of characteristic not dividing some positive integer n. Then $\mathrm{H}^1(k; \mu_n) = k^\times / k^{\times n}$ and $\mathrm{H}^2(k; \mu_n) = \mathrm{Br}(k)[n]$.

Proof. We use the short exact sequence

$$1 \to \mu_n \to \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \to 1$$

of étale group schemes on k. Taking Galois cohomology, we receive a long exact sequence

$$k^\times \xrightarrow{n} k^\times \to \mathrm{H}^1(k; \mu_n) \to \mathrm{H}^1(k; \mathbb{G}_m).$$

The last group vanishes by Hilbert's theorem 90, so $\mathrm{H}^1(k; \mu_n) \cong k^\times / k^{\times n}$ follows. Similarly, we have a long exact sequence

$$\mathrm{H}^2(k; \mathbb{G}_m) \to \mathrm{H}^2(k; \mu_n) \to \mathrm{H}^2(k; \mathbb{G}_m) \xrightarrow{n} \mathrm{H}^(k; \mathbb{G}_m),$$

and the left group vanishes by Hilbert's theorem 90, so $\mathrm{H}^2(k; \mu_n) \cong \mathrm{Br}(k)[n]$ follows by Proposition 4.23. ■

[1] Technically, injectivity requires the vanishing of $\mathrm{H}^1(k; \mathrm{GL}_n)$ for any twist of GL_n. (One sees this by computing the fiber directly.) These twists are given by central simple algebras.

Remark 4.26. If we use flat cohomology, then the same proof works even if $\operatorname{char} k \mid n$, so the conclusion is recovered.

Having a cohomological interpretation of the Brauer group allows us to do Galois cohomology things to it. Here is an example calculation.

Example 4.27. There is a canonical cup product

$$\mathrm{H}^1(k;\mu_2) \times \mathrm{H}^1(k;\mathbb{Z}/2\mathbb{Z}) \to \mathrm{H}^2(k;\mu_2).$$

Of course, there is a canonical isomorphism $\mathrm{H}^1(k;\mathbb{Z}/2\mathbb{Z}) \cong \mathrm{H}^1(k;\mu_2)$, so we see that this induces a cup product

$$k^\times/k^{\times 2} \times k^\times/k^{\times 2} \to \operatorname{Br}(k)[2]$$

by Proposition 4.25. One can check that this takes a pair (a,b) to the quaternion algebra $H_{a,b}$, which we can do with some explicit cocycle calculations to $\mathrm{H}^2(k;\mu_2)$.

Remark 4.28. For fixed $a \in k^\times$, it turns out that the natural map $k^\times \to \operatorname{Br}(k)$ given by $b \mapsto [H_{a,b}]$ has kernel given by $\mathrm{N}_{k(\sqrt{a})/k}\, k(\sqrt{a})^\times$. Indeed, the idea is to use Tate Galois cohomology: there is an explicit isomorphism

$$\mathrm{H}^0_T(\operatorname{Gal}(k(\sqrt{a})/k);\mathbb{G}_m) \to \mathrm{H}^2(\operatorname{Gal}(k(\sqrt{a})/k);\mathbb{G}_m)$$

given by some cup product, computed in Example 4.27. The source reduces to $k^\times/\mathrm{N}_{k(\sqrt{a})/k}\, k(\sqrt{a})^\times$ by expanding out the Tate cohomology group, and the target can be seen to be the kernel of the natural map $\operatorname{Br}(k) \to \operatorname{Br}(k(\sqrt{a}))$ by the Inflation–restriction exact sequence.

4.1.4 Examples of Brauer Groups

It is as good a time as any to give some examples of Brauer groups.

Example 4.29. If k is separably closed, then Proposition 4.14 shows that $\operatorname{Br}(k)$ is trivial. One can also see this directly from Proposition 4.23.

Example 4.30. By Wedderburn's theorem, all finite division algebras are fields, so there is only one finite central division algebras over $\mathbb{F}_q$ (namely, $\mathbb{F}_q$). Thus, $\operatorname{Br}\mathbb{F}_q = 0$.

Example 4.31. By Tsen's theorem, one has that $\operatorname{Br}\mathbb{C}(t) = 0$. In fact, the same is true for any field of transcendence degree 1 over an algebraically closed field.

Example 4.32. A direct calculation in Galois cohomology, one has that $\operatorname{Br}\mathbb{R} = \frac{1}{2}\mathbb{Z}/\mathbb{Z}$, where $\mathbb{R}$ goes to 0, and $\mathbb{H}$ goes to $\frac{1}{2}$. This is equivalent to Example 4.16.

Example 4.33. For any nonarchimedean local field k_v, it turns out that $\operatorname{Br} k_v \cong \mathbb{Q}/\mathbb{Z}$. There is a way to write down these elements explicitly. This assertion is roughly as hard as local class field theory.

Example 4.34. For any global field k, functoriality gives a map $\operatorname{Br}(k) \to \prod_v \operatorname{Br} k_v$. It turns out that the image lands in $\bigoplus_v \operatorname{Br} k_v$, which one sees because a central simple algebra over k uses finitely many denominators, so it can be trivialized at all but finitely many places v. By global class field theory, it turns out that this map is injective, and it fits into a short exact sequence

$$0 \to \operatorname{Br} k \to \bigoplus_v \operatorname{Br} k_v \to \mathbb{Q}/\mathbb{Z} \to 0,$$

where the last map is given by summing. Similarly, this is roughly as hard as global class field theory.

Remark 4.35. It turns out that the short exact sequence

$$0 \to \operatorname{Br} \mathbb{Q} \to \bigoplus_v \operatorname{Br} \mathbb{Q}_v \to \mathbb{Q}/\mathbb{Z} \to 0$$

encodes quadratic reciprocity, if we apply to the quaternion algebra $H_{p,q}$ where p and q are odd primes.

4.2 April 6

Here we go.

4.2.1 Étale Cohomology

Étale cohomology arises by changing the notion of "open subset" in algebraic geometry from (Zariski) open embeddings to étale morphisms. Thus, we should start by defining what it means for a morphism to be étale. Roughly speaking, an étale morphism should be a local diffeomorphism.

Definition 4.36 (étale)**.** Fix a morphism $f\colon Y \to X$ of schemes, and choose a point $y \in Y$.

(a) If f takes the form $\operatorname{Spec} A[t_1, \ldots, t_n]/(g_1, \ldots, g_n) \to \operatorname{Spec} A$, then f is *étale* at y if and only if the Jacobian matrix $[\partial g_i/\partial t_j]_{ij}$ is an invertible matrix with entries in $\mathcal{O}_{Y,y}/\mathfrak{m}_y$.

(b) In general, we say that f is *étale* at y if and only if there are open neighborhoods $Y' \subseteq Y$ of y and $X' \subseteq X$ of $f(y)$ such that the morphism $Y' \to X'$ is isomorphic to the situation in (a). Explicitly, X' should be some affine scheme $\operatorname{Spec} A$, and Y' should take the form $\operatorname{Spec} A[t_1, \ldots, t_n]/(g_1, \ldots, g_n)$, and the Jacobian at y should be invertible.

We say that f is *étale* if and only if it is étale at every point $y \in Y$.

Remark 4.37. The implicit function theorem tells us that (a) is equivalent to f being

Remark 4.38. It is not totally obvious that (a) and (b) are equivalent. Namely, it is not obvious that one cannot pass to some suitable open subsets of $\operatorname{Spec} A$ and $\operatorname{Spec} A[t_1, \ldots, t_n]/(g_1, \ldots, g_n)$ to force the Jacobian to be invertible when it wasn't before. This doesn't happen, and the best argument gives a more intrinsic definition of being étale (e.g., being finitely presented, flat, and unramified).

Now that we know what our new open subsets are, we would like to build a sheaf theory. This requires a notion of an open covering, for which our definition is unsurprising.

Definition 4.39 (étale covering)**.** An *étale covering* of a scheme X is a collection of étale morphisms $\{f_i\}$ with target X for which $\bigcup_i \operatorname{im} f_i = X$ on the underlying topological spaces.

Definition 4.40 (small étale site)**.** Fix a scheme X. Then the *small étale site* of X is the category $\mathrm{\acute{E}t}(X)$ whose objects are étale morphisms $U \to X$ and morphisms are morphisms over X. We equip this category with coverings given by the collection of étale coverings: namely, a covering of an étale map $U \to X$ is just an étale covering $\{V_i \to U\}$ of U by étale morphisms $V_i \to X$.

Remark 4.41. Part of the coherence here is that if we have two morphisms $X \to Y \to Z$ where $X \to Z$ and $Y \to Z$ are étale, then $X \to Y$ is étale.

With a notion of a covering, we can produce a sheaf theory.

Definition 4.42 (étale sheaf)**.** Fix a scheme X. Then an *étale presheaf* $\mathfrak{F}$ on X is a presheaf on $\mathrm{\acute{E}t}(X)$. An *étale sheaf* $\mathcal{F}$ on X is a presheaf satisfying the sheaf condition; explicitly, for any open covering $\{V_i\}$ of some étale open $U \to X$, there is an equalizer diagram

$$\mathcal{F}(U) \to \prod_i \mathcal{F}(V_i) \rightrightarrows \prod_{i,j} \mathcal{F}(V_i \times_X V_j).$$

We let $\mathrm{Sh}(X_{\text{ét}})$ denote the category of abelian sheaves on $X_{\text{ét}}$.

Example 4.43. There is an étale sheaf $\mathbb{G}_m$ on X which sends $V \to X$ to $\mathcal{O}(V)^\times$. We will not check that this satisfies the sheaf condition.

Remark 4.44. It turns out that $\mathrm{Sh}(X_{\text{ét}})$ is an abelian category. The proof of this is similar to the Zariski case, and it proceeds by defining a notion of sheafification.

With a sheaf theory, we also have a cohomology theory.

Definition 4.45 (étale cohomology)**.** Fix a scheme X. Then the functor $\Gamma\colon \mathrm{Sh}(X_{\text{ét}}) \to \mathrm{Ab}$ given by taking global sections is left-exact, so we define its right-derived functors by $\mathrm{H}^i_{\text{ét}}(X; -)$. These are the *étale cohomology* functors.

We close our discussion of this example.

Example 4.46. Let X be a field $\operatorname{Spec} k$. Then an étale morphism $Y \to X$ must be a zero-dimensional scheme, so the map $Y \to X$ can be seen to force Y to be affine. In fact, using the local criterion, we conclude that Y must be a disjoint union of finite separable extensions of k. Now, for an étale sheaf $\mathcal{F}$ on X, we define

$$\mathcal{F}(k^{\text{sep}}) := \operatorname*{colim}_{L/k} \mathcal{F}(L),$$

where the colimit is taken over finite separable extensions L of k. For each Galois extension L of k, we see that $\mathcal{F}(L)$ admits an action by $\mathrm{Gal}(L/k)$, so $\mathcal{F}(k^{\text{sep}})$ is a continuous module for $G_k := \mathrm{Gal}(k^{\text{sep}}/k)$; thus, we have defined a functor $\mathrm{Sh}(X_{\text{ét}}) \to \mathrm{Mod}_{\mathbb{Z}}(G_k)$. There is also an inverse functor which sends $M \in \mathrm{Mod}_{\mathbb{Z}}(G_k)$ to the sheaf $L \mapsto M^{\mathrm{Gal}(k^{\text{sep}}/L)}$, and one can check that these functors are equivalences. By comparing the notions of global sections on both sides, we see that

$$\mathrm{H}^i_{\text{ét}}(X; \mathcal{F}) = \mathrm{H}^i(G_k; \mathcal{F}(k^{\text{sep}})).$$

4.2.2 Brauer Groups of Schemes

Throughout, we fix a scheme X.[2] We would like to retell our story of the Brauer group of a field to work on general schemes. Accordingly, our first goal will be to redefine the notion of a central simple algebra.

Definition 4.47 (Azumaya algebra)**.** Fix a scheme X. Then an *Azumaya algebra* $\mathcal{A}$ on X is an $\mathcal{O}_X$-algebra $\mathcal{A}$ satisfying the following.

(a) The sheaf $\mathcal{A}$ is coherent over $\mathcal{O}_X$.

(b) The algebra $\mathcal{A}$ is étale-locally isomorphic to a matrix algebra. Explicitly, there is an étale cover $\{U_i\}$ of X and a tuple of ranks $\{r_i\}$ such that there are isomorphisms $\mathcal{A}|_{U_i} \cong M_{r_i}(\mathcal{O}_{U_i})$ for each i.

Remark 4.48. Here, (a) is analogous to requiring that a central simple algebra be finite-dimensional. The second condition is analogous to Proposition 4.14(ii); we remark that (ii) implies that $\mathcal{A}$ is locally free.

Remark 4.49. If X is connected, then it turns out that the rank of $\mathcal{A}$ is constant because it is locally free!

Remark 4.50. The notion of an Azumaya algebra is older than the notion of the étale topology. Instead of the above, one can replace (b) that $\mathcal{A}$ is Zariski-locally free, nonzero at every point, and $\mathcal{A} \otimes \mathcal{A}^{\mathrm{op}} \cong \mathcal{E}nd(\mathcal{A})$. This last condition comes from some non-commutative algebra.

Example 4.51 (quaternion algebra)**.** Fix a scheme X such that 2 is a unit in $\mathcal{O}_X$, and choose $f, g \in \mathcal{O}_X(X)^\times$. Then

$$\mathcal{H}_{f,g} := \frac{\mathcal{O}_X\langle i, j\rangle}{(i^2 - f, j^2 - g, ij + ji)}$$

is an Azumaya algebra. This is analogous to Example 4.17.

We can now use these algebras to define a Brauer group, using a similar notion of equivalence.

Definition 4.52 (Brauer group)**.** Fix a scheme X. Then two Azumaya algebras $\mathcal{A}$ and $\mathcal{A}'$ are *equivalent* if and only if there are locally free coherent $\mathcal{O}_X$-modules $\mathcal{E}$ and $\mathcal{E}'$ of positive rank everywhere on X such that

$$\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{E}nd_{\mathcal{O}_X}(\mathcal{E}) \cong \mathcal{A}' \otimes_{\mathcal{O}_X} \mathcal{E}nd_{\mathcal{O}_X}(\mathcal{E}').$$

Then we define the *Azumaya Brauer group* $\mathrm{Br}_{\mathrm{Az}}(X)$ to be the collection of Azumaya algebras up to equivalence.

Remark 4.53. Similar to Remark 4.21, the Brauer group is a group with operation given by the tensor product, and inverses given by the opposite Azumaya algebra.

Alternatively, one could have tried to generalize the Brauer group by taking étale cohomology.

Definition 4.54 (Brauer group)**.** Fix a scheme X. Then the *cohomological Brauer group* to be $\mathrm{Br}\, X := \mathrm{H}^2_{\text{ét}}(X; \mathbb{G}_m)$.

[2] One could even work with algebraic spaces or stacks, but we will not bother.

Remark 4.55. A direct generalization of the argument from Proposition 4.23 provides a map $\operatorname{Br}_{\mathrm{Az}} X \to \operatorname{Br} X$.

One may hope that the natural map $\operatorname{Br}_{\mathrm{Az}} X \to \operatorname{Br} X$ is an isomorphism, and this is the case in favorable circumstances.

Theorem 4.56 (Gabber, de Jong)**.** Fix a scheme X. If X is quasi-projective over an affine scheme, then the natural map $\operatorname{Br}_{\mathrm{Az}} X \to \operatorname{Br} X$ is injective, and it surjects onto $(\operatorname{Br} X)_{\mathrm{tors}}$.

Remark 4.57. Gabber never published his proof of this result, so it can be difficult to find.

Theorem 4.56 tells us that we would like to know when $\operatorname{Br} X$ is torsion.

Proposition 4.58. Fix a regular integral Noetherian scheme X. Then the map $\operatorname{Br} X \to \operatorname{Br} K(X)$ is injective.

Idea. One can think of elements of the Brauer group as sections of the Brauer–Severi variety. Under regularity assumptions, one can always do this splitting. ■

Remark 4.59. It follows from Proposition 4.58 that $\operatorname{Br} X$ is torsion when X is regular, integral, and Noetherian.

Remark 4.60. There are examples of singular varieties where $\operatorname{Br} X$ fails to be torsion.

4.2.3 Some Computational Tools

Here are some applications of Theorem 4.56.

Example 4.61. Here is a silly example where $\operatorname{Br} X$ has torsion: take $\bigsqcup_{i \in \mathbb{N}} \operatorname{Spec} \mathbb{Q}_2$. Then the cohomological Brauer group has an element $(1/2, 1/3, 1/4, \ldots) \in \prod_{i \in \mathbb{N}} \mathbb{Q}/\mathbb{Z}$, which is not torsion.

Example 4.62. If X is a nice curve over an algebraically closed field k, then $\operatorname{Br} X$ is trivial. Indeed, $\operatorname{Br} K(X)$ is trivial by Tsen's theorem, so this follows from Proposition 4.58.

Remark 4.63. In fact, if X is a nice curve over a separably closed field k

We can even go back and do some calculations for fields.

Proposition 4.64. Fix a complete discrete valuation ring R with residue field k. Then the natural map $\operatorname{Br} R \to \operatorname{Br} k$ is an isomorphism.

Proof. We may pass to Azumaya algebras by Theorem 4.56. This is an application of Hensel's lemma. For injectivity, the main point is to check that if $\mathcal{A}_k$ is equivalent to a matrix algebra, then $\mathcal{A}$ is equivalent to a matrix algebra, which one does by finding idempotents in k and using Hensel's lemma to lift them back to $\mathcal{A}$. For surjectivity, one uses some deformation theory. The idea is to successively lift the Azumaya algebra from k to $R/\mathfrak{p}$ and then $R/\mathfrak{p}^2$ and so on, where here $\mathfrak{p}$ is the maximal ideal of R. All these lifts turn out to be unique, so the lifting can be taken all the way to R. ■

Corollary 4.65. Fix a global field k. Then the image of the map

$$\operatorname{Br} k \to \prod_v \operatorname{Br} k_v$$

lands in $\bigoplus_v \operatorname{Br} k_v$.

Proof. For a finite set of places S of k including the infinite ones, let $\mathcal{O}_{k,S}$ be the ring of S-integers, which is

$$\mathcal{O}_{k,S} = \{x \in k : v(x) \geq 0 \text{ for finite } v \in S\}.$$

Now, there is a map $\mathcal{O}_{k,S} \to \prod_{v \in S} k_v \times \prod_{v \notin S} \mathcal{O}_v$, so $\operatorname{Br} \mathcal{O}_{k,S}$ embeds into $\prod_v \operatorname{Br} k_v$ because $\operatorname{Br} \mathcal{O}_v = 0$ by Proposition 4.64. By taking the direct limit, we see that

$$\operatorname*{colim}_S \operatorname{Br} \mathcal{O}_{k,S} \to \bigoplus_v \operatorname{Br} k_v.$$

To complete the proof, we should show that this map factors through $\operatorname{Br} k$, which holds by passing to Azumaya algebras (via Theorem 4.56): indeed, any Azumaya algebra on k is defined with finitely many elements (e.g., on a basis with some explicit multiplication law), which have finitely many denominators, so it can be lifted to $\operatorname{Br} \mathcal{O}_{k,S}$ for some potentially large S. It follows that the natural map $\operatorname{colim}_S \operatorname{Br} \mathcal{O}_{k,S} \to \operatorname{Br} k$ is surjective, so we are done! ■

In the sequel, it will be helpful to have a spectral sequence to compute our Brauer groups.

Theorem 4.66 (Hochschild–Serre spectral sequence)**.** Fix a nice k-variety X, and set $G := \operatorname{Gal}(k^{\operatorname{sep}}/k)$. Then there is a spectral sequence E such that

$$E_2^{pq} = \operatorname{H}^p\left(G; \operatorname{H}^q_{\text{ét}}(X_{k^{\operatorname{sep}}}; \mathbb{G}_m)\right) \Rightarrow \operatorname{H}^{p+q}_{\text{ét}}(X; \mathbb{G}_m).$$

Proof. This follows from Grothendieck's composition-of-functors spectral sequence ■

Corollary 4.67. Fix a nice k-variety X, and set $G := \operatorname{Gal}(k^{\operatorname{sep}}/k)$. Let $\operatorname{Br}_1 X$ be the kernel of the map $\operatorname{Br} X \to \operatorname{Br} X_{k^{\operatorname{sep}}}$. Then there is an exact sequence

$$0 \to \operatorname{Pic} X \to (\operatorname{Pic} X_{k^{\operatorname{sep}}})^G \to \operatorname{Br} k \to \operatorname{Br}_1 X \to \operatorname{H}^1(G; \operatorname{Pic} X_{k^{\operatorname{sep}}}) \to \operatorname{H}^3(G; k^{\operatorname{sep}\times}).$$

Proof. This follows from the exact sequence associated to the spectral sequence, in addition to the following calculations.

- One has $\operatorname{H}^0_{\text{ét}}(X_{k^{\operatorname{sep}}}; \mathbb{G}_m) = k^{\operatorname{sep}\times}$ because the global sections of a projective variety are the constant functions.
- One has $\operatorname{H}^1_{\text{ét}}(X_{k^{\operatorname{sep}}}; \mathbb{G}_m) = \operatorname{H}^1_{\text{Zar}}(X_{k^{\operatorname{sep}}}; \mathbb{G}_m) = \operatorname{Pic} X_{k^{\operatorname{sep}}}$. Indeed, these groups classify étale or Zariski $\mathbb{G}_m$-bundles, which one can show are all realized as line bundles, and $\operatorname{Pic} X_{k^{\operatorname{sep}}}$ is the group of line bundles up to isomorphism.
- By definition, $\operatorname{H}^2_{\text{ét}}(X_{k^{\operatorname{sep}}}; \mathbb{G}_m) = \operatorname{Br} X_{k^{\operatorname{sep}}}$ and similarly for X. ■

Remark 4.68. The cohomological proofs of class field theory show that $\operatorname{H}^3(G; k^{\operatorname{sep}\times}) = 0$ when k is local or global.

4.3 April 8

Here we go.

4.3.1 Purity

Another property of Brauer groups is purity. For motivation, note that any normal integral Noetherian scheme X admits an exact sequence

$$1 \to \mathcal{O}(X)^\times \to K(X)^\times \to \bigoplus_{x \in X^{(1)}} \mathbb{Z},$$

where $X^{(1)}$ is the set of codimension-1 points. Namely, we are asserting that a meromorphic function $f \in K(X)^\times$ is a regular function if and only if it has vanishing valuation at all codimension-1 points. The analog of this for Brauer groups is as follows.

Theorem 4.69. Fix a regular integral Noetherian scheme X. Then there is an exact sequence

$$0 \to \operatorname{Br} X \to \operatorname{Br} K(X) \to \bigoplus_{x \in X^{(1)}} \mathrm{H}^1(k(x); \mathbb{Q}/\mathbb{Z}),$$

where one implicitly removes all the p-primary parts if any of the $k(x)$ are imperfect of characteristic p.

Remark 4.70. We will not prove this or even define the maps on the right, but we will say that this recovers the left-exactness of Example 4.34 when k is a function field.

Corollary 4.71. Fix a regular integral Noetherian variety X over a field of characteristic zero. Given a Zariski open over $\{U_i\}$ of a regular integral Noetherian scheme X,

$$\operatorname{Br} X = \bigcap_i \operatorname{Br} U_i,$$

where the intersection takes place in $\operatorname{Br} K(X)$.

Proof. To begin, use Theorem 4.69 to present $\operatorname{Br} X$ as elements of $\operatorname{Br} K(X)$ which vanish in the various groups $\mathrm{H}^1(k(x); \mathbb{Q}/\mathbb{Z})$. Then the right-hand side is also presented as elements of $\operatorname{Br} X$ which vanish in the various groups $\mathrm{H}^1(k(x); \mathbb{Q}/\mathbb{Z})$ where $x \in U_i$, so we recover the result by looping over all U_is. ■

One can remove the characteristic zero assumption, but it is harder. The following was conjectured by Grothendieck in 1968 but only proven in 2019 with perfectoid methods.

Theorem 4.72 (Česnavičius)**.** Fix a regular integral Noetherian scheme X. For a closed subscheme $Z \subseteq X$ of codimension at least 2, set $U := X \setminus Z$, and the natural map $\operatorname{Br} X \to \operatorname{Br} U$ is an isomorphism.

Remark 4.73. This follows from Theorem 4.69, except for the p-primary parts.

Corollary 4.74. Fix nice birational varieties X and Y. Then $\operatorname{Br} X \cong \operatorname{Br} Y$.

Proof. Choose our birational map $f\colon X \to Y$. Then f induces an isomorphism $K(X) \to K(Y)$, so

$$\operatorname{Br} K(X) = \operatorname{Br} K(Y).$$

Now, f is defined on some open subset $U \subseteq X$, and we may even assume that the codimension of $X \setminus U$ is at least 2. Thus, there is a well-defined map $\operatorname{Br} Y \to \operatorname{Br} U$, which is then isomorphic to $\operatorname{Br} X$ (by Theorem 4.72!), which then embeds back into $\operatorname{Br} K(X) = \operatorname{Br} K(Y)$. Thus, $\operatorname{Br} Y \subseteq \operatorname{Br} X$ as subgroups of the Brauer group of the function field. The opposite inclusion holds by symmetry, so we are done. ■

4.3.2 Examples of Brauer Groups of Schemes

Let's do an example.

Example 4.75. For a field k, we claim that $\operatorname{Br} \mathbb{P}^1_k = \operatorname{Br} k$. Indeed, $\operatorname{Pic} \mathbb{P}^1_k = \mathbb{P}^1_{k^{\mathrm{sep}}} = \mathbb{Z}$, so Corollary 4.67 gives us the exact sequence

$$0 \to \mathbb{Z} \to \mathbb{Z}^G \to \operatorname{Br} k \to \operatorname{Br}_1 \mathbb{P}^1_k \to \mathrm{H}^1(G; \mathbb{Z}).$$

Now, because G acts trivially on $\mathbb{Z}$, we see $\mathrm{H}^1(G;\mathbb{Z}) = \operatorname{Hom}_{\mathrm{cts}}(G, \mathbb{Z})$; because G is compact, any map $G \to \mathbb{Z}$ has compact image, but the only compact subgroup of $\mathbb{Z}$ is trivial, so we conclude $\mathrm{H}^1(G;\mathbb{Z}) = 0$. Thus, we see that we have an isomorphism $\operatorname{Br} k \to \operatorname{Br}_1 \mathbb{P}^1_k$. However, $\operatorname{Br}_1 \mathbb{P}^1_k$ is the kernel of the map $\operatorname{Br} \mathbb{P}^1_k \to \operatorname{Br} \mathbb{P}^1_{k^{\mathrm{sep}}}$, and the latter vanishes by a classical calculation in étale cohomology. For example, if k is perfect, this follows from Tsen's theorem because $\operatorname{Br} K\left(\mathbb{P}^1_{\bar{k}}\right)$ already vanishes.

Remark 4.76. This argument does not work as well for other curves or other varieties. The problem is that Pic is no longer so simple.

We can upgrade this as follows.

Proposition 4.77. Fix a regular integral Noetherian scheme B. Then the natural map $\operatorname{Br} B \to \operatorname{Br} \mathbb{P}^1_B$ is an isomorphism.

Proof. There is a pullback square

$$\begin{tikzcd} \mathbb{P}^1_{K(B)} \arrow[r] \arrow[d] & \mathbb{P}^1_B \arrow[d] \\ \operatorname{Spec} K(B) \arrow[r] & B \end{tikzcd}$$

which on Brauer groups produces the following commutative diagram.

$$\begin{tikzcd} \operatorname{Br} B \arrow[r] \arrow[d] & \operatorname{Br} K(B) \arrow[d] & \\ \operatorname{Br} \mathbb{P}^1_B \arrow[r] & \operatorname{Br} \mathbb{P}^1_{K(B)} \arrow[r] & \operatorname{Br} K\left(\mathbb{P}^1_{K(B)}\right) \end{tikzcd}$$

Here, the top arrow and the bottom composite are both embeddings, so we conclude that all horizontal maps are embeddings. Now, the right vertical map is an isomorphism by Example 4.75. The commutativity of the diagram then implies that $\operatorname{Br} B$ must be embedded in $\operatorname{Br} \mathbb{P}^1_B$.

Conversely, note that the projection $\mathbb{P}^1_B \to B$ admits easy sections s, which then must give the inverse map $\operatorname{Br} \mathbb{P}^1_{K(B)} \to \operatorname{Br} K(B)$. Thus, the inverse isomorphism of the right vertical arrow also fits into a commutative diagram

$$\begin{tikzcd} \operatorname{Br} B \arrow[r] & \operatorname{Br} K(B) & \\ \operatorname{Br} \mathbb{P}^1_B \arrow[r] \arrow[u, "s"] & \operatorname{Br} \mathbb{P}^1_{K(B)} \arrow[r] \arrow[u, "s"] & \operatorname{Br} K\left(\mathbb{P}^1_{K(B)}\right) \end{tikzcd}$$

so it follows that $\operatorname{Br} \mathbb{P}^1_B$ embeds into $\mathbb{P}^1_{K(B)}$, as required! ■

Example 4.78. It follows from Proposition 4.77 that $\operatorname{Br}\left(\mathbb{P}^1_k \times X\right) \cong \operatorname{Br} X$ for a nice k-variety X. For example, $\mathbb{P}^n_k$ is birational to iterated products of $\mathbb{P}^1_k$, so Corollary 4.74 allows us to conclude that $\operatorname{Br} \mathbb{P}^n_k = \operatorname{Br} k$.

Here is another example.

Definition 4.79 (quadric)**.** A *quadric hypersurface* X over a field k is a degree-2 hypersurface of some projective space $\mathbb{P}^n_k$.

Proposition 4.80. Fix a smooth quadric hypersurface X over a field k. Then the natural map $\operatorname{Br} k \to \operatorname{Br} X$ is surjective.

Proof. The main point is to use the exact sequence

$$\operatorname{Br} k \to \operatorname{Br}_1 X \to \mathrm{H}^1(G; \operatorname{Pic} X_{k^{\mathrm{sep}}})$$

from Corollary 4.67.

- We claim $\operatorname{Br}_1 X = \operatorname{Br} X$. One can show (by putting the quadratic equation in a standard form) that $X_{k^{\mathrm{sep}}}$ is birational to $\mathbb{P}^{n-1}_{k^{\mathrm{sep}}}$. Thus, by Corollary 4.74 and Example 4.78, we see that $\operatorname{Br} X_{k^{\mathrm{sep}}} = 0$, so $\operatorname{Br}_1 X = \operatorname{Br} X$.
- We claim that $\mathrm{H}^1(G; \operatorname{Pic} X_{k^{\mathrm{sep}}})$. One can compute (it's [Har77, Exercise II.6.5]) that

$$\operatorname{Pic} X_{k^{\mathrm{sep}}} = \begin{cases} \mathbb{Z}^2 & \text{if } \dim X = 2, \\ \mathbb{Z} & \text{if } \dim X = 2. \end{cases}$$

  In the case where the group is $\mathbb{Z}$, the Galois group should act trivially: the Galois group needs to preserve the fact that a line bundle is ample, so it needs to preserve the positives, so it needs to act by identity. In this case, one concludes that $\mathrm{H}^1(G; \mathbb{Z}) = 0$ as in Example 4.75. Even if the Picard group is $\mathbb{Z}^2$, one can compute the Galois cohomology must vanish, so we are done; indeed, in this case, $X_{k^{\mathrm{sep}}}$ is isomorphic to $\mathbb{P}^1_{k^{\mathrm{sep}}} \times \mathbb{P}^1_{k^{\mathrm{sep}}}$, so the Galois action splits into each $\mathbb{Z}$, where it acts trivially on both. ■

We can also do a relative version.

Definition 4.81 (quadric bundle)**.** A *quadric bundle* is a morphism $\pi\colon X \to B$ is a flat morphism of regular integral k-varieties satisfying the following.

- Quadric: the generic fiber $X_{K(B)}$ is a smooth quadric.
- Every fiber of π has an irreducible component of multiplicity 1 that is geometrically irreducible.

Remark 4.82. The second point basically prevents a fiber from looking like $X^2 - 2Y^2$ for a quadric bundle $\pi\colon X \to \operatorname{Spec} \mathbb{Z}$. The moral is that

Proposition 4.83. Fix a quadric bundle $\pi\colon X \to B$ over a field k of characteristic zero. Then the natural map $\operatorname{Br} B \to \operatorname{Br} X$ is surjective.

Proof. We use Theorem 4.69, which gives us a morphism

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Br} B \arrow[r] \arrow[d] & \operatorname{Br} K(B) \arrow[r, "\mathrm{res}"] \arrow[d] & \bigoplus_{b \in B^{(1)}} \mathrm{H}^1(k(b); \mathbb{Q}/\mathbb{Z}) \\
0 \arrow[r] & \operatorname{Br} X \arrow[r] & \operatorname{Br} K(X) \arrow[r, "\mathrm{res}"] & \bigoplus_{x \in X^{(1)}} \mathrm{H}^1(k(x); \mathbb{Q}/\mathbb{Z})
\end{tikzcd}$$

of exact sequences. In fact, we can compare the analogous exact sequence for $\operatorname{Br} X_{K(B)}$, which we see is then cut out by those codimension-1 points $x \in X^{(1)}$ which surject onto $\operatorname{Spec} B$. Thus, by looping over the remaining points in $X^{(1)}$, we get a morphism

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Br} B \arrow[r] \arrow[d] & \operatorname{Br} K(B) \arrow[r, "\mathrm{res}"] \arrow[d] & \bigoplus_{b \in B^{(1)}} \mathrm{H}^1(k(b); \mathbb{Q}/\mathbb{Z}) \arrow[d] \\
0 \arrow[r] & \operatorname{Br} X \arrow[r] & \operatorname{Br} K(X) \arrow[r, "\mathrm{res}"] & \bigoplus_{\substack{b \in B^{(1)} \\ \pi(x) = b}} \mathrm{H}^1(k(x); \mathbb{Q}/\mathbb{Z})
\end{tikzcd}$$

of exact sequences. By some diagram chasing (with the Four lemma), it is enough to show that the right vertical morphism is injective because we already know that the middle vertical morphism is surjective by Proposition 4.80.

To this end, for a given $b \in B^{(1)}$ we choose a generic point x of the hypersurface granted from the second assumption on our quadric bundle. Then $k(x)$ is an extension of $k(b)$ of some transcendence degree, so there is a surjection of Galois groups $\operatorname{Gal}(k(x)^{\mathrm{sep}}/k(x)) \to \operatorname{Gal}(k(b)^{\mathrm{sep}}/k(b))$, which then induces an embedding on Galois cohomology. The result follows. ■

4.3.3 The Brauer–Manin Obstruction

It will be helpful to view Br as a function we can evaluate.

Definition 4.84. Fix a morphism $\pi\colon X \to B$. For $\alpha \in \operatorname{Br} X$, we define the *evaluation map* $\mathrm{ev}_\alpha\colon X(B) \to \operatorname{Br} B$ by

$$\mathrm{ev}_\alpha(x) := x^*\alpha,$$

where we view $x \in X(k)$ as a morphism $\operatorname{Spec} k \to \operatorname{Spec} X$.

Remark 4.85. In most applications, we will take $B = \operatorname{Spec} k$ so that ev_α is basically "evaluating" α on the k-points of X.

Proposition 4.86. Fix a nice k-variety X over a global field k, and choose some $\alpha \in \operatorname{Br} X$. Then the evaluation map $\mathrm{ev}_\alpha\colon X(k_v) \to \operatorname{Br} k_v$ vanishes for all but finitely many v.

Proof. Choose a set S of finite places large enough so that X spreads out to some nice k-variety $\mathfrak{X}$. By expanding S, we may assume that α extends to some $\widetilde{\alpha} \in \operatorname{Br} \mathfrak{X}$: indeed, we may simply add to S the denominators needed to be able to define the Azumaya algebra corresponding to α.

We now claim that ev_α vanishes for $v \notin S$. Indeed, note that the diagram

$$\begin{tikzcd}
\mathfrak{X}(\mathcal{O}_v) \arrow[r, "\mathrm{ev}_{\widetilde{\alpha}}"] \arrow[d] & \operatorname{Br} \mathcal{O}_v \arrow[d] \\
X(k_v) \arrow[r, "\mathrm{ev}_\alpha"] & \operatorname{Br} k_v
\end{tikzcd}$$

commutes, so it is enough to recall that $\mathfrak{X}(\mathcal{O}_v) = X(k_v)$ by properness, and $\operatorname{Br} \mathcal{O}_v = 0$ by Proposition 4.64. ■

We are now ready to describe the Brauer–Manin obstruction. We continue with a nice k-variety X and $\alpha \in \operatorname{Br} X$. By the valuative criterion for properness, $X(\mathbb{A}_k) = \prod_v X(k_v)$. Then Proposition 4.86 tells us that $\mathrm{ev}_\alpha \colon X(\mathbb{A}_k) \to \prod_v \operatorname{Br} k_v$ actually lands in $\bigoplus_v \operatorname{Br} k_v$, so the diagram

$$\begin{tikzcd}
 & X(k) \arrow[r] \arrow[d, "\mathrm{ev}_\alpha"'] & X(\mathbb{A}_k) \arrow[d, "\mathrm{ev}_\alpha"] & & \\
0 \arrow[r] & \operatorname{Br} k \arrow[r] & \bigoplus_v \operatorname{Br} k_v \arrow[r] & \mathbb{Q}/\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

commutes, where the bottom row comes from global class field theory. Notably, a rational point $x \in X(k)$ not only gives to a point in $X(\mathbb{A}_k)$, but we see that class must also vanish in $\mathbb{Q}/\mathbb{Z}$ by the commutative diagram.

Definition 4.87 (Brauer set)**.** Fix a nice variety X over a global field k. For each $\alpha \in \operatorname{Br} X$, let $X(\mathbb{A}_k)^\alpha$ be the collection of $x \in X(\mathbb{A}_k)$ which vanish along the composite

$$X(\mathbb{A}_k) \overset{\mathrm{ev}_\alpha}{\to} \bigoplus_v \operatorname{Br} k_v \to \mathbb{Q}/\mathbb{Z}.$$

We then define the *Brauer set* $X(\mathbb{A}_k)^{\mathrm{Br}} := \bigcap_{\alpha \in \operatorname{Br} X} X(\mathbb{A}_k)^\alpha$. If $X(\mathbb{A}_k) \neq \varnothing$ while $X(\mathbb{A}_k)^{\mathrm{Br}} = \varnothing$, then we may say that there is a Brauer–Manin obstruction to the Hasse principle.

Remark 4.88. Explicitly, $x \in X(\mathbb{A}_k)$ lives in $X(\mathbb{A}_k)^\alpha$ if and only if

$$\sum_v \operatorname{inv}_v \alpha(x_v) = 0,$$

where $\operatorname{inv}_v \colon \operatorname{Br} k_v \to \mathbb{Q}/\mathbb{Z}$ is the local invariant map.

Remark 4.89. The preceding discussion shows that $X(k) \subseteq X(\mathbb{A}_k)^{\mathrm{Br}}$. Thus, $X(\mathbb{A}_k)^{\mathrm{Br}} = \varnothing$ tells us that $X(k) = \varnothing$, so the Brauer–Manin obstruction possibly refines the Hasse principle.

Next time, we will see an example where the Brauer–Manin obstruction sees $X(\mathbb{A}_k)^{\mathrm{Br}} = \varnothing$ even though $X(\mathbb{A}_k) = \varnothing$.

4.4 April 13

Today, we continue discussing the Brauer–Manin obstruction.

4.4.1 Iskovskikh's Conic Bundle

This subsection will be interested in the following conic bundle.

Example 4.90 (Iskovskikh)**.** Consider the subvariety $U \subseteq \mathbb{A}^3_{\mathbb{Q}}$ cut out by

$$U \colon y^2 + z^2 = \left(3 - x^2\right)\left(x^2 - 2\right).$$

We will show that U admits no rational points using the Brauer–Manin obstruction.

Remark 4.91. Iskovskikh's original proof that U admits no rational points merely used quadratic reciprocity. This is not so surprising because the Brauer–Manin obstruction amounts to quadratic reciprocity in this situation. Namely, the fundamental exact sequence over $\mathbb{Q}$ (for quaternion algebras) can be checked to be equivalent to quadratic reciprocity.

Our first task is to turn U into a nice variety, so we need to find a smooth projective model. For this, we should explain why we keep calling U a conic bundle.

Definition 4.92 (conic)**.** A *conic* over a field k is a variety cut out by $s = 0$ in $\mathbb{P}^2_k$, where s is some homogeneous polynomial in $k[x_0, x_1, x_2]$ of degree 2.

Remark 4.93. For motivation, we give a coordinate-free definition: for a vector space E over k, we may view $\mathbb{P}^2_k = \mathbb{P}E$ as hyperplanes in E, and we note that homogeneous polynomials s of degree 2 are elements in $\operatorname{Sym}^2 E$.

Definition 4.94 (conic bundle)**.** A *conic bundle* over a scheme B is a scheme $C \to B$ which is cut out in $\mathbb{P}\mathcal{E}$, where $\mathcal{E}$ is a vector bundle of rank 3, by a nowhere vanishing section $s \in \Gamma\left(B; \operatorname{Sym}^2 \mathcal{E}\right)$.

Remark 4.95. We are not requiring our conic bundles to have any smoothness, so the fibers could easily fail to be irreducible (e.g., by being the union of two lines).

Example 4.96. If $\mathcal{E}$ is the direct sum of three line bundles $\mathcal{L}_0 \oplus \mathcal{L}_1 \oplus \mathcal{L}_2$, then choosing nowhere-vanishing sections $s_i \in \Gamma\left(B; \mathcal{L}_i^{\otimes 2}\right)$ for each i allows us to define a "diagonal" conic bundle with the section $s := s_0 + s_1 + s_2$.

Example 4.97 (Châtelet surface)**.** Fix a field k of characteristic not two, and choose a unit $a \in k^\times$ and a separable quartic $P(x) \in k[x]$ with homogenization $\widetilde{P}(x, w) \in k[x, w]$. Then we can define the Châtelet surface $U \colon y^2 - az^2 = P(x)$ in $\mathbb{A}^3$, which completes to a conic bundle $X \to \mathbb{P}^1$ as follows: let our line bundle be $\mathcal{E} := \mathcal{O} \oplus \mathcal{O} \oplus \mathcal{O}(2)$, and choose our section to be $s := (1, -a, -\widetilde{P})$, which then cut out a diagonal conic bundle. Then X contains U as a dense open subscheme.

We now turn to our surface.

Definition 4.98 (Iskovskikh's surface)**.** We define *Iskovskikh's surface* to be the Châtelet surface $X \to \mathbb{P}^1$ (over $\mathbb{Q}$) which contains

$$U \colon y^2 + z^2 = \left(3 - x^2\right)\left(x^2 - 2\right)$$

as a dense open subscheme.

Remark 4.99. The fiber at infinity is cut out by $y^2 + z^2 = \left(3x^2 - 1\right)\left(1 - 2x^2\right)$; namely, $x = \infty$ is $y^2 + z^2 = -w^2$.

Let's start by showing that X admits local points.

Proposition 4.100. Iskovskikh's surface X over $\mathbb{Q}$ has a $\mathbb{Q}_v$-point for each place v.

Proof. A general approach would be to use Lang–Weil estimates to find a smooth $\mathbb{F}_p$-points for large primes p,[3] and then we use Hensel's lemma to lift these $\mathbb{F}_p$-points to $\mathbb{Q}_p$-points. Once we have handled large primes, we can explicitly find points for small p, and we can handle the archimedean place separately.

Here is a more general argument: we can show that the subset

$$U\colon y^2 + z^2 = \left(3 - x^2\right)\left(x^2 - 2\right)$$

already admits $\mathbb{Q}_v$-points. We now have the following cases.

- For finite primes $p \geq 3$, we may set $x := 1$ and then hunt for $(y, z) \in \mathbb{Z}_v^2$ for which $y^2 + z^2 = -2$. It is enough to find a smooth $\mathbb{F}_p$-point, but by inspection, we see that all $\mathbb{F}_p$-points will be smooth (because y and z cannot simultaneously vanish). For this, we note that y^2 takes $\frac{p+1}{s}$ values as $y \in \mathbb{F}_p$ varies, and the same is true for z^2, so there must be some intersection between the values of y^2 and $-2 - z^2$ as y and z varies.

- For $p = 2$, we have the point $(0, 1, \sqrt{-7})$

- For the real place, we have the point $(\sqrt{2}, 0, 0)$. ■

We now begin the more difficult task of showing that X admits no rational point. To start, we need to find an interesting class in $\operatorname{Br} X$.

Lemma 4.101. Let X be Iskovskikh's surface over $k := \mathbb{Q}$, and set $K := k(X)$. Then the quaternion algebra $A := \left(3 - x^2, -1\right) \in \operatorname{Br} K$ is in $\operatorname{Br} X$.

Proof. The idea is to rewrite A in a way that gives it a definition everywhere in X. Define $B := \left(x^2 - 2, -1\right)$ and $C := \left(3/x^2 - 1, -1\right)$. Then A, B, and C are all two-torsion elements (because they are quaternion algebras). Some arithmetic with quaternion algebras (using the fact that the sum in the Brauer group is given by the cup product of Example 4.27) then lets us write

$$A + B = \left(\left(3 - x^2\right)\left(2 - x^2\right), -1\right) = \left(y^2 + z^2, -1\right),$$

but $y^2 + z^2 = -1$ is a norm from $K(\sqrt{-1})/K$, so this quaternion algebra vanishes. Similarly,

$$A - C = \left(x^2, -1\right),$$

which also vanishes for the same reason.

We now complete the proof.

- Note that A is automatically defined over $X \setminus (X_\infty \cup X_{3-x^2})$, where X_∞ is the fiber over $\infty \in \mathbb{P}^1$ and X_{3-x^2} is the fiber over the closed point cut out by $3 - x^2 \in \mathbb{A}^1$.

- Similarly, B is defined over $X \setminus (X_\infty \cup X_{x^2-2})$, where X_{x^2-2} is as above.

- Lastly, C is defined over $X \setminus (X_0 \cup X_{3-x^2})$.

The above subsets of definition cover X, so $A \in \operatorname{Br} K$ has no residues anywhere on X, so it extends to $\operatorname{Br} X$ by Corollary 4.71. ■

We now want to compute $X(\mathbb{A}_k)^A$.

[3] Explicitly, the Lang–Weil estimates show that geometrically irreducible varieties X over $\mathbb{F}_q$ of dimension d have $\#X(\mathbb{F}_q) = q^d + O_X(q^{d-1/2})$, where the constant in O_X depends explicitly on the geometry of X.

Lemma 4.102. Let X be Iskovskikh's surface over $\mathbb{Q}$. For each place v, we compute the evaluation maps

$$X(\mathbb{Q}_v) \xrightarrow{A} \operatorname{Br} \mathbb{Q}_v \hookrightarrow \mathbb{Q}/\mathbb{Z}$$

as

$$\operatorname{inv}_v A(P) = \begin{cases} 0 & \text{if } v \neq 2, \\ 1/2 & \text{if } v = 2. \end{cases}$$

Proof. We are just testing if the quaternion algebra $A(P)_v$ is a trivial algebra over $\mathbb{Q}_v$. Using Example 4.27, one can see that the quaternion algebra $(3 - x^2, -1)$ is trivial at a point $P = (x, y, z)$ in U if and only if $3 - x^2$ is a norm in $\operatorname{im} \mathrm{N}_{\mathbb{Q}(\sqrt{-1})/\mathbb{Q}}$. Of course, presenting A as $(3 - x^2, -1)$ does not work for all points P, but it does work for $P \in U$, which will be enough for now.

- Suppose $p \notin \{2, \infty\}$, and choose a point $P \in X$ with x-coordinate x. We have two cases.
  - If $x = \infty$ or $v(x) < 0$, then $3/x^2 - 1 \in \mathbb{Z}_p^\times$.
  - If $v(x) \geq 0$, then $3 - x^2$ or $x^2 - 2$ is in $\mathbb{Z}_p^\times$ (because these two elements sum to 1).

  It follows that $A(P)_v$ can be presented to live in $\operatorname{Br} \mathbb{Z}_p$ (in all cases!), which vanishes for finite odd primes p, so $\operatorname{inv}_p A(P) = 0$.
- Suppose $p = \infty$. By Remark 4.99, our point P cannot live over $\infty \in \mathbb{P}^1$ because the conic $y^2 + z^2 = -w^2$ has no real points. Thus, $P \in U$ has some finite x-coordinate x. Then $3 - x^2$ and $x^2 - 2$ sum to 1, so at least one is positive and hence a square in $\mathbb{R}^\times$.[4] Thus, $\operatorname{inv}_\infty A(P) = 0$.
- Suppose $p = 2$. We have the following cases.
  - If $v(x) > 0$, then $3 - x^2 \equiv -1 \pmod 4$.
  - If $v(x) = 0$, then $x^2 - 2 \equiv -1 \pmod 4$.
  - Lastly, if $v(x) > 0$, then $3/x^2 - 1 \equiv -1 \pmod 4$.

  Now, any element which is $-1 \pmod 4$ is not a norm in the image of $\mathbb{Q}_2(\sqrt{-1}) \to \mathbb{Q}_2$; explicitly, we are saying that $-1 \pmod 4$ cannot be written as a sum of two squares, which can be checked $\pmod 4$. Thus, the above three cases have shown that $\operatorname{inv}_2 A(P) = 1/2$ for all points P. ■

Remark 4.103. Morally, what is going on here is that the place v where X has good reduction automatically kill A because we can then go define A over $\mathbb{Z}_p$. It is only at the place $p = 2$ where we have a chance of finding some nontrivial invariants.

The above lemma now feeds into the following result.

Theorem 4.104. Let X be Iskovskikh's surface over $\mathbb{Q}$. Then $X(\mathbb{A}_\mathbb{Q})$ is nonempty, but $X(\mathbb{A}_\mathbb{Q})^{\mathrm{Br}}$ is empty.

Proof. Proposition 4.100 shows that X has $\mathbb{Q}_v$-points for every place v, so $X(\mathbb{A}_\mathbb{Q}) \neq \varnothing$ follows by the properness of X. However, using the quaternion algebra A from Lemma 4.101, we find that $X(\mathbb{A}_\mathbb{Q})^A$ is empty: indeed, Lemma 4.102 implies that

$$\sum_v \operatorname{inv}_v A(P) = \frac{1}{2}$$

for all $P \in X(\mathbb{A}_\mathbb{Q})$. ■

[4] This is now well-known, called the "Etingof principle."

Remark 4.105. It remains a question where the magical quaternion algebra A comes from. The idea is that $3 - x^2$ and $x^2 - 2$ sum to 1, so they are relatively prime. However, they sum to a norm $y^2 + z^2$, so one expects both of them to individually be norms. Thus, the quaternion algebra A should be found to be trivial most of the time, with some notable problems at 2.

Remark 4.106. It may have been reasonable to test with Brauer elements from $\operatorname{Br}\mathbb{Q}$ for our Brauer–Manin obstruction. It turns out that such elements never provide an obstruction to rational points. Instead, for Iskovskikh's surface X, one can use the techniques from Section 4.3.2 to show that there is a split exact sequence

$$0 \to \operatorname{Br}\mathbb{Q} \to \operatorname{Br} X \to \mathbb{Z}/2\mathbb{Z} \to 0,$$

where the $\mathbb{Z}/2\mathbb{Z}$ lifts to $A \in \operatorname{Br} X$; thus, A turns out to have been the only possibility for X to have the Brauer–Manin obstruction! As a corollary of this remark, we note that the Brauer–Manin obstruction never says anything interesting if $\operatorname{Br} k \to \operatorname{Br} X$ is surjective.

Remark 4.107. One does not expect the Brauer–Manin obstruction to be sharp. However, if we pass to covers, then one can show that the obstruction is sharp. Explicitly, one needs to test the Brauer–Manin obstruction on all open covers $X = \bigcup_i X_i$, where the idea is that

$$X(k) \subseteq \bigcup_i X_i(\mathbb{A}_k)^{\operatorname{Br} X_i},$$

where $X_i(\mathbb{A}_k)^{\operatorname{Br} X_i}$ is the Brauer set for X_i.

4.4.2 The Descent Obstruction

Our next goal is to provide an example where there are no rational points, but this is not seen even by the Brauer–Manin obstruction. To this end, we need a different way to show that there are no rational points, for which we use descent obstruction. The starting point is the following classification.

Definition 4.108 (algebraic group)**.** Fix a field k. Then an *algebraic group* G over k is a smooth affine group scheme of finite type over k.

Theorem 4.109. Fix a field k, and let G be an algebraic group. For a k-scheme X. The G-torsors over X are parameterized by $\mathrm{H}^1(X; G)$.

Proof. Omitted! ■

We now retell our Brauer–Manin obstruction.

Definition 4.110. Fix a nice variety X over a field k. Fix an algebraic group G and a G-torsor $f\colon Y \to X$. Then we define the *evaluation map* $\operatorname{ev}_Y\colon X(k) \to \mathrm{H}^1(k; G)$ as taking the point $x \in X(k)$ the pullback of $Y \in \mathrm{H}^1(X; G)$ along the map $x\colon \operatorname{Spec} k \to X$. Equivalently, $\operatorname{ev}_Y(x)$ is the fiber $f^{-1}(\{x\}) \to \operatorname{Spec} k$.

Remark 4.111. There is a decomposition

$$X(k) = \bigsqcup_{\sigma \in \mathrm{H}^1(k;G)} \{x \in X(k) : \mathrm{ev}_Y(x) = \sigma\}.$$

Because a torsor is trivial if and only if it admits a rational point, we see that the set of $x \in X(k)$ where $\mathrm{ev}_Y(x)$ is trivial is simply $f(Y(k))$. More generally, for a class σ, there is a way to twist the torsor $f\colon Y \to X$ to the torsor $f^\sigma\colon Y^\sigma \to X$ (which is basically subtraction fiber-wise). Tracking around the subtraction, we find that

$$X(k) = \bigsqcup_{\sigma \in \mathrm{H}^1(k;G)} f^\sigma(Y^\sigma(k)).$$

The descent obstruction now observes that

$$X(k) \subseteq \bigsqcup_{\sigma \in \mathrm{H}^1(k;G)} f^\sigma(Y^\sigma(\mathbb{A}_k)).$$

It turns out that $Y^\sigma(\mathbb{A}_k)$ is nonempty for only finitely many σ, and one can even effectively compute which σ we have to study.

Definition 4.112 (descent set)**.** Fix a nice variety X over a field k and an algebraic group G. For a G-torsor $f\colon Y \to X$, we define $X(\mathbb{A}_k)^f$ to be the union

$$X(\mathbb{A}_k)^f := \bigsqcup_{\sigma \in \mathrm{H}^1(k;G)} f^\sigma(Y^\sigma(\mathbb{A}_k)).$$

We define the *descent set* $X(\mathbb{A}_k)^{\mathrm{descent}}$ to be the intersection of $X(\mathbb{A}_k)^f$ as f varies over all G-torsors and G varies over all smooth algebraic groups.

Remark 4.113. One can recover the weak Mordell–Weil theorem for an elliptic curve E by taking $G = E[2]$.

4.5 April 15

The homework has been posted.

4.5.1 Comparison of Obstructions

While we're here, we define some refinements of the descent obstruction.

Notation 4.114. Fix a nice variety X over a field k and an algebraic group G. Then we define $X(\mathbb{A}_k)^{\text{ét},\mathrm{Br}}$ to be the intersection of

$$\bigsqcup_{\sigma \in \mathrm{H}^1(k;G)} f^\sigma\left(Y^\sigma(\mathbb{A}_k)^{\mathrm{Br}}\right)$$

as f varies over all G-torsors $f\colon Y \to X$ and finite étale group schemes G. We similarly define the $X(\mathbb{A}_k)^{\text{ét},\mathrm{descent}}$.

Proposition 4.115. Fix a nice variety X over a global field k. Then

$$X(\mathbb{A}_k)^{\mathrm{descent}} = X(\mathbb{A}_k)^{\text{ét},\mathrm{descent}} = X(\mathbb{A}_k)^{\text{ét},\mathrm{Br}} \subseteq X(\mathbb{A}_k)^{\mathrm{Br}}.$$

Proof. Let's show inclusions in sequence.

- It turns out that the Brauer–Manin obstruction is a special case of the descent obstruction for the groups PGL_n. (This is basically Theorem 4.56.) This implies that
$$X(\mathbb{A}_k)^{\mathrm{descent}} \subseteq X(\mathbb{A}_k)^{\mathrm{Br}}.$$
Similarly, $X(\mathbb{A}_k)^{\text{ét},\mathrm{descent}} \subseteq X(\mathbb{A}_k)^{\text{ét},\mathrm{Br}}$.
- Using the structure theory of algebraic groups, one can show that $X(\mathbb{A}_k)^{\mathrm{Br}}$ is the same as using obstructions from all connected groups. By further adding in some component group, one finds that
$$X(\mathbb{A}_k)^{\text{ét},\mathrm{descent}} \subseteq X(\mathbb{A}_k)^{\text{ét},\mathrm{Br}}.$$
- It further turns out that $X(\mathbb{A}_k)^{\mathrm{descent}} \subseteq X(\mathbb{A}_k)^{\text{ét},\mathrm{descent}}$, though this is not obvious. This is due to Skorobogatov and Stoll. ■

Remark 4.116. One could try to further define $X(\mathbb{A}_k)^{\mathrm{descent},\mathrm{descent}}$, but it turns out to equal $X(\mathbb{A}_k)^{\mathrm{descent}}$.

Remark 4.117. These obstructions are not sufficient.

- Skorobagatov found a nice variety X where $X(\mathbb{A}_k)^{\mathrm{Br}}$ is nonempty while $X(\mathbb{A}_k)^{\mathrm{descent}}$ is empty. The example is found among bi-elliptic surfaces.
- Bjorn Poonen showed that $X(\mathbb{A}_k)^{\mathrm{descent}}$ may be nonempty when $X(k)$ is empty. His original counterexample was three-dimensional, but there may be two-dimensional counterexamples.

Remark 4.118. Bjorn Poonen has a heuristic that $X(\mathbb{A}_k)^{\mathrm{Br}}$ is sufficient to detect whether curves have rational points, which explains why these counterexamples are found among surfaces and three-folds.

Remark 4.119. In class, we discussed what might happen if we try to add in abelian varieties. It is not obvious if this refines the descent obstruction.

4.5.2 Descent Is Not Enough

For the rest of class, we will focus on giving an example of some nice X where $X(\mathbb{A}_k)^{\mathrm{descent}}$. Poonen's original example was a bundle of Châtelet surfaces, but we will present an easier example (due to Coliot-Thélène, Pál, and Skorobogatov) of a bundle of quadrics.

Theorem 4.120. There is a nice variety X over $\mathbb{Q}$ for which $X(k) = \varnothing$ while $X(\mathbb{A}_k)^{\mathrm{descent}} \neq \varnothing$.

Proof. We proceed in steps. Let's start with the construction.

1. Start with a nice $\mathbb{Q}$-curve C such that $C(\mathbb{Q})$ is a single point c. For example, a smooth projective model of $y^2 = x^3 - 3$ will work, which one shows by doing a 2-Selmer group calculation to show rank zero and then computing the torsion group by computing $C(\mathbb{F}_p)$ for various primes p.

   The smoothness (and properness) of C allows us to find a map $f\colon C \to \mathbb{P}^1$ which is étale at c, and we may as well assume that $f(c) = \infty$. For $C\colon y^2 = x^3 - 3$, the map $y\colon C \to \mathbb{P}^1$ will do. While we're here, we also choose a simply connected open neighborhood $U \subseteq C(\mathbb{R})$ of c, and we choose it small enough so that it remains simply connected and open as $f(U) \subseteq \mathbb{P}^1_{\mathbb{R}}$. By changing coordinate again, we may assume that $1 \in f(U)$ and that f is étale above 0 and 1.

2. Now, glue the two bundles $Y^{(t)} \subseteq \mathbb{P}^4 \times \mathbb{A}^1_t$ and $Y^{(T)} \subseteq \mathbb{P}^4 \times \mathbb{A}^1_T$ given by

$$\begin{cases} Y^{(t)} \colon t(t-1)x_0^2 + \left(x_1^2 + x_2^2 + x_3^2 + x_4^2\right) = 0, \\ Y^{(T)} \colon (1-T)X_0^2 + \left(x_1^2 + x_2^2 + x_3^2 + x_4^2\right) = 0. \end{cases}$$

   Namely, these may be glued along the equations $t = 1/T$ and $x_0 = X_0 T$, which gives a quadric bundle $Y \to \mathbb{P}^1_t$. This has two degenerate fibers, which are above 0 and 1; namely, we get some degenerate cones above those points.

3. We construct our X. By pulling back $Y \to \mathbb{P}^1$ along $f \colon C \to \mathbb{P}^1$, we obtain a quadric bundle $\pi \colon X \to C$. Note that $Y \to \mathbb{P}^1$ is certainly projective with geometrically irreducible fibers, so we receive the same for $X \to C$. In fact, $Y \to \mathbb{P}^1$ is smooth away from the fibers of 0 and 1, and $C \to \mathbb{P}^1$ is smooth above 0 and 1, so X is smooth over $\mathbb{P}^1$ and thus over $\mathbb{Q}$.

We now run our checks on X.

4. Let's show that $X(\mathbb{Q}) = \varnothing$. On one hand, any $x \in X(\mathbb{Q})$ provides a $\mathbb{Q}$-point of C, so $\pi(x) = c$. But then $f(x) = \infty$. On the other hand, x must also map to a rational point of Y, which now must be in the fiber $Y_\infty \subseteq Y$. But taking $T = 0$ in $Y^{(T)}$ finds no real points, so there are no rational points here.

5. We now want to show that there are local points which survive the descent obstruction. Let's start by showing that there are local points.

   - We handle finite places. To begin, we recall that every quadratic form over $\mathbb{Q}_p$ in at least five variables has a nontrivial zero.[5] Indeed, this is classical. It is basically Hensel's lemma. For example, avoiding $p = 2$ (which is handled separately), we may diagonalize and reorganize the coefficients, so we may assume that the quadratic form looks like $u_1x_1^2 + u_2x_2^2 + u_3x_3^2 + u_4x_4^2 + u_5x^5$, where u_1, u_2, and u_3 are units. Then a counting argument finds a smooth point over $\mathbb{F}_p$, which then lifts to $\mathbb{Z}_p$.

     Now, we define $x_p := (y_p, c) \in X(\mathbb{Q}_p)$, where y_p is a distinguished point in $y_p \in Y_\infty(\mathbb{Q}_p)$.
   - At the infinite place, we let $y_\mathbb{R}$ be the unique real point of $Y_1(\mathbb{R})$, and we choose $c_\mathbb{R} \in U$ for which $f(c_\mathbb{R}) = 1$ in $\mathbb{P}^1(\mathbb{R})$. Thus, we may define $x_\mathbb{R} := (y_\mathbb{R}, c_\mathbb{R})$ to be a point in $X(\mathbb{R})$.

   For future use, we set $x := (x_\mathbb{R}, x_2, x_3, \ldots)$ to be this point of $X(\mathbb{A}_\mathbb{Q})$.

6. As a warm-up, we claim that $x \in X(\mathbb{A}_\mathbb{Q})^{\mathrm{Br}}$. Well, select any $A \in \operatorname{Br} X$, and we want to show that $x \in X(\mathbb{A}_\mathbb{Q})^A$. For this, recall from Proposition 4.83 that $\operatorname{Br} C \to \operatorname{Br} X$ is surjective, so we may find $A_C \in \operatorname{Br} C$ which maps to A. Thus,

$$\sum_v \operatorname{inv}_v \operatorname{ev}_A(x_v) = \sum_v \operatorname{inv}_v \operatorname{ev}_{A_C}(\pi(x_v))$$

   by unwinding the evaluations. Now, for finite v, we see that $\pi(x_v) = c$. Even at ∞, one knows that ev_{A_C} is some continuous map $C(\mathbb{R}) \to \frac{1}{2}\mathbb{Z}/\mathbb{Z}$, so $\operatorname{ev}_{A_C}(c_\mathbb{R}) = \operatorname{ev}_{A_C}(c)$ still. Thus, this sum is

$$\sum_v \operatorname{inv}_v \operatorname{ev}_{A_C}(c),$$

   which vanishes because c is actually a rational point on C!

7. We now claim that $x \in X(\mathbb{A}_\mathbb{Q})^{\text{ét},\mathrm{Br}}$, which will complete the proof by Proposition 4.115. Note that the fibers of $X \to C$ are all simply connected (they are quadric hypersurfaces), so every étale over of X comes from an étale cover of C. Now, choose a finite étale group scheme G, and choose a G-torsor $X' \to X$. Because G is étale, the previous few sentences explain that this torsor must come from a G-torsor $C' \to C$. The descent obstruction does not care if we replace X' with a twist, so we may twist

[5] This is false for four variables because one can use a quadratic form in four variables to cut out a quaternion algebra.

C' so that $c \in C(\mathbb{Q})$ lifts to some $c'' \in C'$; accordingly, we let C'' be the irreducible component of C'.[6] All of this fits into the following pullback squares.

$$\begin{tikzcd} X'' \arrow[r] \arrow[d] & X' \arrow[r, "G"] \arrow[d] & X \arrow[r] \arrow[d] & Y \arrow[d] \\ C'' \arrow[r] & C' \arrow[r, "G"] & C \arrow[r] & \mathbb{P}^1 \end{tikzcd}$$

Now, C'' is smooth and projective, and it has a rational point, so it follows that C'' is a nice curve. (In particular, it is geometrically irreducible.) By pulling back along $X \to C$, we see that X'' is also nice.

It is now enough to show that $x \in X(\mathbb{A}_k)$ lifts to some $x'' \in X''(\mathbb{A}_k)^{\mathrm{Br}}$. To start, let's show that it lifts to $X''(\mathbb{A}_k)$.

- For finite prime p, we define $x''_p := (x_p, c'')$.
- At ∞, we note that U is simply connected, so its inverse image in $C''(\mathbb{R})$ must be a disjoint union of copies of U; accordingly, we let $c''_{\mathbb{R}}$ be the pre-image of $c_{\mathbb{R}} \in U$ in the same copy of U in $C''(\mathbb{R})$ as c''.

These local points now glue to $x'' \in X''(\mathbb{A}_k)$, so it remains to check that x'' is in $X'(\mathbb{A}_k)^{\mathrm{Br}}$. Because X'' is just a connected component of X', new Azumaya algebras gained from going from X'' to X' will not provide a meaningful obstruction, so it is enough to check that $x'' \in X''(\mathbb{A}_k)^{\mathrm{Br}}$. Well, one can just repeat the same argument from the previous step word-for-word, so we are done! ■

Remark 4.121. One could try to perturb this example by replacing C with a surface, but nobody knows an example of a simply connected surface with exactly one rational point.

We end the course with a conjecture.

Conjecture 4.122 (Colliot-Thélène, Karevsky, Sansuc)**.** Fix a nice variety X over a number field which is geometrically rationally connected. If $X(\mathbb{A}_k)^{\mathrm{Br}} \neq \varnothing$, then $X(k) \neq \varnothing$.

Remark 4.123. Rationally connected means that any two points can be connected by some map $\mathbb{P}^1 \to X$.

Remark 4.124. Basically nothing is known about Conjecture 4.122. It is known for Châtelet surfaces.

Remark 4.125. One expects that $\operatorname{Br} X = \operatorname{Br} k$ for varieties of general type, so $X(\mathbb{A}_k)^{\mathrm{Br}} = X(\mathbb{A}_k)$. On the other hand, it is rather rare for such varieties to admit rational points.

[6] There is a classification of the étale covers of elliptic curves, which tells us that C'' is an elliptic curve with identity c'', and the map $C'' \to C$ is an isogeny.

MINICOURSE 5

CLUSTER ALGEBRAS AND CLUSTER VARIETIES

These talks were given by Robin Zhang.

5.1 April 22

We only have three lectures, so we will have to be fast. We will try to introduce the basic notions, and then we will try to explain their applications to Diophantine equations. Here is a rough plan.

1. We'll start with some history and introduce some combinatorics which give rise to cluster algebras $\mathcal{A}$. The combinatorics arises by studying maps $\mathcal{A} \to \mathbb{Z}$.
2. Then we will introduce the basic notions of cluster algebras.
3. We will explain the classification of cluster algebras.
4. With cluster algebras in hand, we will define cluster varieties, which look like $\operatorname{Spec} \mathcal{A}$.
5. From here, we will be able to discuss applications to positive integral points on the affine varieties $\operatorname{Spec} \mathcal{A}$.
6. Lastly, we will explain applications to arithmetic dynamics (in "Donaldson–Thomas dynamics") on the cluster varieties.

5.1.1 Classical Friezes

Cluster algebras began their life in the theory of total positivity and representations of quivers. They later found applications in Teichmüller theory and symplectic/Poisson geometry.[1] They can also be found in Grassmannians, in quantum field theory, and cosmology. We will say nothing about any of these applications. Instead, we will explain connections to counting (positive) integral points on log Calabi–Yau varieties and arithmetic dynamics, originally discovered through the Markov surface.

Cluster algebras arise from freizes. Here is the definition of a classical frieze.

Definition 5.1 (frieze)**.** A *frieze* is the central section of an entablature above a Greek temple.

Example 5.2. The Parthenon featured a frieze, which was some horizontally periodic geometric pattern.

This definition is slightly too classical.

[1] Pavel Etingof points out that the cluster algebras found in Poisson geometry are algorithmically quantizable, which provide an interesting source of examples of quantization.

Definition 5.3 (frieze)**.** A *frieze pattern* is an array of positive integers with infinitely many columns and some SL_2-unimodular conditions. More explicitly, a frieze pattern with n rows is an array $\{x_{ij}\}_{ij}$ where $i \in \{1, 2, \ldots, n\}$ and $j \in \mathbb{Z}^+$ satisfying the condition

$$x_{ij}x_{i,j+1} - x_{i-1,j+1}x_{i+1,j} = 1.$$

Remark 5.4. One typically arranges the frieze as follows.

$$\begin{array}{ccccccccccccc}
\cdots & 1 & & 1 & & 1 & & 1 & & 1 \cdots & & & \\
\cdots & & x_{11} & & x_{12} & & x_{13} & & x_{14} & & \cdots & & \\
\cdots & x_{2,-1} & & x_{21} & & x_{22} & & x_{23} & & x_{24} & \cdots & & \\
\cdots & & \ddots & & \ddots & & \ddots & & \ddots & & \cdots & & \\
\cdots & 1 & & 1 & & 1 & & 1 & & 1 & & 1 & \cdots
\end{array}$$

With this convention, we see that $x_{ij}x_{i,j+1} - x_{i-1,j+1}x_{i+1,j} = 1$ is asking for each "diamond" to have determinant 1.

Example 5.5. Here is a Frieze pattern with 2 rows.

$$\begin{array}{ccccccccccccc}
\cdots & 1 & & 1 & & 1 & & 1 & & 1 & & 1 & \cdots \\
\cdots & & 3 & & 1 & & 2 & & 2 & & 1 & & 3 \ \cdots \\
\cdots & 1 & & 2 & & 1 & & 3 & & 1 & & 2 & \cdots \\
\cdots & & 1 & & 1 & & 1 & & 1 & & 1 & & \cdots
\end{array}$$

(We do not count the row of ones in the number of rows.) We note that there is a "fundamental domain" with $(3, 1, 2)$ in the first row and $(2, 1)$ in the bottom row.

Gauss (roughly speaking) proved the following classification.

Theorem 5.6 (Gauss 1866)**.** All friezes with two rows is 5-periodic.

Remark 5.7. This was proven "posthumously" as Gauss's theorem on the pentagramma minificum; Gauss proved some purely geometric fact which turns out to imply that any three of the equations

$$\begin{cases} 1 + \alpha = \gamma\delta, \\ 1 + \beta = \delta\varepsilon, \\ 1 + \gamma = \alpha\varepsilon, \\ 1 + \delta = \alpha\beta, \\ 1 + \varepsilon = \beta\gamma \end{cases}$$

implies the other two, which Coxeter recognized classifies friezes.

Coxeter would go on to try to classify these friezes.

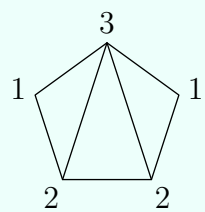
Theorem 5.8 (Coxeter 1971)**.** All friezes with n rows are periodic with period $n + 3$.

Remark 5.9. Periodicity continues to hold, even if we do not require that the entries of the frieze are integers.

Theorem 5.10 (Conway–Coxeter 1973)**.** Fix n. Then there are exactly $\frac{1}{n+2}\binom{2n+2}{n+1}$ friezes with n rows.

Sketch. Note that $\frac{1}{n+2}\binom{2n+2}{n+1}$ is a Catalan number, so in Catalan tradition, we will try to find a bijection with one of the many other ways to count Catalan numbers. It turns out that we can friezes with n rows can be found in bijection with triangulations of $(n+3)$-gons. In particular, given a triangulation of $(n+3)$-gons with labeled vertices $\{1, 2, \ldots, n+3\}$, define $\{x_{1j}\}_j$ to be the periodic sequence of the number of triangles which hit vertex j. Then one can use the determinant condition to complete the frieze. Injectivity of this map follows because one can recover the triangulation from the sequence, and this turns out to hit all friezes, which one can check by induction on n. ■

Example 5.11. For $n = 2$, there is only one triangulation of the pentagon (up to rotation), so we find that Example 5.5 gives the only frieze up to translation.



Example 5.12. There is a triangulation of the nonagon given by the sequence $(4, 2, 1, 4, 2, 1, 4, 2, 1)$. This provides an example of a frieze with six rows with this sequence as the first row.

5.1.2 General Friezes

Here are some more general friezes; this definition is due to Caldero and Chapoton.

Definition 5.13 (frieze)**.** Fix a Dynkin type Δ with (generalized) Cartan matrix $C = \{c_{ij}\}_{i,j\le n}$. Then a *frieze of type* Δ is an array $\{x_{ij}\}$ with $i \in \{1, 2, \ldots, n\}$ and $j \in \mathbb{N}$, satisfying the condition

$$x_{ij}x_{i,j+1} - \prod_{k<i} x_{k,j+1}^{|c_{ki}|} \prod_{k>i} x_{kj}^{|c_{ki}|} = 1.$$

Example 5.14. Take $\Delta = A_n$ recovers a frieze with n rows.

Example 5.15. Type E_8 has the following Dynkin diagram.

This means that the frieze should have seven rows, but the third row will have extra numbers in it ("filling in the diamond"), which must then satisfy the condition $ae - bcd$, where these numbers fit into the following diamond.

$$\begin{array}{ccc} & b & \\ a & c & e \\ & d & \end{array}$$

There is a frieze with

$$(x_{11}, \ldots, x_{18}) = (4, 11, 29, 5, 21, 13, 5, 2),$$

where the vertices are colored as above.

We now try to prove the same theorems as Coxeter (and Conway).

Theorem 5.16 (Assem–Reataver–Smith)**.** Fix a Dynkin type Δ. Then all friezes of type Δ are periodic if and only if Δ is of finite type.

Remark 5.17. The periods are known for all Dynkin types. For example, it is 16 for type E_8.

With some periodicity, we can try to count these things. We have little to say about infinite types.

Theorem 5.18. If Δ is a Dynkin type of infinite type, then there are infinitely many friezes of type Δ.

Remark 5.19. It turns out that the coefficients of a frieze are unbounded in infinite types.

For finite types, we can count them.

Theorem 5.20. Fix a Dynkin type Δ of finite type. Then the friezes of type Δ can be enumerated as follows.

Δ	A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
$\#$	$\frac{1}{n+2}\binom{2n+2}{n+1}$	$\sum_{m=1}^{\lfloor\sqrt{n+1}\rfloor}\binom{2n-m^2+1}{n}$	$\binom{2n}{n}$	$\sum_{m=1}^{n} d(m)\binom{2n-m-1}{n-m}$	868	4400	26952	112	9

Remark 5.21. The counts in type E_7 and E_8 were conjectured in 2016 (by computer search) but only proven recently.

Remark 5.22. We will say basically nothing about the proof, but we do say that many of the arguments here find bijections between these friezes and some kind of polygon triangularizations, perhaps with punctures and so on.

Remark 5.23. The enumerations for E_7 and E_8 are due to Robin Zhang. He used cluster algebras (and Diophantine aspects of cluster varieties).

Of course, we have yet to define a cluster algebra, but we can give some motivation for them via the following counting.

Proposition 5.24. Fix a Dynkin type Δ, and let $\mathcal{A}$ be a cluster algebra of type Δ. Then the friezes of type Δ are in bijection with ring homomorphisms $f\colon \mathcal{A} \to \mathbb{Z}$ which map the cluster variables in $\mathcal{A}$ to positive integers.

5.1.3 Cluster Algebras

We are now allowed to define cluster algebras. The most general definition is technical, so we will content ourselves with the following notion.

Definition 5.25 (cluster algebra)**.** A *cluster algebra* $\mathcal{A}$ over a field k is a commutative subring of $\mathbb{Q}(x_1, \dots, x_n)$ equipped with *cluster variables* which are constructed from some mutation $n \times n$ matrix B as follows (with zeroes along the diagonals).

1. Start with $(x_1, \dots, x_n)$. Then define

$$x_1' := \frac{\prod_{\substack{1 \le i \le n \\ b_{i1} < 0}} x_i^{|b_{i1}|} + \prod_{\substack{1 \le i \le n \\ b_{i1} > 0}} x_i^{|b_{i1}|}}{x_1}.$$

   In words, in the first column, we take a weighted product of the x_is with positive coefficients, and the weighted product of the x_is with negative coefficients. Lastly, we divide by x_1. We then change B by flipping the sign on the first row and first column. This gives a new tuple of rational functions $(x_1', x_2, \dots, x_n)$.

2. We then repeat the process at x_2, then x_3, and so on. Eventually we go back and repeat.

At the end, $\mathcal{A}$ is generated by the resulting rational functions.

Example 5.26. Over the field $\mathbb{Q}$, we define a cluster algebra $\mathcal{A} \subseteq \mathbb{Q}(x_1, x_2)$ with the mutation matrix $B := \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. Then $\mathcal{A}$ is defined as follows.

1. We take the pair (x_1, x_2) and apply the following process: take x_1, and define a numerator to be the product of all variables with a positive coefficient in B (in the first column) plus the product of the variables with a negative coefficient, and then we define out by x_1. This gives $x_1' = (x_2+1)/x_1$. Lastly, we adjust B by flipping signs in the second row and second column, giving $B = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$.

2. Then we repeat the process with the second column. This gives $x_2' = (x_1' + 1)/x_2 = (x_1 + x_2 + 1)/(x_1 x_2)$ and $B'' = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$.

3. We now go back to do the process to x_1, which gives $x_1'' = (x_2' + 1)/x_1' = (x_1 + 1)/x_2$.

4. Repeating the process for the second column gives $x_2'' = (x_1'' + 1)/x_2' = x_1$.

5. Going back to the first column gives $x_1''' = x_2$.

Thus, we see that our process is periodic! The resulting cluster algebra is thus generated by five rational functions (which are the cluster variables), each of which is a cluster variable.

Remark 5.27. One can play the same game with any skew-symmetric matrix B with integral coefficients. In fact, it is even enough if B is skew-symmetrizable, meaning that it is the product of a skew-symmetric matrix and a diagonal matrix with positive entries.

We will classify cluster algebras next time. The above example will turn out to be type A_2; the number of cluster variables turns out to correspond to the period of the frieze.

5.2 April 27

Today, we classify cluster algebras.

5.2.1 Cluster Algebras, Again

Last time, we constructed a cluster algebra attached to a skew-symmetrizable matrix. Let's return to this definition, as follows.

Definition 5.28 (mutation)**.** Fix a positive integer n. A *seed* is a pair (x, B) of an n-tuple $x = (x_1, \ldots, x_n)$ of algebraically independent variables over $\mathbb{Q}$, and a *mutation matrix* B is an $n \times n$ skew-symmetrizable matrix. Given this data, we define the kth *mutation* $\mu_k(x, B)$ to be the seed given by

$$\mu_k(x)_i := \begin{cases} x_i & \text{if } i = k, \\ \dfrac{\prod_{b_{jk}>0} x_j^{|b_{jk}|} + \prod_{b_{jk}<0} x_j^{|b_{jk}|}}{x_k} & \text{if } i \neq k, \end{cases}$$

together with the mutation matrix given by the matrix B' with coefficients

$$B'_{ij} = \begin{cases} -b_{ij} & \text{if } i = k \text{ or } j = k, \\ b_{ij} + \frac{|b_{ik}|b_{kj} + b_{ik}|b_{kj}|}{2} & \text{if } i \neq k \text{ and } j \neq k. \end{cases}$$

Remark 5.29. By construction, applying the kth mutation twice does nothing. This property explains the rule for mutating the mutation matrix. Roughly speaking, the mysterious second equation in the mutation of B should be split up into two cases: if b_{ik} and b_{kj} have the same sign, then the two pieces combine; and if they have different signs, then we have cancellation.

Definition 5.30 (cluster algebra)**.** Fix a positive integer n, and choose a seed (x, B). Then the associated *cluster algebra* $\mathcal{A}$ is the $\mathbb{Z}$-subalgebra of $\mathbb{Q}(x)$, generated by all possible sequence of mutations. The collection of generators is known as the set of *cluster variables*.

Remark 5.31. Our cluster algebras $\mathcal{A}$ will have $\mathcal{A} \cap \mathbb{Q} = \mathbb{Z}$, which follows from Theorem 5.37.

It is not at all obvious if the associated cluster algebra is finitely generated. In light of our cluster variables, the correct notion is as follows.

Definition 5.32 (finite type)**.** A cluster algebra $\mathcal{A}$ has *finite type* if and only if there are only finitely many distinct cluster variables.

Here is how we will test this.

Definition 5.33. A cluster algebra $\mathcal{A}$ is *acyclic* if and only if the associated seed (x, B) (and its mutations) produce directed graphs with no oriented cycles. Here, the directed graph associated to an $n \times n$ skew-symmetric B is the one with vertices $\{v_1, \ldots, v_n\}$ and b_{ij} edges from v_i to v_j; for a skew-symmetrizable matrix DB where D is diagonal, then we also label the edges by the diagonal elements: the edge $i \to j$ for $b_{ij} > 0$ is labeled by the pair $(|b_{ij}|, |b_{ji}|)$ when $b_{ij} > 0$.

Remark 5.34. It turns out that $\mathcal{A}$ is finitely generated if and only if $\mathcal{A}$ is acyclic.

Example 5.35. The matrix $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ induces the graph $1 \to 2$. The only mutation of this matrix is its negative, so we see that this cluster algebra is acyclic.

Non-Example 5.36. The matrix

$$\begin{bmatrix} 0 & 2 & -2 \\ -2 & 0 & 2 \\ 2 & -2 & 0 \end{bmatrix}$$

produces the graph

$$\begin{tikzcd} 1 \arrow[r, shift left] & 2 \arrow[l, shift left] \arrow[d, shift left] \\ & 3 \arrow[ul, shift left] \arrow[u, shift left] \end{tikzcd}$$

which is notably not acyclic. The mutations of this matrix correspond to Vieta jumps. Namely, specializing the seed variables $(x_1, x_2, x_3) \mapsto (1, 1, 1)$ discovers all solutions to the Markov equation $x^2+y^2+z^2 = 3xyz$.

5.2.2 Theorems on Cluster Algebras

Here are the main theorems on cluster algebras. The following is due to Fomin and Zelevinsky.

Theorem 5.37 (Laurent phenomenon)**.** Fix a cluster algebra $\mathcal{A}$ with initial seed $((x_1, \ldots, x_n), B)$. Then any cluster variable in $\mathcal{A}$ is a Laurent polynomial in the variables $(x_1, \ldots, x_n)$. In other words, $\mathcal{A} \subseteq \mathbb{Z}[x_1, \ldots, x_n, 1/x_1, \ldots, 1/x_n]$.

Sketch. The key ingredient is the following intersection lemma: if x mutates to x' via the ith mutation, then by construction, $x_i x_i'$ can be expanded into a sum of two disjoint monomials. Thus, setting $R := \mathbb{Z}[x_1^{\pm}, \ldots, x_{i-1}^{\pm}, x_{i+1}^{\pm}, \ldots, x_n^{\pm}]$, we find

$$R\left[x_i^{\pm}\right] \cap R\left[(x_i')^{\pm}\right] = R[x_i, x_i'].$$

For example, the inclusion from the right to the left follows from the expansion of $x_i x_i'$. The moral is that having a cluster variable z which is a Laurent polynomial in the variables $(x_1, \ldots, x_n)$ and also a Laurent polynomial in all adjacent mutated seeds, then $z \in \mathbb{Z}[x_1, \ldots, x_n, x_1', \ldots, x_n']$, so z is Laurent.

One can then do an induction in the "distance" of a cluster variable from the initial seed. Namely, at distance zero, there is nothing to do. At distance one, this follows from the construction. At distance two, one has a cluster variable which is Laurent in distance one, and one can check by hand that the resulting polynomials are also Laurent in all the other mutations in distance two (and hence polynomial), which again uses the given mutation formula explicitly. After this, the induction kicks in: for example, in distance three, one can use the distance-two version to show that a given cluster variable is a polynomial in cluster variables that are distance zero and one. ■

Remark 5.38. Fomin called this proof "elementary but technical but elementary."

Remark 5.39. Here is an application: if we set $x_1 = \cdots = x_n = 1$, then all the cluster variables will be sent to integers. This remains true as long as the x_is are sent to elements of a ring which are units.

The following is due to many people, starting with Lee–Schiffler and ending with Gross–Hacking–Keel–Kontesevich. It uses scattering diagrams from mirror symmetry, which we do not have time for, so we will say nothing about its proof.

Theorem 5.40 (Positivity phenomenon)**.** Fix a cluster algebra $\mathcal{A}$ with initial seed $((x_1, \ldots, x_n), B)$. Then any cluster variable in $\mathcal{A}$ is a Laurent polynomial in the variables $(x_1, \ldots, x_n)$ with nonnegative coefficients.

Remark 5.41. The recursion can directly show that these polynomials take positive values on positive points.

Our last result is classification.

Definition 5.42 (Cartan companion)**.** Fix a skew-symmetric matrix B. Then we define the *Cartan companion* C to be the matrix C with coefficients

$$c_{ij} := \begin{cases} 2 & \text{if } i = j, \\ -|b_{ij}| & \text{if } i \neq j. \end{cases}$$

Example 5.43. The Cartan companion of the matrix $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ is the matrix $\left[\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right]$.

Theorem 5.44. Fix a cluster algebra $\mathcal{A}$ with initial seed $((x_1, \ldots, x_n), B)$. Then $\mathcal{A}$ has finite type if and only if B and all of its iterated mutations has positive-definite Cartan companion.

Sketch. In rank 2, our B looks like $\left[\begin{smallmatrix} 0 & b \\ -c & 0 \end{smallmatrix}\right]$, and we are interested in what happens to (x_1, x_2) when we alternatively apply the two mutations μ_1 and μ_2. This gives a sequence $\{x_1, x_2, x_3, \ldots\}$ of cluster variables. They satisfy

$$x_{m-1}x_{m+1} = \left[\begin{smallmatrix} x_m^b + 1 & \text{if } m \text{ is odd,} \\ x_m^c + 1 & \text{if } m \text{ is even.} \end{smallmatrix}\right]$$

One finds that the sequence $\{x_n\}$ is periodic (which is equivalent to finite type in this case because there are not so many mutations) if and only if $|bc| \leq 3$: the backward direction is a direct calculation, and the forward direction follows by some degree argument.

We now move on to higher rank. Let's start with the forward direction. We will show that if the Cartan companion avoids one of the finite Dynkin types fails to have finite type. Well, one should start by checking that B does not admit one of the rank-2 situations. It turns out that all affine Dynkin types (away from the finite type Dynkin types) and other infinite types have one of these infinite-type rank-2 subalgebras.

Lastly, we must show that B having finite Dynkin type implies that $\mathcal{A}$ has finite type, which is some direct calculation. Let Φ be the associated finite root system, and we choose a polarization with simple roots $\Pi = \{\alpha_1, \ldots, \alpha_n\}$. Now, for any cluster variable z, Theorem 5.37 tells us that

$$z = \frac{P(x_1, \ldots, x_n)}{x_1^{d_1} \cdots x_n^{d_n}},$$

where P is not divisible by any x_i with $d_i \neq 0$. We can now define a map which sends the cluster variables to $\operatorname{span} \Phi$ which sends $x_i \mapsto -\alpha_i$ for each i and in general sends z to $\sum_i d_i \alpha_i$. It turns out that z gets sent to a positive root by this map, and it turns out that this map is injective, so there are only finitely many cluster variables. In total, we find that the cluster variables are in bijection with $\Phi^+ \sqcup -\Pi$, so we being finite type follows. ■

Example 5.45. One can check that Example 5.26 is the cluster algebra from type A_3.

Non-Example 5.46. The seed $\left(x_1, x_2, \left[\begin{smallmatrix} 0 & -3 \\ 3 & 0 \end{smallmatrix}\right]\right)$ has all mutations of the mutation matrix with Cartan companion $\left[\begin{smallmatrix} 2 & 3 \\ 3 & 2 \end{smallmatrix}\right]$, which fails to be positive-definite. Indeed, this matrix admits a negative eigenvalue.

Remark 5.47. It turns out that the Cartan companions being positive-definite can be checked from the initial seed. This follows from the property of its mutation.

Remark 5.48. If $\mathcal{A}$ is acyclic, then $\mathcal{A}$ is finite type if and only if the initial seed's mutation matrix B has Cartan companion matrix of finite type. One needs to check the mutations only when $\mathcal{A}$ fails to be acyclic.

5.3 April 29

Here we go.

5.3.1 Cluster Varieties

For today, we will choose a finite type acyclic cluster algebra $\mathcal{A}$, for which we may assume has initial seed (x, B) with B whose lower-triangular submatrix is given by the associated Cartan matrix.

Definition 5.49 (cluster variety)**.** An *acyclic cluster variety* is an affine variety isomorphic to $\operatorname{Spec} \mathcal{A}$ for some acyclic cluster algebra $\mathcal{A}$. In the sequel, we will abbreviate this to a *cluster variety*.

Remark 5.50. There is a more general definition due to Fock–Goncharov and Gross–Hacking–Keel: namely, one can define a cluster variety to be covered by some algebraic tori T_Σ, which we glue together via some exchange relations. For example, there exist projective cluster varieties.

It will be helpful to have a version of a cluster variety which is manifestly integral and finite type.

Definition 5.51 (bounded cluster algebra)**.** Fix a cluster algebra $\mathcal{A}$ with initial seed (x, B). Then the *lower bound cluster algebra* $\mathcal{L}$ of $\mathcal{A}$ is the subalgebra generated over $\mathbb{Z}[x_1, \ldots, x_n]$ by the cluster variables which are one mutation away. Similarly, we define the *upper bound cluster algebra* $\mathcal{U}$ of $\mathcal{A}$ is the intersection

$$\mathcal{U} := \bigcap_z \mathbb{Z}\left[z_1^{\pm}, \ldots, z_n^{\pm}\right],$$

where z varies over all seeds.

Remark 5.52. Note that $\mathcal{L} \subseteq \mathcal{A} \subseteq \mathcal{U}$. Indeed, $\mathcal{L} \subseteq \mathcal{A}$ has no content, and $\mathcal{A} \subseteq \mathcal{U}$ follows from Theorem 5.37.

Remark 5.53. One can glue the tori $\operatorname{Spec} \mathbb{Z}\left[z_1^{\pm}, \ldots, z_n^{\pm}\right]$ together to produce $\mathcal{A}$ in some cases. This explains how Remark 5.50 generalizes our definition of a cluster variety.

In fact, in the acyclic case, these bounds collapse.

Theorem 5.54 (Berenstein–Fomin–Zelevinsky)**.** If $\mathcal{A}$ is an acyclic cluster algebra, then $\mathcal{L} = \mathcal{A} = \mathcal{U}$.

Sketch. One shows $\mathcal{U} \subseteq \mathcal{L}$ using the intersection lemma discussed in Theorem 5.37. ■

Example 5.55. From Example 5.26, one has that $\mathcal{A} = \mathbb{Z}[x_1, x_2, (x_2+1)/x_1, (x_1+1)/x_2]$. Indeed, the last cluster variable is $(x_1 + x_2 + 1)/(x_1x_2)$, which is $(x_2 + 1)/x_1 \cdot (x_1 + 1)/x_2 - 1$.

For example, one can now say something about friezes.

Definition 5.56. Fix an $n \times n$ generalized Cartan matrix. Then for each i, define the polynomial

$$f_{C,i} := x_i y_i - \prod_{j<o} x_j^{|c_{ji}|} - \prod_{j>i} x_j^{|c_{ji}|}$$

in $\mathbb{Z}[x_1, \ldots, x_n, y_1, \ldots, y_n]$. We now define $Y_1^{\text{frieze}}(C)$ to be the n-dimensional subvariety cut out by these functions. We may abbreviate $Y_1^{\text{frieze}}(C)$ to $Y_1^{\text{frieze}}(\Delta)$, where Δ denotes the generalized Dynkin type.

Remark 5.57. The notation Y_1 is intended to be similar to the notation of a modular curve.

Remark 5.58. If $\mathcal{A}$ is an acyclic cluster algebra, then it turns out that $\operatorname{Spec} \mathcal{A} \cong Y_1^{\text{frieze}}(C)$.

In fact, the remark upgrades as follows.

Theorem 5.59 (de Saint Germain–Huang–Lin)**.** Fix a generalized $n \times n$ generalized Cartan matrix C. Then there is a bijection between $Y_1^{\text{frieze}}(C)(\mathbb{N})$ and $\operatorname{Frieze}(C, \mathbb{N})$, which sends the variables $(x_1, \ldots, x_n)$ to the first column of the frieze. Here, $Y_1^{\text{frieze}}(C)(\mathbb{N})$ means that all the variables $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ must get sent to positive integers.

Remark 5.60. The values of the other variables $y_1, \ldots, y_n$ can also be seen in other entries of the corresponding frieze

Thus, $Y_1^{\text{frieze}}(C)$ can be understood as a moduli space of friezes.

5.3.2 Diophantine Geometry

We will get some motivation from the following comparison of trichotomies.

- Fano varieties have $-K_X$ ample.
- Calabi–Yau varieties have K_X trivial.
- General type varieties have K_X big.

On the other hand, one has some results for curves.

- Fano cures have genus zero, which have a dense subset of points after a field extension.
- Calabi–Yau curves have genus one, which are elliptic curves (once attaining a point), which remain hard to understand.
- General type curves have genus at least two, which have finitely many points.

The first point produces the following conjecture.

Conjecture 5.61 (Campana)**.** A Fano variety X over a number field K admits a finite extension L for which $X(L)$ is Zariski dense.

The last point produces the following conjecture.

Conjecture 5.62 (Bombieri–Lang)**.** Fix a variety X of general type over a number field K. Then $X(K)$ is not Zariski dense.

Remark 5.63. There are similar statements for integral points.

The Calabi–Yau case remains hard to understand; for example, one can hunt for infinite orbits, but there are other ways to find rational points. For example, there is a conjecture that a K3 surface admitting local points and no Brauer–Manin obstruction admit rational points.

This trichotomy is relevant for us because our cluster varieties turn out to live in the interesting Calabi–Yau case.

Theorem 5.64. Every cluster variety is log Calabi–Yau.

Remark 5.65. Here, we mean that one can glue the tori together with the "boundary" of varieties cut out by $x_i = 0$ along $\Omega_\Sigma = dx_1/x_1 \wedge \cdots \wedge dx_n/x_n$. The adjective "log" is present because Ω_Σ is basically the wedge of the $d \log x_i$s.

5.3.3 Counting Friezes

Note that it is not so hard to find rational points on $\operatorname{Spec} \mathcal{A}$, so we will focus on the harder question of finding integral points or even positive integral points, where a positive integral point is one in which the cluster variables go to positive integers.

It will help to have the following automorphism. The following is due to Kontsevich–Soiberlman, Keller, Fock–Goncharov, Gross–Hacking–Keel, and Fomin–Zelevinsky.

Theorem 5.66. Fix a generalized Dynkin type Δ.

(a) There is a volume-preserving automorphism $\tau \colon Y_1^{\text{frieze}}(\Delta) \to Y_1^{\text{frieze}}(\Delta)$.

(b) Furthermore, τ is periodic if and only if Δ is finite; and if Δ is finite, then the period divides $h + 2$, where h is the Coxeter number of Δ.

Remark 5.67. This τ is given by horizontally translating a frieze one column. Thus, the existence of the automorphism and the fact that it preserves volumes is not hard to check. The difficulty is the period; for example, for $\mathfrak{sl}_3$, this follows from the work of Gauss.

Remark 5.68. This τ has many names: it can be the Donaldson–Thomas transformation, the maximal green sequence, the Auslander–Reiten translation, or the Fomin–Zelevinsky twist.

We can use this τ to understand positive integral points.

Theorem 5.69 (Zhang)**.** Fix a generalized Dynkin type Δ. Then $Y_1^{\text{frieze}}(\Delta)(\mathbb{N})$ is finite if and only if Δ is finite type, and the values are given by the table in Theorem 5.20.

Sketch. In the case where Δ is infinite, the result is due to Morier–Genoud. Here are two arguments.

- One can construct many maps $\mathcal{A} \to \mathbb{Z}$, basically by sending one collection of cluster variables to 1. There are infinitely many sets of cluster variables, so this produces infinitely many friezes.

- Alternatively, as soon as we have a single positive integral point (by sending cluster variables to 1), one can use Theorem 5.66 to show that the orbit of this point has infinite size. Roughly speaking, one knows that τ preserves positivity and integrality, and one can show that the size of the point grows.

We now turn to the finite type case. Define $Y_0^{\text{frieze}}(\Delta)$ to be the quotient of $Y_1^{\text{frieze}}(\Delta)$ by $\langle\tau\rangle$, which is a quotient by a finite group because τ now has finite order. We let $\mathcal{O}_\tau(x)$ denote the image of $x \in Y_1^{\text{frieze}}(\Delta)$.

Now, view $\log \mathcal{O}_\tau(x)$ as a vector and then apply the matrix C to it. It was shown by Muller that $C \cdot \log \mathcal{O}_\tau(x)$ is in a bounded set. However, Lusztig–Tits showed that C^{-1} has positive rational coefficients if and only if C is finite type, so one finds that the coordinates of $\log \mathcal{O}_\tau(x)$ are bounded by the coordinates of $C^{-1}(1, \ldots, 1)$. This shows that there can only be finitely many orbits $\mathcal{O}_\tau(x)$, so there are only finitely many points because τ has finite order!

It remains to actually count the number of friezes. For example, for $\Delta = E_8$, the bound provided by the previous paragraph is on the order of 10^{50} for the height of the orbit, which is much too large. We will do this with the following trick.

Lemma 5.70. Fix a finite Dynkin type Δ. Let i be a vertex in Δ of degree 1. Then the intersection of $Y_1^{\text{frieze}}(\Delta)$ with $x_i = 1$ is isomorphic to $Y_1^{\text{frieze}}(\Delta \setminus \{1\})$.

Proof. One can see this explicitly on the level of the equations. Alternatively, this amounts to saying that a frieze whose bottom row is all 1s is simply a frieze for the smaller Dynkin type. ■

Thus, we may assume that all nodes i of degree 1 have $x_i \geq 2$. From here, a long naïve calculation over a few days can complete the proof. For example, in E_8, there are only four new friezes not from a smaller Dynkin type. ■

Remark 5.71. It is an interesting question to find the number of τ-orbits among the friezes. Professor Zhang has threatened to put this on the homework.

Example 5.72. It turns out that $Y_1^{\text{frieze}}\left(\left[\begin{smallmatrix} 2 & -a \\ -b & 2 \end{smallmatrix}\right]\right)$ can be presented with the equation

$$xyz = (x^a + 1)^b + y.$$

It follows that there are 5 points if $ab = 1$, 6 points if $ab = 2$, and 9 points of $ab = 9$, and there are infinitely many points if $ab \geq 4$. More generally, if $G(x, y)$ has positive integral coefficients, we are able to determine when $xyz = G(x, y)$ should have infinitely many positive integral solutions. For example, there are infinitely many when $\deg_x G$ and $\deg_y G$ are both at least three and has non-vanishing constant term.

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